

Siemens PTI Report R165-20

MISO DPP 2019 West Area Phase 1 Study

Prepared for

MISO

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Executive Summary

This report presents results of a System Impact Study (SIS) performed to evaluate the interconnection of the DPP 2019 Phase 1 West Area Group (DPP West Area) generating facilities. The results for 2024 scenario are summarized below.

1.1 Project List

The DPP West Area study group has fifty (50) generation projects with a combined nameplate rating of 8126 MW. The DPP West Area generating facilities are listed in Table ES-1. The modeling details and projects' slider diagrams are shown in Appendix B.

Table ES-1: Generating Facilities in DPP 2019 West Area Group

MISO Project #	Service Type	TO	County	State	Point of Interconnection	Fuel Type	ERIS Output	NRIS Output	SH MW	SPK MW
J1190	NRIS	ITCM	Clinton	IA	Calamus East 161 kV	Solar	200	200	0.0	200.0
J1212	NRIS	XCEL	Murray	MN	Chanarambie-Ellsborough 115 kV	Solar	60	60	0.0	60.0
J1218	NRIS	MEC	Muscatine	IA	Sub 92 345 kV	Solar	200	200	0.0	200.0
J1221	NRIS	MP	Wadena	MN	Riverton 230 kV	Solar	150	150	0.0	150.0
J1229	NRIS	MEC	Muscatine	IA	Sub 92 345 kV	Battery	50	50	±50	50.0
J1230	NRIS	MP	Wadena	MN	Riverton 230 kV	Battery	40	40	±40	40.0
J1270	NRIS	MEC	Palo Alto	IA	Kossuth - O'Brien 345 kV	Wind	200	200	200.0	31.2
J1284	NRIS	ITCM	Linn	IA	Duane Arnold - Tiffin 345 kV	Solar	250	250	0.0	250.0
J1285	NRIS	ITCM	Linn	IA	Duane Arnold - Tiffin 345 kV	Battery	50	50	±50	50.0
J1298	NRIS	SMPMPA	Freeborn	MN	Hayward - Murphy Creek 161 kV	Solar	150	150	0.0	150.0
J1300	NRIS	ITCM	Nobles	MN	Brewster - Elk 161 kV	Solar	120	120	0.0	120.0
J1312	NRIS	XCEL	McCook	SD	Grant 115 kV	Solar	50	50	0.0	50.0
J1313	NRIS	ITCM	Greene	IA	Karma 161 kV	Solar	100	100	0.0	100.0
J1314	NRIS	DPC	Polk	WI	Apple River 161 kV	Solar	100	100	0.0	100.0
J1315	NRIS	GRE	Cottonwood, Murray, Redwood	MN	Lyon County - Cedar Mountain 345 kV	Wind	600	600	600.0	93.6
J1321	NRIS	MEC	Palo Alto	IA	O'Brien - Kossuth 345 kV	Battery	65	65	±65	65.0
J1325	NRIS	ITCM	Rock	MN	Magnolia 161 kV	Solar	200	200	0.0	200.0
J1328	NRIS	OTP	Grant	SD	Big Stone South 345 kV	Solar	100	100	0.0	100.0

MISO Project #	Service Type	TO	County	State	Point of Interconnection	Fuel Type	ERIS Output	NRIS Output	SH MW	SPK MW
J1337	NRIS	XCEL	Sherburne	MN	Sherburne County 345 kV	Solar	300	300	0.0	300.0
J1341	NRIS	ITCM	Buchanan	IA	Hazleton 345 kV	Wind	398	398	398.0	62.1
J1349	NRIS	XCEL	Cottonwood	MN	Crandall 345 kV	Hybrid	335 (60 PV, 275 Wind)	335	0 (PV) 275 (Wind)	60 (PV) 42.9 (Wind)
J1359	NRIS	ITCM	Linn	IA	Oakridge 69 kV	Gas	95	95	0.0	95.0
J1365	NRIS	ITCM	Tama	IA	Traer 161 kV	Wind	169.4	169.4	169.4	26.4
J1371	NRIS	OTP	Yellow Medicine	MN	Astoria 345 kV	Wind	150	150	150.0	23.4
J1395	NRIS	GRE	Lyon	MN	Nest Lake - Lyon County 345 kV	Solar	200	200	0.0	200.0
J1413	NRIS	ITCM	Howard, Mitchell	IA	Hazleton - Mitchell County 345 kV (5.4 mi from Mitchell Co, 65 mi from Hazleton)	Solar	300	300	0.0	300.0
J1416	NRIS	ITCM	Worth	IA	Worth County - Glenworth 161 kV	Solar	149	149	0.0	149.0
J1417	NRIS	ITCM	Worth	IA	Ledyard - Colby 345 kV	Battery	75	75	±75	75.0
J1426	NRIS	GRE	Benton	MN	Benton County 115 kV	Solar	100	100	0.0	100.0
J1427	NRIS	ITCM	Benton	IA	Duane Arnold 345 kV	Wind	400	400	400.0	62.4
J1428	NRIS	OTP	Cass	ND	Buffalo 345 kV	Solar	200	200	0.0	200.0
J1432	NRIS	ITCM	Clinton	IA	Salem - Rock Creek 345 kV	Wind	250	250	250.0	39.0
J1435	NRIS	ITCM	Dubuque	IA	Center Grove 161 kV	Battery	100	100	±100	100.0
J1438	NRIS	ITCM	Fayette	IA	Windsor - Hazleton 161 kV	Solar	100	100	0.0	100.0
J1444	NRIS	ITCM	Linn	IA	Duane Arnold 345 kV	Solar	200	200	0.0	200.0
J1445	NRIS	XCEL	Benton	MN	Mayhew Lake 115 kV	Solar	100	100	0.0	100.0
J1446	NRIS	XCEL	Wright	MN	Lake Pulaski 115 kV	Solar	150	150	0.0	150.0
J1456	NRIS	OTP	Sheridan	ND	Harvey - Underwood 230 kV	Wind	300	300	300.0	46.8
J1461	NRIS	XCEL	Carver	MN	McLeod - Blue Lake 230 kV	Solar	50	50	0.0	50.0
J1467	NRIS	ITCM	Fayette	IA	Windsor - Hazleton 161 kV	Wind	150	150	150.0	23.4
J1468	NRIS	XCEL	Dakota	MN	West Hastings 115 kV	Battery	100	100	±100	100.0
J1471	NRIS	ITCM	Lee	IA	Wever - Burlington Generator 161 kV	Solar	200	200	0.0	200.0
J1473	NRIS	XCEL	Chisago	MN	Lawrence Creek - Shafer 115 kV	Solar	100	100	0.0	100.0
J1474	NRIS	XCEL	Chippewa	WI	Pine Lake - Eagle Point 115 kV	Solar	200	200	0.0	200.0

MISO Project #	Service Type	TO	County	State	Point of Interconnection	Fuel Type	ERIS Output	NRIS Output	SH MW	SPK MW
J1476	NRIS	ITCM	Union	IA	Creston East 69 kV (J1132)	Battery	25	25	±25	25.0
J1477	NRIS	MEC	Scott	IA	Sub 56 161 kV (J1131)	Battery	50	50	±50	50.0
J1478	NRIS	ITCM	Des Moines	IA	Agency 69 kV (J1135)	Battery	25	25	±25	25.0
J1479	NRIS	ITCM	Des Moines	IA	Burlington 161 kV	Battery	100	100	±100	100.0
J1485	NRIS	OTP	Deuel	SD	Astoria 345 kV (J493/J510)	Wind	250	250	250.0	39.0
J1487	NRIS	ITCM	Freeborn	MN	Freeborn 161 kV	Wind	120	120	120.0	18.7

1.2 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

Per FERC Order 827, non-synchronous generation projects in the DPP 2019 West Area study group are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

The reactive power requirement analysis results are summarized as following:

- J1190, J1218, J1221, J1270, J1284, J1298, J1300, J1328, J1341, J1365, J1371, J1395, J1456, J1471, and J1485 do not meet the reactive power requirements per FERC Order 827.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection Service as of the System Impact Study report date. The total cost of network upgrades in the interconnection plan required for each generation project is listed in Table ES-2. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies.

Table ES-2: Total Cost of Network Upgrades for DPP 2019 West Area Generation Projects

Project Num	ERIS Network Upgrades (\$)					NRIS Network Upgrades (\$)	Interconnection Substation TO NUs (\$)	TO's Interconnection Facilities (TOIF)	SNU (\$)	Total Network Upgrade Cost (Exclude TOIF & Affected System) (\$)	M2 Received (\$)	M3 (\$)
	MWEX Voltage Stability	MISO Thermal & Voltage	GRE LPC	OTP LPC	DPC LPC							
J1190	\$0	\$0	\$0	\$0	\$0	\$0	\$1,500,000	NA	\$0	\$1,500,000	\$800,000	\$0
J1212	\$0	\$0	\$0	\$0	\$0	\$0	\$7,500,000	NA	\$0	\$7,500,000	\$240,000	\$510,000
J1218	\$0	\$0	\$0	\$0	\$0	\$1,583,881	\$11,200,000	NA	\$0	\$12,783,881	\$800,000	\$478,388
J1221	\$0	\$0	\$16,657,895	\$0	\$0	\$4,028,944	\$7,000,000	NA	\$0	\$27,686,838	\$600,000	\$2,168,684
J1229	\$0	\$0	\$0	\$0	\$0	\$395,970	\$2,800,000	NA	\$0	\$3,195,970	\$200,000	\$119,597
J1230	\$0	\$11,903,900	\$4,442,105	\$0	\$0	\$1,074,385	\$0	NA	\$0	\$17,420,390	\$160,000	\$1,582,039
J1270	\$0	\$165,517	\$0	\$0	\$0	\$2,498,119	\$15,750,000	NA	\$0	\$18,413,636	\$800,000	\$1,041,364
J1284	\$0	\$0	\$0	\$0	\$0	\$2,146,315	\$8,330,000	NA	\$0	\$10,476,315	\$1,000,000	\$47,631
J1285	\$0	\$13,155	\$0	\$0	\$0	\$429,263	\$1,770,000	NA	\$0	\$2,212,418	\$200,000	\$21,242
J1298	\$0	\$0	\$0	\$0	\$0	\$14,867,116	\$0	NA	\$0	\$14,867,116	\$600,000	\$886,712
J1300	\$0	\$40,901,203	\$0	\$0	\$0	\$1,357,116	\$5,000,000	NA	\$0	\$47,258,319	\$480,000	\$4,245,832
J1312	\$0	\$0	\$0	\$0	\$0	\$300,428	\$5,000,000	NA	\$0	\$5,300,428	\$200,000	\$330,043
J1313	\$0	\$0	\$0	\$0	\$0	\$0	\$2,000,000	NA	\$0	\$2,000,000	\$400,000	\$0
J1314	\$0	\$2,636,000	\$0	\$0	\$0	\$776,665	\$750,000	NA	\$0	\$4,162,665	\$400,000	\$16,267
J1315	\$0	\$7,330,000	\$1,566,000	\$0	\$0	\$3,649,869	\$12,800,000	NA	\$0	\$25,345,869	\$2,400,000	\$134,587
J1321	\$0	\$53,711	\$0	\$0	\$0	\$748,500	\$100,000	NA	\$0	\$902,211	\$260,000	\$0
J1325	\$0	\$82,848,797	\$0	\$0	\$0	\$15,505,910	\$7,500,000	NA	\$0	\$105,854,706	\$800,000	\$9,785,471
J1328	\$0	\$0	\$0	\$0	\$0	\$836,898	\$1,250,000	NA	\$0	\$2,086,898	\$400,000	\$0
J1337	\$0	\$0	\$21,743,890	\$0	\$0	\$2,311,981	\$10,000,000	NA	\$0	\$34,055,871	\$1,200,000	\$2,205,587
J1341	\$0	\$11,199,554	\$0	\$0	\$0	\$8,066,173	\$6,000,000	NA	\$0	\$25,265,727	\$1,600,000	\$926,573
J1349	\$0	\$202,939	\$0	\$0	\$0	\$4,247,082	\$10,000,000	NA	\$0	\$14,450,022	\$1,340,000	\$105,002
J1359	\$0	\$450,000	\$0	\$0	\$0	\$735,559	\$2,500,000	NA	\$0	\$3,685,559	\$380,000	\$0
J1365	\$0	\$0	\$0	\$0	\$0	\$11,597,084	\$5,000,000	NA	\$0	\$16,597,084	\$680,000	\$979,708
J1371	\$0	\$0	\$0	\$0	\$0	\$1,262,324	\$2,000,000	NA	\$0	\$3,262,324	\$600,000	\$0
J1395	\$0	\$0	\$0	\$0	\$0	\$1,323,416	\$12,800,000	NA	\$0	\$14,123,416	\$800,000	\$612,342
J1413	\$0	\$0	\$0	\$0	\$0	\$14,327,941	\$10,000,000	NA	\$0	\$24,327,941	\$1,200,000	\$1,232,794
J1416	\$0	\$0	\$0	\$0	\$0	\$10,177,176	\$5,000,000	NA	\$0	\$15,177,176	\$596,000	\$921,718
J1417	\$0	\$55,008	\$0	\$0	\$0	\$966,694	\$10,000,000	NA	\$0	\$11,021,702	\$300,000	\$802,170
J1426	\$0	\$0	\$0	\$0	\$0	\$728,039	\$4,000,000	NA	\$0	\$4,728,039	\$400,000	\$72,804

Executive Summary

Project Num	ERIS Network Upgrades (\$)					NRIS Network Upgrades (\$)	Interconnection Substation TO NUs (\$)	TO's Interconnection Facilities (TOIF)	SNU (\$)	Total Network Upgrade Cost (Exclude TOIF & Affected System) (\$)	M2 Received (\$)	M3 (\$)
	MWEX Voltage Stability	MISO Thermal & Voltage	GRE LPC	OTP LPC	DPC LPC							
J1427	\$0	\$1,612,479	\$0	\$0	\$0	\$3,261,011	\$2,000,000	NA	\$0	\$6,873,490	\$1,600,000	\$0
J1428	\$0	\$11,000,000	\$0	\$0	\$0	\$7,852,193	\$1,250,000	NA	\$0	\$20,102,193	\$800,000	\$1,210,219
J1432	\$0	\$0	\$0	\$0	\$0	\$1,580,847	\$10,000,000	NA	\$0	\$11,580,847	\$1,000,000	\$158,085
J1435	\$0	\$8,700,000	\$0	\$0	\$0	\$615,617	\$5,000,000	NA	\$0	\$14,315,617	\$400,000	\$1,031,562
J1438	\$0	\$0	\$0	\$0	\$0	\$5,460,559	\$5,000,000	NA	\$0	\$10,460,559	\$400,000	\$646,056
J1444	\$0	\$0	\$0	\$0	\$0	\$1,630,505	\$2,000,000	NA	\$0	\$3,630,505	\$800,000	\$0
J1445	\$0	\$0	\$0	\$0	\$0	\$615,477	\$5,000,000	NA	\$0	\$5,615,477	\$400,000	\$161,548
J1446	\$0	\$0	\$152,690	\$0	\$0	\$1,100,572	\$5,000,000	NA	\$0	\$6,253,262	\$600,000	\$25,326
J1456	\$0	\$0	\$0	\$2,500,000	\$0	\$3,314,937	\$5,000,000	NA	\$0	\$10,814,937	\$1,200,000	\$0
J1461	\$0	\$0	\$0	\$0	\$0	\$318,721	\$10,000,000	NA	\$0	\$10,318,721	\$200,000	\$831,872
J1467	\$0	\$12,454,813	\$0	\$0	\$0	\$10,327,374	\$5,000,000	NA	\$0	\$27,782,186	\$600,000	\$2,178,219
J1468	\$0	\$0	\$0	\$0	\$0	\$0	\$5,000,000	NA	\$0	\$5,000,000	\$600,000	\$0
J1471	\$0	\$4,288,268	\$0	\$0	\$0	\$10,289,777	\$1,500,000	NA	\$0	\$16,078,046	\$800,000	\$807,805
J1473	\$0	\$0	\$0	\$0	\$0	\$556,632	\$7,500,000	NA	\$0	\$8,056,632	\$400,000	\$405,663
J1474	\$0	\$0	\$0	\$0	\$0	\$3,183,312	\$7,500,000	NA	\$0	\$10,683,312	\$800,000	\$268,331
J1476	\$0	\$23,860,149	\$0	\$0	\$0	\$0	\$0	NA	\$0	\$23,860,149	\$100,000	\$2,286,015
J1477	\$0	\$0	\$0	\$0	\$0	\$359,073	\$100,000	NA	\$0	\$459,073	\$200,000	\$0
J1478	\$0	\$19,952	\$0	\$0	\$0	\$1,059,769	\$0	NA	\$0	\$1,079,722	\$100,000	\$7,972
J1479	\$0	\$171,779	\$0	\$0	\$0	\$5,874,521	\$7,000,000	NA	\$0	\$13,046,300	\$400,000	\$904,630
J1485	\$0	\$0	\$0	\$0	\$0	\$1,992,848	\$2,750,000	NA	\$0	\$4,742,848	\$1,000,000	\$0
J1487	\$0	\$71,404	\$0	\$0	\$0	\$6,528,906	\$1,500,000	NA	\$0	\$8,100,309	\$480,000	\$330,031
Total (\$)	\$0	\$219,938,628	\$44,562,580	\$2,500,000	\$0	\$171,865,500	\$255,650,000	NA	\$0	\$694,516,708	\$32,716,000	\$40,479,886

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

1.4 Per Project Summary

This section provides estimated cost of Network Upgrades on a per project basis for the 2024 scenario. The shared cost of Network Upgrades for all the generation projects are listed below.

The Interconnection Customers are required to mitigate the constraints observed from the 2024 scenario.

1.4.1 J1190 Summary

Network Upgrade	Cost	J1190	NUs Type	Self Funding?
No Network Upgrades		\$0		
Total Cost Per Project for Actual NRIS Elections for each Project		\$0		

1.4.2 J1212 Summary

Network Upgrade	Cost	J1212	NUs Type	Self Funding?
No Network Upgrades		\$0		
Total Cost Per Project for Actual NRIS Elections for each Project		\$0		

1.4.3 J1218 Summary

Network Upgrade	Cost	J1218	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$65,500	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$1,242,773	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$275,608	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,583,881		

1.4.4 J1221 Summary

Network Upgrade	Cost	J1221	NUs Type	Self Funding?
Benton Co-Mud Lake 230 kV	\$21,100,000	\$16,657,895	J1221&J1230 LPC	GRE: No MP: Yes
Stone Lake 345/161 Transformer Upgrade	\$4,000,000	\$3,157,895	NRIS	No
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$714,377	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$32,289	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$21,535	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$93,141	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$9,707	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$20,686,838		

1.4.5 J1229 Summary

Network Upgrade	Cost	J1229	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$16,375	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$310,693	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$68,902	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$395,970		

1.4.6 J1230 Summary

Network Upgrade	Cost	J1230	NUs Type	Self Funding?
Riverton-Mud Lake 230 kV	\$600,000	\$600,000	MISO SH	MP: Yes GRE: No
Brainerd-Riverton 115 kV	\$2,361,600	\$2,361,600	MISO SH	Yes
Inman-Wing River 230 kV	\$125,000	\$125,000	MISO SH	No
Wing River-Riverton 230 kV	\$3,237,300	\$3,237,300	MISO SH	GRE: No MP: Yes
Wing River-Verndale 115 kV	\$1,500,000	\$1,500,000	MISO SH	GRE: No MP: Yes
Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C	\$4,080,000	\$4,080,000	MISO SH	Yes
Benton Co-Mud Lake 230 kV	\$21,100,000	\$4,442,105	J1221&J1230 LPC	GRE: No MP: Yes
Stone Lake 345/161 Transformer Upgrade	\$4,000,000	\$842,105	NRIS	No
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$190,500	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$8,610	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$5,743	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$24,838	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$2,588	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$17,420,390		

1.4.7 J1270 Summary

Network Upgrade	Cost	J1270	NUs Type	Self Funding?
Montgomery-J1145 POI 345 kV	\$28,000,000	\$165,517	MISO SH	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$2,072,883	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$42,258	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$17,991	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$12,777	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$352,210	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$2,663,636		

1.4.8 J1284 Summary

Network Upgrade	Cost	J1284	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$72,842	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$1,382,082	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$307,572	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$296,100	NRIS	ITCM: Yes MEC: Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$87,718	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$2,146,315		

1.4.9 J1285 Summary

Network Upgrade	Cost	J1285	NUs Type	Self Funding?
Arnold-J1284 POI 345 kV	\$240,000	\$13,155	MISO SH	Yes
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$14,568	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$276,416	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$61,514	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$59,220	NRIS	ITCM: Yes MEC: Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$17,544	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$442,418		

1.4.10 J1298 Summary

Network Upgrade	Cost	J1298	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$27,248	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$516,984	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$115,499	NRIS	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$767,062	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$51,866	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$40,926	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$15,542	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$33,105	NRIS	Yes
Murphy Creek Transformer Upgrade	\$3,500,000	\$2,533,557	NRIS	No

Network Upgrade	Cost	J1298	NUs Type	Self Funding?
Austin - Adams 161 kV Line Rebuild	\$15,750,000	\$6,669,140	NRIS	ITCM: Yes SMMPA: No
Pleasant Valley-Byron 161 kV Rebuild	\$24,900,000	\$4,096,187	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$14,867,116		

1.4.11 J1300 Summary

Network Upgrade	Cost	J1300	NUs Type	Self Funding?
Split Rock-Rock Co. 161 kV	\$20,550,000	\$7,706,250	MISO PK	Yes
Rock Co.-Magnolia 161 kV	\$18,000,000	\$6,742,868	MISO PK	XEL: Yes ITCM: Yes
Split Rock 161-115-36.2 kV xfmr #6	\$3,000,000	\$1,123,811	MISO PK	Yes
Lakefield Corner station and J1300 - Lakefield Corner new line	\$62,600,000	\$25,328,274	MISO PK	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$876,540	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$25,159	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$12,281	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$7,603	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$61,879	NRIS	Yes
Brewster-Heron Lake 161 Structure Remediation	\$960,000	\$373,653	NRIS	GRE: No ITCM: Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$42,258,319		

1.4.12 J1312 Summary

Network Upgrade	Cost	J1312	NUs Type	Self Funding?
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$284,262	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$8,814	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$4,688	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$2,664	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$300,428		

1.4.13 J1313 Summary

Network Upgrade	Cost	J1313	NUs Type	Self Funding?
No Network Upgrades		\$0		
Total Cost Per Project for Actual NRIS Elections for each Project		\$0		

1.4.14 J1314 Summary

Network Upgrade	Cost	J1314	NUs Type	Self Funding?
Apple River 161-69 kV xfmr #2	\$2,636,000	\$2,636,000	MISO PK	Yes
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$16,673	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$316,354	NRIS	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$402,432	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$20,919	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$14,011	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$6,276	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$3,412,665		

1.4.15 J1315 Summary

Network Upgrade	Cost	J1315	NUs Type	Self Funding?
Franklin-Cedar Mountain 115 kV	\$1,000,000	\$1,000,000	MISO SH	XEL: Yes GRE: No
Lyon Co 115-69 kV xfmr	\$1,500,000	\$1,500,000	MISO SH	Yes
Crystal Foods Tap-Arlington 69 kV	\$4,830,000	\$4,830,000	MISO SH	Yes
Sheridan Tap-Wabasso 69 kV	\$1,566,000	\$1,566,000	J1426 LPC	GRE: No ITCM: Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$2,989,667	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$133,639	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$85,355	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$256,409	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$40,181	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$144,618	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$12,545,869		

1.4.16 J1321 Summary

Network Upgrade	Cost	J1321	NUs Type	Self Funding?
Montgomery-J1145 POI 345 kV	\$28,000,000	\$53,711	MISO SH	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$724,378	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$14,037	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$5,839	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$4,245	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$802,211		

1.4.17 J1325 Summary

Network Upgrade	Cost	J1325	NUs Type	Self Funding?
Split Rock-Rock Co. 161 kV	\$20,550,000	\$12,843,750	MISO PK	Yes
Rock Co.-Magnolia 161 kV	\$18,000,000	\$11,257,132	MISO PK	XEL: Yes ITCM: Yes
Split Rock 161-115-36.2 kV xfmr #6	\$3,000,000	\$1,876,189	MISO PK	Yes
Lakefield Corner station and J1300 - Lakefield Corner new line	\$62,600,000	\$37,271,726	MISO PK	Yes
Magnolia-Elk 161 kV	\$19,600,000	\$19,600,000	MISO PK	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$1,348,982	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$39,410	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$19,802	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$11,904	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$64,966	NRIS	Yes
Elk-J1300 POI 161 kV Rebuild	\$12,700,000	\$12,700,000	NRIS	Yes
Wolf Tap-Rushmore 69 kV Rebuild	\$217,500	\$217,500	NRIS	Yes
Magnolia Transformer Terminal upgrade with ERIS Upgrade	\$517,000	\$517,000	NRIS	Yes
Brewster-Heron Lake 161 Structure Remediation	\$960,000	\$586,347	NRIS	GRE: No ITCM: Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$98,354,706		

1.4.18 J1328 Summary

Network Upgrade	Cost	J1328	NUs Type	Self Funding?
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$536,973	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$20,429	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$12,546	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$55,266	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$6,148	NRIS	Yes
Granite Falls-Minn Valley 230 terminal upgrades	\$1,000,000	\$205,536	NRIS	XEL: Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$836,898		

1.4.19 J1337 Summary

Network Upgrade	Cost	J1337	NUs Type	Self Funding?
Bunker Lake-Blaine 230 kV	\$6,962,690	\$6,962,690	J1426 LPC	No
Bunker Lake-Elk River 230 kV	\$8,158,000	\$8,158,000	J1426 LPC	No
Elk River-Monticello 230 kV	\$6,623,200	\$6,623,200	J1426 LPC	GRE: No XEL: Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$1,462,687	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$68,991	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$47,395	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$645,602	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$20,723	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$66,584	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$24,055,871		

1.4.20 J1341 Summary

Network Upgrade	Cost	J1341	NUs Type	Self Funding?
Hazleton-Dundee 161 kV	\$13,830,000	\$9,150,638	MISO SH	ITCM: Yes CIPCO: No
Liberty-Hickory Creek 161 kV	\$3,190,000	\$1,866,086	MISO SH	ITCM: Yes CIPCO: No
Arnold-J1284 POI 345 kV	\$240,000	\$77,670	MISO SH	Yes
Morgan Valley-Tiffin 345 kV	\$300,000	\$105,160	MISO SH	ITCM: Yes MEC: Yes
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$94,857	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$1,799,769	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$398,057	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$272,075	NRIS	ITCM: Yes MEC: Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$81,039	NRIS	Yes
Pleasant Valley-Byron 161 kV Rebuild	\$24,900,000	\$5,420,377	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$19,265,727		

1.4.21 J1349 Summary

Network Upgrade	Cost	J1349	NUs Type	Self Funding?
Montgomery-J1145 POI 345 kV	\$28,000,000	\$202,939	MISO SH	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$1,943,254	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$61,123	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$31,487	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$18,416	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$192,803	NRIS	Yes
ITCM Lakefield-XEL Lakefield 345 Structure Replacements	\$2,000,000	\$2,000,000	NRIS	ITCM: Yes XEL: Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$4,450,022		

1.4.22 J1359 Summary

Network Upgrade	Cost	J1359	NUs Type	Self Funding?
PCI J2-Oakridge 69 kV	\$450,000	\$450,000	MISO PK	ITCM: Yes CIPCO: No
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$27,176	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$515,631	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$114,583	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$60,240	NRIS	ITCM: Yes MEC: Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$17,928	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,185,559		

1.4.23 J1365 Summary

Network Upgrade	Cost	J1365	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$39,026	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$740,463	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$166,536	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$69,569	NRIS	ITCM: Yes MEC: Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$21,491	NRIS	Yes
Laurel - Jasper 161 kV Rebuild	\$10,560,000	\$10,560,000	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$11,597,084		

1.4.24 J1371 Summary

Network Upgrade	Cost	J1371	NUs Type	Self Funding?
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$819,747	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$30,503	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$18,551	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$76,031	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$9,188	NRIS	Yes
Granite Falls-Minn Valley 230 terminal upgrades	\$1,000,000	\$308,304	NRIS	XEL: Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,262,324		

1.4.25 J1395 Summary

Network Upgrade	Cost	J1395	NUs Type	Self Funding?
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$1,108,475	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$42,912	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$26,807	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$89,165	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$12,912	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$43,145	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,323,416		

1.4.26 J1413 Summary

Network Upgrade	Cost	J1413	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$60,263	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$1,143,402	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$253,587	NRIS	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$1,066,207	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$134,620	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$182,683	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$40,204	NRIS	Yes
Pleasant Valley-Byron 161 kV Rebuild	\$24,900,000	\$11,446,975	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$14,327,941		

1.4.27 J1416 Summary

Network Upgrade	Cost	J1416	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$27,115	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$514,474	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$115,365	NRIS	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$529,550	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$52,877	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$24,676	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$15,876	NRIS	Yes
Austin - Adams 161 kV Line Rebuild	\$15,750,000	\$6,624,679	NRIS	ITCM: Yes SMMPA: No
Pleasant Valley-Byron 161 kV Rebuild	\$24,900,000	\$2,154,113	NRIS	Yes
Glenworth - Hayward 161 kV terminal upgrades	\$130,000	\$118,453	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$10,177,176		

1.4.28 J1417 Summary

Network Upgrade	Cost	J1417	NUs Type	Self Funding?
Montgomery-J1145 POI 345 kV	\$28,000,000	\$55,008	MISO SH	Yes
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$12,704	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$241,039	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$54,447	NRIS	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$589,807	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$23,779	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$8,087	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$7,145	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$18,139	NRIS	Yes
Glenworth - Hayward 161 kV terminal upgrades	\$130,000	\$11,547	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,021,702		

1.4.29 J1426 Summary

Network Upgrade	Cost	J1426	NUs Type	Self Funding?
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$488,157	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$22,670	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$15,441	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$173,430	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$6,810	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$21,531	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$728,039		

1.4.30 J1427 Summary

Network Upgrade	Cost	J1427	NUs Type	Self Funding?
Liberty-Hickory Creek 161 kV	\$3,190,000	\$1,323,914	MISO SH	ITCM: Yes CIPCO: No
Arnold-J1284 POI 345 kV	\$240,000	\$126,253	MISO SH	Yes
Morgan Valley-Tiffin 345 kV	\$300,000	\$162,312	MISO SH	ITCM: Yes MEC: Yes
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$112,305	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$2,130,827	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$473,363	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$419,968	NRIS	ITCM: Yes MEC: Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$124,547	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$4,873,490		

1.4.31 J1428 Summary

Network Upgrade	Cost	J1428	NUs Type	Self Funding?
Buffalo 345-115-41.6 kV xfmr	\$3,000,000	\$3,000,000	MISO PK	Yes
Maple River 345-230-13.8 kV xfmr #1	\$4,000,000	\$4,000,000	MISO PK	OTP: Yes
Maple River 345-230-13.8 kV xfmr #2	\$4,000,000	\$4,000,000	MISO PK	OTP: Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$967,981	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$40,764	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$26,069	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$118,405	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$12,265	NRIS	Yes
Buffalo-Castleton 115 kV Reconductor	\$4,250,000	\$3,343,355	NRIS	Yes
Mapleton-Castleton 115 kV Reconductor	\$4,250,000	\$3,343,355	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$18,852,193		

1.4.32 J1432 Summary

Network Upgrade	Cost	J1432	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$65,633	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$1,245,289	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$269,925	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,580,847		

1.4.33 J1435 Summary

Network Upgrade	Cost	J1435	NUs Type	Self Funding?
Hickory Creek - Turkey River 13 miles 161 kV, DC with MVP 5	\$8,700,000	\$8,700,000	MISO SH	Yes
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$24,629	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$467,298	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$101,861	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$21,830	NRIS	ITCM: Yes MEC: Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$9,315,617		

1.4.34 J1438 Summary

Network Upgrade	Cost	J1438	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$23,237	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$467,298	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$97,457	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$56,103	NRIS	ITCM: Yes MEC: Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$16,464	NRIS	Yes
J1438 POI - Hazelton 161 kV Rebuild	\$12,000,000	\$4,800,000	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$5,460,559		

1.4.35 J1444 Summary

Network Upgrade	Cost	J1444	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$56,152	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$1,065,414	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$236,682	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$209,984	NRIS	ITCM: Yes MEC: Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$62,273	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,630,505		

1.4.36 J1445 Summary

Network Upgrade	Cost	J1445	NUs Type	Self Funding?
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$489,943	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$22,647	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$15,393	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$59,202	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$6,802	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$21,489	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$615,477		

1.4.37 J1446 Summary

Network Upgrade	Cost	J1446	NUs Type	Self Funding?
Dickinson-Dickinson SS 115 kV	\$152,690	\$152,690	J1426 LPC	No
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$742,059	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$34,881	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$23,966	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$254,843	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$10,474	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$34,350	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,253,262		

1.4.38 J1456 Summary

Network Upgrade	Cost	J1456	NUs Type	Self Funding?
1×15 Mvar switched reactor at J1456 POI 230 kV	\$2,500,000	\$2,500,000	OTP LPC	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$1,391,249	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$57,364	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$35,743	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$17,291	NRIS	Yes
Buffalo-Castleton 115 kV Reconductor	\$4,250,000	\$906,645	NRIS	Yes
Mapleton-Castleton 115 kV Reconductor	\$4,250,000	\$906,645	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$5,814,937		

1.4.39 J1461 Summary

Network Upgrade	Cost	J1461	NUs Type	Self Funding?
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$256,580	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$11,732	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$8,006	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$26,950	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$3,525	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$11,927	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$318,721		

1.4.40 J1467 Summary

Network Upgrade	Cost	J1467	NUs Type	Self Funding?
Windsor 161-69 kV xfmr	\$3,500,000	\$3,500,000	MISO SH	SMPMA: No ITCM: Yes
Luana-Monona 69 kV	\$1,200,000	\$1,200,000	MISO SH	Yes
Luana-Postville 69 kV	\$3,020,000	\$3,020,000	MISO SH	ITCM: Yes DPC: Yes
Hazleton-Dundee 161 kV	\$13,830,000	\$4,679,362	MISO SH	ITCM: Yes CIPCO: No
Arnold-J1284 POI 345 kV	\$240,000	\$22,922	MISO SH	Yes
Morgan Valley-Tiffin 345 kV	\$300,000	\$32,528	MISO SH	ITCM: Yes MEC: Yes
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$34,706	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$658,494	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$145,760	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$82,719	NRIS	ITCM: Yes MEC: Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$24,696	NRIS	Yes
J1438 POI - Hazelton 161 kV Rebuild	\$12,000,000	\$7,200,000	NRIS	Yes
J1467 POI-J1438 POI 161 kV Rebuild	\$2,181,000	\$2,181,000	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$22,782,186		

1.4.41 J1468 Summary

Network Upgrade	Cost	J1468	NUs Type	Self Funding?
No Network Upgrades		\$0		
Total Cost Per Project for Actual NRIS Elections for each Project		\$0		

1.4.42 J1471 Summary

Network Upgrade	Cost	J1471	NUs Type	Self Funding?
Niota 161-69 kV xfmr	\$4,000,000	\$4,000,000	MISO PK	Yes
Denmark-Burlington 161 kV #1	\$480,000	\$288,268	MISO PK	Yes
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$86,118	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$1,633,970	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$366,246	NRIS	Yes
Burlington-Denmark 161 kV Rebuild	\$13,850,000	\$8,203,444	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$14,578,046		

1.4.43 J1473 Summary

Network Upgrade	Cost	J1473	NUs Type	Self Funding?
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$513,161	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$21,923	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$14,964	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$6,584	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$556,632		

1.4.44 J1474 Summary

Network Upgrade	Cost	J1474	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$34,010	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$645,286	NRIS	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$728,664	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$38,616	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$25,163	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$11,573	NRIS	Yes
Holcombe Transformer and Terminal Equipment Upgrade	\$1,700,000	\$1,700,000	NRIS	XEL: Yes DPC: Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$3,183,312		

1.4.45 J1476 Summary

Network Upgrade	Cost	J1476	NUs Type	Self Funding?
Montgomery-J1145 POI 345 kV	\$28,000,000	\$30,149	MISO SH	Yes
Greenfield Tap-Greenfield 69 kV	\$300,000	\$300,000	MISO SH	No
Greenfield Tap-Greenfield Municipal 69 kV	\$1,000,000	\$1,000,000	MISO SH	No
Greenfield-MENLOTP 69 kV	\$4,000,000	\$4,000,000	MISO SH	No
Anita-Casey REC 69 kV	\$6,750,000	\$6,750,000	MISO SH	No
Orient M8-Orient REC 69 kV	\$1,700,000	\$1,700,000	MISO SH	No
Orient M8-North Summit Lake 69 kV	\$4,000,000	\$4,000,000	MISO SH	No
Orient REC-Greenfield Muni 69 kV	\$3,000,000	\$3,000,000	MISO SH	No
North Summit Lake-Creston J2 69 kV	\$100,000	\$100,000	MISO SH	ITCM: Yes CIPCO: No
South Summit Lake-Creston 69 kV	\$250,000	\$250,000	MISO SH	CIPCO: No
Casey REC-MENLOTP 69 kV	\$2,000,000	\$2,000,000	MISO SH	No
East Creston Tap-Creston J1 69 kV	\$730,000	\$730,000	MISO PK	ITCM: Yes CIPCO: No
Total Cost Per Project for Actual NRIS Elections for each Project		\$23,860,149		

1.4.46 J1477 Summary

Network Upgrade	Cost	J1477	NUs Type	Self Funding?
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$14,883	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$282,391	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$61,799	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$359,073		

1.4.47 J1478 Summary

Network Upgrade	Cost	J1478	NUs Type	Self Funding?
Denmark-Burlington 161 kV #1	\$480,000	\$19,952	MISO PK	Yes
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$10,442	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$198,114	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$44,431	NRIS	Yes
Burlington-Denmark 161 kV Rebuild	\$13,850,000	\$806,783	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,079,722		

1.4.48 J1479 Summary

Network Upgrade	Cost	J1479	NUs Type	Self Funding?
Denmark-Burlington 161 kV #1	\$480,000	\$171,779	MISO PK	Yes
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$42,694	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$810,067	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$181,986	NRIS	Yes
Burlington-Denmark 161 kV Rebuild	\$13,850,000	\$4,839,774	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$6,046,300		

1.4.49 J1485 Summary

Network Upgrade	Cost	J1485	NUs Type	Self Funding?
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$1,282,902	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$50,838	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$30,918	NRIS	Yes
Coon Creek-Terminal 345 Terminal Equipment Upgrade	\$2,000,000	\$126,719	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$15,313	NRIS	Yes
Granite Falls-Minn Valley 230 terminal upgrades	\$1,000,000	\$486,159	NRIS	XEL: Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,992,848		

1.4.50 J1487 Summary

Network Upgrade	Cost	J1487	NUs Type	Self Funding?
Montgomery-J1145 POI 345 kV	\$28,000,000	\$71,404	MISO SH	Yes
J1363 POI - J1352 POI 345 kV Rebuild	\$1,000,000	\$20,843	NRIS	Yes
J1338 POI - J1363 POI 345 kV Rebuild	\$19,000,000	\$395,474	NRIS	Yes
Spencer-J1338 POI 345 kV Rebuild	\$4,000,000	\$88,820	NRIS	Yes
Beaver Creek - Grimes 345 Line Rebuild	\$26,000,000	\$711,519	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	\$2,700,000	\$38,523	NRIS	ITCM: Yes MEC: Yes
J1181 POI - Hazelton 345 Structure Replacements	\$800,000	\$23,934	NRIS	Yes
Tiffin-Hills 345 kV Structure Replacements	\$800,000	\$11,567	NRIS	Yes
Remson-Raun Structure Upgrades	\$1,100,000	\$33,255	NRIS	Yes
Murphy Creek Transformer Upgrade	\$3,500,000	\$966,443	NRIS	No
Austin - Adams 161 kV Line Rebuild	\$15,750,000	\$2,456,181	NRIS	ITCM: Yes SMMPA: No
Pleasant Valley-Byron 161 kV Rebuild	\$24,900,000	\$1,782,348	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$6,600,309		

1.5 Study Compliance with NERC FAC-002-2 Standard

This DPP 2019 West Area study was completed in compliance with NERC FAC-002-2:

R1.1: The reliability impact of the new interconnection, or materially modified existing interconnection, on affected system(s).

Section 3 covers summer peak steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 4 covers summer shoulder steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 5.1 covers reliability impact of the generating facilities per GRE Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.2 covers reliability impact of the generating facilities per OTP Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.3 covers reliability impact of the generating facilities per DPC Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 6 covers voltage stability (PV) analysis on the MWEX System Operating Limit (SOL). Network Upgrades required for MWEX voltage stability are identified.

Section 7 covers Deliverability reliability impact of the new NRIS generating facilities.

R1.2: Adherence to applicable NERC Reliability Standards; regional and Transmission Owner planning criteria; and Facility interconnection requirements.

Sections 2.2-2.4 cover NERC Reliability Standard TPL-001-4.

Section 5.1 covers GRE Local Planning Criteria (LPC).

Section 5.2 covers OTP LPC.

Section 5.3 covers DPC LPC.

Section 6 (voltage stability analysis) covers individual system planning criteria (ATC).

Section 7 covers MISO system planning criteria.

R1.3: Steady-state, short-circuit, and dynamics studies, as necessary, to evaluate system performance under both normal and contingency conditions.

Section 3 and Section 4 cover MISO steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.1 covers GRE's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.2 covers OTP's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.3 covers DPC's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6 covers steady-state voltage stability assessment.

Section 7 covers MISO deliverability study (steady-state assessment) including NERC category P0 to P1 contingencies (TPL-001-4).

R1.4: Study assumptions, system performance, alternatives considered, and coordinated recommendations. While these studies may be performed independently, the results shall be evaluated and coordinated by the entities involved.

Section 2.1, Section 2.2, Section 2.3, and Section 2.4 cover study assumptions and system performance criteria.

Jointly coordinated recommendations can be found in Section 5.1 (MISO and GRE), Section 5.2 (MISO and OTP), Section 5.3 (MISO and DPC), and Section 6 (MISO and ATC). Results in Section 3, 4, 6, and 7 have also been reviewed by PJM, SPP, and ATC.

Introduction

Fifty (50) generation projects, listed in Table A-1 (Appendix A.1), have requested to interconnect to the MISO transmission network in the West Area and have advanced to the Definitive Planning Phase (DPP) 2019 Phase 1 study (DPP West Area). All generating facilities have requested both Energy Resource Interconnection Service (ERIS) and Network Resource Interconnection Service (NRIS).

This report presents the study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generating facilities in the DPP West Area Phase 1 study.

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

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Model Development and Study Criteria

2.1 Model Development

2.1.1 Benchmark Cases

DPP 2019 West area power flow benchmark cases representing 2024 summer shoulder and summer peak conditions were developed from the MTEP19 models with LBA dispatch.

The benchmark cases for DPP 2019 study were created as follows:

- MISO Prior queued generation projects and their associated Network Upgrades (NU) were modeled. Appendix A.2 lists all DPP 2018 April West Area Phase 2 MISO ERIS Network Upgrades included in the models, and Appendix A.3 lists all DPP 2017 August West Area Phase 2 MISO ERIS Network Upgrades included in the models.
- DPP 2019 generation projects in the West Area (DPP West Area, Table A-1), Central Area (Table A-3), Michigan Area (Table A-4), and ATC Area (Table A-5) were modeled with offline status.
- Prior queued solar projects were turned off in summer shoulder cases and dispatched at 100% of their Pmax in summer peak case.
- For MISO North generation projects, their output was sunk to the MISO North (Appendix A.5, Table A-9), where generation was scaled uniformly;
- For MISO South generation projects, their output was sunk to the MISO South (Appendix A.5, Table A-9), where generation was scaled uniformly;
- PJM generation projects were modeled and dispatched. The generation output was sunk to the PJM market (Appendix A.6, Table A-10), where generation was scaled uniformly.
- SPP generation projects were modeled and dispatched. The generation output was sunk to the SPP market (Appendix A.7, Table A-11), where generation was scaled uniformly.
- The Hickory Creek–Cardinal 345 kV project (MVP project 3127) was included in the 2024 models; the Hickory Creek-Cardinal 345 kV project has an in-service date of 12/31/2023.
- Models were further reviewed by the Ad Hoc study members (transmission owners and customers). Model corrections and changes were made based on the comments and feedback. These modeling changes are listed in Appendix A.4.
- Adjusted Square Butte DC to match the total output of the Bison (Bison 1 to 5) and Oliver County (Oliver County 1 and 2) wind farms.
- Adjusted CU DC to match the total output of Coal Creek generation units #1 and #2.
- MHEX interface transfer level is at 1530 MW in summer shoulder and 1800 MW in summer peak cases.

2.1.2 Study Cases

Summer peak study discharging case was created by dispatching the DPP 2019 generation projects at the specified summer peak level from the benchmark case. Battery projects were dispatched in discharging mode.

Summer shoulder study discharging case was created by dispatching the DPP 2019 generation projects at the specified summer shoulder level from the benchmark case. Battery projects were dispatched in discharging mode, and PV projects were turned off.

Summer shoulder study charging case was created by dispatching the DPP 2019 generation projects at the specified summer shoulder level from the benchmark case. Battery projects were dispatched in charging mode, and PV projects were turned off.

For the DPP 2019 generation projects in MISO North, the MISO North was used for power balance, where generation was scaled uniformly.

Both study and benchmark power flow cases were solved with transformer tap adjustment enabled, area interchange disabled, phase shifter adjustment enabled and switched shunt adjustment enabled.

The interface transfer levels in the 2024 study cases are summarized in Table 2-1.

Table 2-1: Interface Transfer Levels in 2024 Steady State Study Cases

Interface	2024 SH Discharging Case (MW)	2024 SH Charging Case (MW)	2024 SPK Discharging Case (MW)
MHEX	1531	1530	1799
MWEX	1217.8	1296.6	307.5
Arrowhead PST	485.1	516.5	6.9
J732 POI – Stone Lake 345 kV	715.8	769.7	417.9

2.2 Contingency Criteria

A variety of contingencies were considered for steady-state analysis:

- NERC Category P0 with system intact (no contingencies)
- NERC Category P1 contingencies
 - NERC Category P1 contingencies, at buses with a nominal voltage of 69 kV and above, in the following areas: CWLD (area 333), AMMO (area 356), AMIL (area 357), CWLP (area 360), SIPC (area 361), WEC (area 295), WEC MI (area 296), XCEL (area 600), MP (area 608), SMMPA (area 613), GRE (area 615), OTP (area 620), ITCM (area 627), MPW (area 633), MEC (area 635), MDU (area 661), BEPC-MISO (area 663), MHEB (area 667), DPC (area 680), ALTE (area 694), WPS (area 696), MGE (area 697), UPPC (area 698), CE(area 222), NPPD (area 640), OPPD (area 645), LES (area 650), WAPA (area 652), BEPC-SPP (area 659), AECI (area 330), MIPU(area 540), KCPL (area 541), KACY (area 542), INDN (area 545).

- Multiple-element NERC Category P1 contingencies, in Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P1 contingencies are listed in Appendix A.8.
- NERC Category P2-P7 contingencies
 - Selected NERC Category P2-P7 contingencies provided by the Ad Hoc Study Group, in the study region of Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P2-P7 contingencies are listed in Appendix A.8.

For all contingency and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

2.3 Monitored Elements

The study area is defined in Table 2-2. Facilities in the study area were monitored for system intact and contingency conditions. Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal (PSS[®]E rate A) rating. Under NERC category P1-P7 conditions, branches were monitored for loading as shown in the column labeled "Post-Disturbance Thermal Limits".

Table 2-2: Monitored Elements

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
AECI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
AMIL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
AMMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
ATCLLC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
BEPC-MISO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
BEPC-SPP	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CBPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CMMPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.07/0.90
CWLD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CWLP	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
CE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
DPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GRE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.92/0.90
INDN	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
ITCM	69 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.95	1.10/0.93
KACY	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
KCPL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
LES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MDU	57 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MEC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.96/0.95	1.05/0.96/0.95/0.94/0.93 ³
MHEB	69 kV and above	100% of Rate A	100% of Rate B	1.12/1.1/1.07/1.05/1.04/0.99/0.97/0.96/0.95	1.15/1.10/0.94/0.90
MP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.10/0.95
MPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.97	1.10/0.92
MPW	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.06/0.92
MRES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.97	1.05/0.92
NPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OTP	40 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.97	1.10/0.92
PPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
RPU	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.92
SIPC	69 kV and above	100% of Rate A	100% of Rate B	1.07/0.95	1.09/0.91
SMMPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
WAPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
WPPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
XEL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.92

Notes

- 1: PSS®E Rate A, Rate B or Rate C
- 2: Limits dependent on nominal bus voltage
- 3: For facilities in Cedar Falls Utilities or Ames Municipal Utilities, post-contingency voltage limits are 1.05/0.94 for >200 kV, and 1.05/0.93 for others.

2.4 Performance Criteria

A branch is considered as a thermal injection constraint if the branch is loaded above its applicable normal or emergency rating for the post-change case, and any of the following conditions are met:

- 1) the generator (NR/ER) has a larger than 20% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, or
- 2) the megawatt impact due to the generator is greater than or equal to 20% of the applicable rating (normal or emergency) of the overloaded facility, or

- 3) the overloaded facility or the overload-causing contingency is at generator's outlet, or
- 4) for any other constrained facility, where none of the study generators meet one of the above criteria in 1), 2), or 3), however, the cumulative megawatt impact of the group of study generators (NR/ER) is greater than 20% of the applicable rating, then only those study generators whose individual MW impact is greater than 5% of the applicable rating and has DF greater than 5% (OTDF or PTDF) will be responsible for mitigating the cumulative MW impact constraint.

A bus is considered a voltage constraint if both of the following conditions are met. All voltage constraints must be resolved before a project can receive interconnection service.

- 1) the bus voltage is outside of applicable normal or emergency limits for the post-change case, and
- 2) the change in bus voltage is greater than 0.01 per unit.

All DPP 2018 April West Area study generators must mitigate thermal injection constraints and voltage constraints in order to obtain unconditional Interconnection Service.

Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the deliverability algorithm, to meet the system performance criteria for NERC category P0-P1 events, if the constraint demonstrates an incremental flow caused by the generator equal to or greater than 5% of the generator's maximum dispatch level in each case.

2.5 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

Per FERC Order 827, non-synchronous generation projects in the DPP 2019 West Area study group are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

Collector system and shunt compensation of DPP West projects are modeled, which are listed in Appendix A.1, Table A-2. An analysis was performed to study the reactive power requirements (FERC Order 827) for the non-synchronous generation projects in the DPP 2019 West study group. The analysis was performed as follows:

Step 1: Verify whether total dynamic reactive power (reactive power from generators and dynamic compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the generator terminal bus. The verification in Step 1 was performed when generator data was submitted and modeled.

Step 2: Verify whether total reactive power (reactive power from generators, dynamic compensation devices, and static compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the high-side of the generator substation. The testing procedure in Step 2 is described in the following:

- Lock the high-side of the generator substation at 1.0 pu voltage by adding a fictitious SVC. This is to ensure that the test result is not affected by the system condition.
- Lock the reactive power output of the generator to the maximum limit (Q_{max}). Make sure all shunt compensation devices within the substation are at the maximum capacitive output. Adjust transformer tap to ensure bus voltages within the substation

are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 lagging requirement.

- Lock the reactive power output of the generator to the minimum limit (Q_{min}). Make sure all shunt compensation devices within the substation are at the maximum inductive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 leading requirement.

Appendix C lists reactive power requirement analysis results for the DPP West generation projects. The results are summarized as following:

- J1190, J1218, J1221, J1270, J1284, J1298, J1300, J1328, J1341, J1365, J1371, J1395, J1456, J1471, and J1485 do not meet the reactive power requirements per FERC Order 827.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

Summer Peak Steady-State Analysis

Summer peak steady-state analysis was performed in summer peak scenario to identify thermal and voltage upgrades required interconnecting the generating facilities in the DPP 2019 West Area group to the transmission system. Battery projects are in discharging mode in the summer peak scenario.

3.1 Study Procedure

3.1.1 Computer Programs

Steady-state analyses were performed using PSS®E version 33.12.1 and PSS®MUST version 12.4.0.

3.1.2 Study Methodology

Summer peak power flow case was created in the procedure as described in Section 2.1. Non-linear (AC) contingency analysis was first performed for single critical contingencies (contingencies in ≥ 200 kV system in MISO West area) to identify voltage collapse and thermal overloads. Based on the results, the summer peak case can converge under post-contingency conditions. Therefore, Base Case NUs are not required for summer peak scenario.

Full AC contingency analysis was further performed on the benchmark and study cases without Base Case NUs. The incremental impact of the DPP West Area generating facilities was evaluated by comparing the steady-state performance of the transmission system in the benchmark and study cases. Network upgrades were identified to mitigate any summer peak constraints.

3.2 Contingency Analysis Results for Summer Peak Discharging Condition

The incremental impact of the proposed interconnection on individual facilities was evaluated by comparing flows and voltages between benchmark case (without DPP West Area projects) and study case (with DPP West Area projects). All battery projects in DPP 2019 cycle were in discharging condition.

3.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, no thermal constraints (Table D-1) or voltage constraints (Table D-2) were identified.

3.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All category P1 contingencies were converged.

For P1 contingencies, thermal constraints are listed in Table D-3. No voltage constraints were identified (Table D-4).

Two category P2-P7 contingencies (Table D-7) were not converged, and their dc thermal results are listed in Table D-8. The contingency was not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these contingencies.

For P2-P7 contingencies, thermal constraints are listed in Table D-5. No voltage constraints were identified (Table D-6).

3.3 Summer Peak Worst Thermal Constraints

Table 3-1 lists worst thermal constraints and Network Upgrades in the 2024 summer peak discharging scenario.

Table 3-1: 2024 Summer Peak Thermal Constraints and Network Upgrades, Maximum Screened Loading

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J1479	Henry Co. 161-69 kV xfmr	56.0	AECI	62.5	111.5	CEII Redacted	P1	NU is not required unless it is identified as constraint in affected system study.
J1471	Amoco-Niota 69 kV	67.0	Ameren	71.7	107.0	CEII Redacted	P1	Incorrect rating. Rating should be 72 MVA.
J1471	Amoco-Niota 69 kV	67.0	Ameren	71.6	106.9	CEII Redacted	P2-P7	Incorrect rating. Rating should be 72 MVA.
J1471	Niota 161-69 kV xfmr	84.0	Ameren	98.2	117.0	CEII Redacted	P1	Replace with new 112 MVA 161/69 kV Transformer.
J1471	Niota 161-69 kV xfmr	84.0	Ameren	97.8	116.5	CEII Redacted	P2-P7	Replace with new 112 MVA 161/69 kV Transformer.
J1300,J1325	Split Rock-Rock Co. 161 kV	224.2	XEL	269.8	120.3	CEII Redacted	P1	Rebuild 13.7 miles of 477 ACSR to 795 ACSR

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J1300,J1325	Split Rock-Rock Co. 161 kV	224.2	XEL	269.9	120.4	CEII Redacted	P2-P7	Rebuild 13.7 miles of 477 ACSR to 795 ACSR
J1300,J1325	Rock Co.-Magnolia 161 kV	171.0	XEL ITCM	263.1	153.8	CEII Redacted	P1	XEL: Rebuild XEL portion 5.27 miles of 477 ACSR to 795 ACSR. \$6M ITCM: line rebuild ITCM portion 11.4 miles. \$12M
J1300,J1325	Rock Co.-Magnolia 161 kV	171.0	XEL ITCM	263.2	153.9	CEII Redacted	P2-P7	XEL: Rebuild XEL portion 5.27 miles of 477 ACSR to 795 ACSR. \$6M ITCM: line rebuild ITCM portion 11.4 miles. \$12M
J1300,J1325	Split Rock 161-115-36.2 kV xfmr #6	187.0	XEL	268.7	143.7	CEII Redacted	P1	Replace existing with 336 MVA transformer

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J1300,J1325	Split Rock 161-115-36.2 kV xfmr #6	187.0	XEL	268.8	143.8	CEII Redacted	P2-P7	Replace existing with 336 MVA transformer
J1300,J1325	Brewster-J1300 POI 161 kV	219.0	GRE ITCM	266.5	121.7	CEII Redacted	P1	GRE: Owned by ITCM. \$0 ITCM: Lakefield Corner station and J1300 - Lakfield Corner new line. \$62.6M
J1300,J1325	Brewster-J1300 POI 161 kV	219.0	GRE ITCM	266.5	121.7	CEII Redacted	P2-P7	GRE: Owned by ITCM. \$0 ITCM: Lakefield Corner station and J1300 - Lakfield Corner new line. \$62.6M
J1300,J1325	Brewster-Heron Lake 161 kV	167.0	GRE ITCM	261.0	156.3	CEII Redacted	P1	GRE: Owned by ITCM. \$0 ITCM: Lakefield Corner station and J1300 - Lakfield Corner new line. \$62.6M
J1300,J1325	Brewster-Heron Lake 161 kV	167.0	GRE ITCM	261.0	156.3	CEII Redacted	P2-P7	GRE: Owned by ITCM. \$0 ITCM: Lakefield Corner station and J1300 - Lakfield Corner new line. \$62.6M

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J1428	Buffalo 345-115-41.6 kV xfmr	140.0	OTP	151.2	108.0	CEII Redacted	P2-P7	Replace the Buffalo #2 transformer with a larger transformer.
J1428	Maple River 345-230-13.8 kV xfmr #1	420.0	OTP MPC	422.6	100.6	CEII Redacted	P1	Replace the Maple River #1 transformer with a larger transformer.
J1428	Maple River 345-230-13.8 kV xfmr #2	420.0	OTP MPC	427.0	101.7	CEII Redacted	P1	Replace the Maple River #2 transformer with a larger transformer.
J1428	Maple River 345-230-13.8 kV xfmr #2	420.0	OTP MPC	423.2	100.8	CEII Redacted	P2-P7	Replace the Maple River #2 transformer with a larger transformer.
J1300,J1325	Heron Lake 161-69 kV xfmr	56.0	ITCM	80.1	143.1	CEII Redacted	P1	Lakefield Corner station and J1300 - Lakefield Corner new line
J1300,J1325	Heron Lake 161-69 kV xfmr	56.0	ITCM	80.1	143.1	CEII Redacted	P2-P7	Lakefield Corner station and J1300 - Lakefield Corner new line
J1476	East Creston Tap-Creston J1 69 kV	48.0	ITCM CIPCO	53.3	111.1	CEII Redacted	P1	CIPCO: Not a constraint ITCM: Rebuild 1 mile, new rating 72 MVA. \$730K

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J1476	East Creston Tap-Creston J1 69 kV	48.0	ITCM CIPCO	53.7	111.9	CEII Redacted	P2-P7	CIPCO: Not a constraint ITCM: Rebuild 1 mile, new rating 72 MVA. \$730K
J1359	PCI J2-Oakridge 69 kV	109.0	ITCM CIPCO	128.7	118.0	CEII Redacted	P1	CIPCO: Not a constraint. \$0 ITCM: Relay thermal, replace sub taps (.08 miles), new rating 143 MVA. \$450K
J1359	PCI J2-Oakridge 69 kV	109.0	ITCM CIPCO	123.5	113.3	CEII Redacted	P2-P7	CIPCO: Not a constraint. \$0 ITCM: Relay thermal, replace sub taps (.08 miles), new rating 143 MVA. \$450K
J1325	Magnolia-Elk 161 kV	151.0	ITCM	211.8	140.3	CEII Redacted	P1	Rebuild 18.56 miles, new rating 366 MVA
J1325	Magnolia-Elk 161 kV	151.0	ITCM	211.8	140.3	CEII Redacted	P2-P7	Rebuild 18.56 miles, new rating 366 MVA
J1300,J1325	Heron Lake-Lakefield 161 kV	167.0	ITCM	188.6	112.9	CEII Redacted	P1	MTEP15 project 8106, limit 233 MVA
J1300,J1325	Heron Lake-Lakefield 161 kV	167.0	ITCM	188.6	112.9	CEII Redacted	P2-P7	MTEP15 project 8106, limit 233 MVA

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J1471,J1478,J1479	Denmark-Burlington 161 kV #1	240.0	ITCM	253.3	105.5	CEII Redacted	P2-P7	Sag mitigation, 4 spans brings rating to 263 MVA
J1471,J1478,J1479	Denmark-Burlington 161 kV #2	229.0	ITCM	239.1	104.4	CEII Redacted	P1	Current rating 254 MVA
J1471,J1478,J1479	Denmark-Burlington 161 kV #2	229.0	ITCM	237.5	103.7	CEII Redacted	P2-P7	Current rating 254 MVA
J1471	Wever-Lee 69 kV	84.0	ITCM AECI	93.6	111.4	CEII Redacted	P1	ITCM: AECI constraint AECI: NU is not required unless it is identified as constraint in affected system study.
J1471	Wever-Lee 69 kV	84.0	ITCM AECI	94.9	112.9	CEII Redacted	P2-P7	ITCM: AECI constraint AECI: NU is not required unless it is identified as constraint in affected system study.
J1314	Apple River 161-69 kV xfmr #2	82.0	DPC	87.9	107.2	CEII Redacted	P2-P7	Upgrade Apple River Transformer #2

3.4 Network Upgrades Identified in MISO ERS Analysis for 2024 Summer Peak Scenario

Based on the MISO 2024 summer peak discharging steady state analyses, the MISO thermal NUs and cost are listed in Table 3-2. No reactive power NUs are required.

Table 3-2: Summer Peak Discharging Thermal NUs

Constraint	Owner	Mitigation	Cost (\$)
Henry Co. 161-69 kV xfmr	AECI	NU is not required unless it is identified as constraint in affected system study.	\$0
Amoco-Niota 69 kV	Ameren	Incorrect rating. Rating should be 72 MVA.	\$0
Niota 161-69 kV xfmr	Ameren	Replace with new 112 MVA 161/69 kV Transformer.	\$4,000,000
Split Rock-Rock Co. 161 kV	XEL	Rebuild 13.7 miles of 477 ACSR to 795 ACSR	\$20,550,000
Rock Co.-Magnolia 161 kV	XEL ITCM	XEL: Rebuild XEL portion 5.27 miles of 477 ACSR to 795 ACSR. \$6M ITCM: line rebuild ITCM portion 11.4 miles. \$12M	\$18,000,000
Split Rock 161-115-36.2 kV xfmr #6	XEL	Replace existing with 336 MVA transformer	\$3,000,000
Brewster-J1300 POI 161 kV	GRE ITCM	GRE: Owned by ITCM. \$0 ITCM: Lakefield Corner station and J1300 - Lakfield Corner new line. \$62.6M	\$62,600,000
Brewster-Heron Lake 161 kV	GRE ITCM	GRE: Owned by ITCM. \$0 ITCM: Lakefield Corner station and J1300 - Lakfield Corner new line. \$62.6M	Incl.
Buffalo 345-115-41.6 kV xfmr	OTP	Replace the Buffalo #2 transformer with a larger transformer.	\$3,000,000
Maple River 345-230-13.8 kV xfmr #1	OTP MPC	Replace the Maple River #1 transformer with a larger transformer.	\$4,000,000
Maple River 345-230-13.8 kV xfmr #2	OTP MPC	Replace the Maple River #2 transformer with a larger transformer.	\$4,000,000
Heron Lake 161-69 kV xfmr	ITCM	Lakefield Corner station and J1300 - Lakefield Corner new line	Incl.
East Creston Tap-Creston J1 69 kV	ITCM CIPCO	CIPCO: Not a constraint ITCM: Rebuild 1 mile, new rating 72 MVA. \$730K	\$730,000
PCI J2-Oakridge 69 kV	ITCM CIPCO	CIPCO: Not a constraint. \$0 ITCM: Relay thermal, replace sub taps (.08 miles), new rating 143 MVA. \$450K	\$450,000

Constraint	Owner	Mitigation	Cost (\$)
Magnolia-Elk 161 kV	ITCM	Rebuild 18.56 miles, new rating 366 MVA	\$19,600,000
Heron Lake-Lakefield 161 kV	ITCM	MTEP15 project 8106, limit 233 MVA	\$0
Denmark-Burlington 161 kV #1	ITCM	Sag mitigation, 4 spans brings rating to 263 MVA	\$480,000
Denmark-Burlington 161 kV #2	ITCM	Current rating 254 MVA	\$0
Wever-Lee 69 kV	ITCM AECI	ITCM: AECI constraint AECI: NU is not required unless it is identified as constraint in affected system study.	\$0
Apple River 161-69 kV xfmr #2	DPC	Upgrade Apple River Transformer #2	\$2,636,000

Section

4

Summer Shoulder Steady-State Analysis

Summer shoulder steady-state analysis was performed in summer shoulder scenario to identify thermal and voltage upgrades required interconnecting the generating facilities in the DPP 2019 West Area group to the transmission system. Two conditions for discharging and charging of battery projects were both studied for summer shoulder scenario.

4.1 Study Procedure

Summer shoulder power flow cases (discharging and charging) were created in the procedure as described in Section 2.1. AC contingency analysis was first performed for single critical contingencies (contingencies in ≥ 200 kV system in MISO West area) to identify voltage collapse and thermal overloads. Based on the results, the summer shoulder discharging and charging cases can converge under post-contingency conditions. Therefore, Base Case NUs are not required for summer shoulder scenario.

Full AC contingency analysis was further performed on the benchmark and study cases for discharging and charging conditions without Base Case NUs. The incremental impact of the DPP West Area generating facilities was evaluated by comparing the steady-state performance of the transmission system in the benchmark and study cases. Network upgrades were identified to mitigate any summer shoulder constraints.

4.2 AC Contingency Analysis for Summer Shoulder Scenario

In the summer shoulder scenario, two conditions for discharging and charging of battery projects were both studied.

4.2.1 Contingency Analysis Results for Summer Shoulder Discharging Condition

The discharging condition has all DPP 2019 battery projects in maximum discharging mode.

4.2.1.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-1. No voltage constraints were identified (Table E-2).

4.2.1.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All NERC Category P1 contingencies were converged.

For P1 contingencies, thermal constraints are listed in Table E-3. No voltage constraints were identified (Table E-4).

Two category P2-P7 contingencies (Table E-7) were not converged in both the benchmark and study cases. No mitigation plan is required for the study projects for these non-converged contingencies.

For the non-converged contingencies in Table E-7, DC contingency analysis was performed to get the dc thermal results in the benchmark case. The dc thermal results for non-converged contingencies are listed in Table E-8.

For P2-P7 contingencies, thermal constraints are listed in Table E-5. No voltage constraints were identified (Table E-6).

4.2.2 Contingency Analysis Results for Summer Shoulder Charging Condition

The charging condition has all DPP 2019 battery projects in maximum charging mode.

4.2.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-9. No voltage constraints were identified (Table E-10).

4.2.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All NERC Category P1 contingencies were converged.

For P1 contingencies, thermal constraints are listed in Table E-11. No voltage constraints were identified (Table E-12).

Two category P2-P7 contingencies (Table E-15) were not converged in both the benchmark and study cases. No mitigation plan is required for the study projects for these non-converged contingencies.

For the non-converged contingencies in Table E-15, DC contingency analysis was performed to get the dc thermal results in the benchmark case. The dc thermal results for non-converged contingencies are listed in Table E-16.

For P2-P7 contingencies, thermal constraints are listed in Table E-13. No voltage constraints were identified (Table E-14).

4.2.3 Summer Shoulder Worst Thermal Constraints

Combining the results for both discharging and charging conditions, Table 4-1 lists worst thermal constraints and Network Upgrades identified in the 2024 summer shoulder scenario.

Table 4-1: 2024 Summer Shoulder Thermal Constraints, Maximum Screened Loading

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1467	J1438 POI-J1467 POI 161 kV	221.0	ITCM	221.8	100.4	CEII Redacted	P1	SH Charge	Sag limit between Windsor and J1467 POI, next limit 226 MVA.
J1467	J1438 POI-J1467 POI 161 kV	221.0	ITCM	221.8	100.3	CEII Redacted	P2-P7	SH Charge	Sag limit between Windsor and J1467 POI, next limit 226 MVA.
J1341,J1365,J1427, J1432,J1467	Electric Jct-Nelson 345 kV	1334.0	PJM	1353.4	101.5	CEII Redacted	P0	SH Charge	NU is not required unless it is identified as constraint in affected system study.
J1476	Montgomery-J1145 POI 345 kV	998.0	Ameren	1866.7	187.0	CEII Redacted	P0	SH Discharge	Rebuild 28 mi of 345 kV line for 3000A SE
J1270,J1321,J1349, J1417,J1487	Montgomery-J1145 POI 345 kV	1193.0	Ameren	1902.2	159.4	CEII Redacted	P1	SH Charge, Discharge	Rebuild 28 mi of 345 kV line for 3000A SE
J1371,J1485	Split Rock-White 345 kV	717.1	XEL WAPA	768.4	107.2	CEII Redacted	P1	SH Charge	XEL: WAPA limiter. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.
J1315,J1371,J1485	Split Rock-White 345 kV	717.1	XEL WAPA	783.6	109.3	CEII Redacted	P2-P7	SH Charge	XEL: WAPA limiter. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.
J1315	Franklin-Cedar Mountain 115 kV	331.2	XEL GRE	341.2	103.0	CEII Redacted	P2-P7	SH Charge	XEL: Fix line limiter, line is 795 ACSS and should have a rating of 397 MVA emergency. \$1,000,000

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
									GRE: Owned by XEL. \$0
J1315	Lyon Co 115-69 kV xfmr	80.5	XEL	82.8	102.9	CEII Redacted	P2-P7	SH Charge	Repalce transformer with 112 MVA transformer
J1315	Crystal Foods Tap-Arlington 69 kV	35.9	XEL	49.3	137.5	CEII Redacted	P2-P7	SH Charge, Discharge	Rebuild 6.9 miles to 477 ACSR
J1230	Riverton-Mud Lake 230 kV	338.8	MP GRE	376.6	111.2	CEII Redacted	P1	SH Discharge	MP: MP limits are 414/414/478/484 SN/SE/WN/WE. \$0 GRE: Remediate with structures and get it up to 383 MVA.
J1230	Riverton-Mud Lake 230 kV	338.8	MP GRE	378.3	111.7	CEII Redacted	P2-P7	SH Discharge	MP: MP limits are 414/414/478/484 SN/SE/WN/WE. \$0 GRE: Remediate with structures and get it up to 383 MVA.
J1230	Brainerd-Riverton 115 kV	78.0	MP	83.9	107.6	CEII Redacted	P1	SH Discharge	Increase conductor clearance to 75C, new limits would be 90/99/114/125. Line is 13.12 miles long.

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1467	Windsor 161-69 kV xfmr	75.0	SMMPA ITCM	83.1	110.8	CEII Redacted	P1	SH Charge, Discharge	New Transformer and associated equipment
J1467	Windsor 161-69 kV xfmr	75.0	SMMPA ITCM	92.3	123.1	CEII Redacted	P2-P7	SH Charge, Discharge	New Transformer and associated equipment
J1230	Inman-Wing River 230 kV	318.6	GRE	320.5	100.6	CEII Redacted	P2-P7	SH Charge	Replace Line CT's
J1230	Wing River-Riverton 230 kV	268.0	GRE MP	269.4	100.5	CEII Redacted	P2-P7	SH Charge	GRE: Owned by MP. \$0 MP: MP's portion would require increase of conductor clearance to 65C, new limits would be 282/311/388/427 SN/SE/WN/WE. MP owns miles of conductor. \$3,237,300

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1230	Wing River-Verndale 115 kV	107.0	GRE MP	123.3	115.3	CEII Redacted	P1	SH Charge	GRE: Owned by MP. \$0 MP: Replace (3) 115 kV switches, replace substation conductor at Verndale Substation, reconductor 1.07-mile line. New limits would be 126/126/126/126. \$1,500,000
J1230	Thomastown Tap-Minnesota Pipeline Staples Tap 115 kV	77.0	GRE MP	92.2	119.8	CEII Redacted	P1	SH Charge	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080.000
J1230	Thomastown Tap-Dog Lake 115 kV	84.7	GRE MP	90.3	106.6	CEII Redacted	P1	SH Charge	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080.000
J1230	Aldrich Tap-Minnesota Pipeline Staples Tap 115 kV	77.0	GRE MP	100.7	130.8	CEII Redacted	P1	SH Charge	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080.000

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1230	Aldrich-Verndale 115 kV	77.0	GRE MP	102.7	133.4	CEII Redacted	P1	SH Charge	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080.000
J1476	Greenfield Tap-Greenfield 69 kV	57.0	CIPCO	68.5	120.1	CEII Redacted	P2-P7	SH Charge	Rebuild 0.51 miles of 69 kV line
J1476	Greenfield Tap-Greenfield Municipal 69 kV	57.0	CIPCO	67.8	118.9	CEII Redacted	P2-P7	SH Charge	Rebuild 2 miles of 69 kV line

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1476	Greenfield-MENLOTP 69 kV	59.0	CIPCO	70.5	119.4	CEII Redacted	P2-P7	SH Charge	Rebuild 7.97 miles of 69 kV line
J1476	Anita-Casey REC 69 kV	59.0	CIPCO	74.7	126.6	CEII Redacted	P2-P7	SH Charge	Rebuild 13.5 miles of 69 kV line
J1435	Liberty 161-69 kV xfmr	84.0	CIPCO	91.4	108.8	CEII Redacted	P2-P7	SH Charge	NU for IR20 in CIPCO Queue. This will not be a constraint.
J1467	Luana-Monona 69 kV	48.0	ITCM	59.3	123.6	CEII Redacted	P1	SH Charge, Discharge	Line rebuild 1.52 miles, relay upgrade

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1467	Luana-Monona 69 kV	48.0	ITCM	56.3	117.2	CEII Redacted	P2-P7	SH Charge, Discharge	Line rebuild 1.52 miles, relay upgrade
J1467	Luana-Postville 69 kV	48.0	ITCM DPC	59.8	124.6	CEII Redacted	P1	SH Charge, Discharge	ITCM: Line rebuild 4.06 miles, \$3M DPC: Upgrade Luana-Postville Line. \$20,000
J1467	Luana-Postville 69 kV	48.0	ITCM DPC	56.7	118.2	CEII Redacted	P2-P7	SH Charge, Discharge	ITCM: Line rebuild 4.06 miles, \$3M DPC: Upgrade Luana-Postville Line. \$20,000
J1487	Albert Lea Westside-South Broadway 69 kV	48.0	ITCM	63.4	132.2	CEII Redacted	P1	SH Charge, Discharge	MTEP13 project 4099
J1487	Albert Lea Westside-South Broadway 69 kV	48.0	ITCM	59.6	124.1	CEII Redacted	P2-P7	SH Charge	MTEP13 project 4099

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1476	Orient M8-Orient REC 69 kV	57.0	CIPCO	61.3	107.5	CEII Redacted	P2-P7	SH Charge	Rebuild 3.35 miles of 69 kV line
J1476	Orient M8-North Summit Lake 69 kV	57.0	CIPCO	60.9	106.9	CEII Redacted	P2-P7	SH Charge	Rebuild 7.71 miles of 69 kV line
J1477	Riverside-Sharon Tap 69 kV	74.0	CIPCO ITCM	89.6	121.1	CEII Redacted	P2-P7	SH Charge	No mitigation required. Contingency would disconnect Sub 56 161 kV bus including J1477.

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1476	Orient REC-Greenfield Muni 69 kV	57.0	CIPCO	62.1	109.0	CEII Redacted	P2-P7	SH Charge	Rebuild 6 miles of 69 kV line
J1476	North Summit Lake-Creston J2 69 kV	58.0	ITCM CIPCO	72.5	125.1	CEII Redacted	P1	SH Charge	ITCM: Relay thermal, new ITCM rating 79 MVA. \$100K CIPCO: ITCM constraint. \$0
J1476	North Summit Lake-Creston J2 69 kV	58.0	ITCM CIPCO	70.4	121.3	CEII Redacted	P2-P7	SH Charge	ITCM: Relay thermal, new ITCM rating 79 MVA. \$100K CIPCO: ITCM constraint. \$0
J1476	South Summit Lake-Creston 69 kV	120.0	CIPCO WAPA	120.6	100.5	CEII Redacted	P2-P7	SH Charge	Reconductor 69 kV tie

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1476	Casey REC-MENLOTP 69 kV	59.0	CIPCO	74.2	125.8	CEII Redacted	P2-P7	SH Charge	Reconductor 3.96 miles of 69 kV line
J1476	Osceola-Osceola REC 69 kV	38.0	CIPCO ITCM	41.5	109.2	CEII Redacted	P2-P7	SH Charge	CIPCO: ITCM CONSTRAINT. \$0 ITCM: Rating 62/66 MVA. \$0
J1477	Sharon Tap-Hills 69 kV	75.0	MEC CIPCO	91.5	122.0	CEII Redacted	P2-P7	SH Charge	No mitigation required. Contingency would disconnect Sub 56 161 kV bus including J1477.

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1467	Hazleton-J1438 POI 161 kV	221.0	ITCM	221.7	100.3	CEII Redacted	P1	SH Charge	Most limiting sag limit between Windsor and J1467 POI, next limit 226 MVA. \$0
J1467	Hazleton-J1438 POI 161 kV	221.0	ITCM	221.6	100.3	CEII Redacted	P2-P7	SH Charge	Most limiting sag limit between Windsor and J1467 POI, next limit 226 MVA. \$0
J1341	Hazleton-Dundee 161 kV	366.0	ITCM CIPCO	395.3	108.0	CEII Redacted	P1	SH Charge	CIPCO: Replace terminal equipment at Dundee. \$600,000 ITCM: reconductor 18 miles, line is DC with Hazleton - Hickory Creek 345. \$13.23M
J1341,J1467	Hazleton-Dundee 161 kV	366.0	ITCM CIPCO	404.8	110.6	CEII Redacted	P2-P7	SH Charge	CIPCO: Replace terminal equipment at Dundee. \$600,000 ITCM: reconductor 18 miles, line is DC with Hazleton - Hickory Creek 345. \$13.23M
J1435	Lore-Hickory Creek 161 kV	366.0	ITCM	496.0	135.5	CEII Redacted	P2-P7	SH Charge	Hickory Creek - Turkey river 13 miles 161 kV, DC with MVP 5
J1435	Lore 161 kV bus tie	335.0	ITCM	385.5	115.1	CEII Redacted	P2-P7	SH Charge	Hickory Creek - Turkey river 13 miles 161 kV, DC with MVP 5
J1365	Traer-Dysart 161 kV	297.0	ITCM	297.6	100.2	CEII Redacted	P2-P7	SH Charge	MTEP19 project 15455

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1341	Liberty-Hickory Creek 161 kV	327.0	ITCM CIPCO	375.4	114.8	CEII Redacted	P1	SH Charge	CIPCO: Replace terminal equipment at Liberty. \$250,000 ITCM: Reconductor 4 miles, DC with Hickory - Hazleton 345. \$2.94M
J1341,J1427	Liberty-Hickory Creek 161 kV	327.0	ITCM CIPCO	409.4	125.2	CEII Redacted	P2-P7	SH Charge	CIPCO: Replace terminal equipment at Liberty. \$250,000 ITCM: Reconductor 4 miles, DC with Hickory - Hazleton 345. \$2.94M
J1285,J1341,J1427, J1467,	Arnold-J1284 POI 345 kV	1043.0	ITCM	1072.4	102.8	CEII Redacted	P1	SH Charge	Sag mitigation, 2 spans brings rating to 1203 MVA
J1285,J1341,J1427, J1467	Arnold-J1284 POI 345 kV	1043.0	ITCM	1166.0	111.8	CEII Redacted	P2-P7	SH Charge	Sag mitigation, 2 spans brings rating to 1203 MVA
J1341,J1427,J1467	Morgan Valley-J1284 POI 345 kV	1043.0	ITCM	1115.7	107.0	CEII Redacted	P2-P7	SH Charge	Sag limits near Arnold end, rating 1321 MVA
J1341,J1427,J1467	Morgan Valley-Tiffin 345 kV	961.0	ITCM MEC	986.7	102.7	CEII Redacted	P2-P7	SH Charge	MEC: Structure Replacements. New rating predicted to be 1079 MVA. ITCM: ITCM terminal 1895 MVA. \$0
J1456	Ward-Bismark 230 kV	352.0	BEPC WAPA	386.0	109.6	CEII Redacted	P1	SH Charge	NU is not required unless it is identified as constraint in affected system study.

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J1456	Pleasant Lake-Leeds 115 kV	121.0	WAPA	121.7	100.6	CEII Redacted	P2-P7	SH Charge	NU is not required unless it is identified as constraint in affected system study.
J1456	Pleasant Lake-Rugby 115 kV	121.0	WAPA	123.6	102.1	CEII Redacted	P2-P7	SH Charge	NU is not required unless it is identified as constraint in affected system study.
J1315	Granite Falls-Marshall 115 kV	103.0	WAPA BEPC	132.0	128.1	CEII Redacted	P2-P7	SH Charge, Discharge	NU is not required unless it is identified as constraint in affected system study.
J1230	Wilton-Winger 230 kV	288.0	MPC	288.4	100.1	CEII Redacted	P1	SH Charge	NU is not required unless it is identified as constraint in affected system study.
J1456	Mandan-Ward 230 kV	391.0	MDU BEPC	410.0	104.9	CEII Redacted	P1	SH Charge	MDU: Limiter is on BEPC. \$0 BEPC: NU is not required unless it is identified as constraint in affected system study.

4.3 Network Upgrades Identified in MISO ERS Analysis for 2024 Summer Shoulder Scenario

Based on the MISO 2024 summer shoulder charging and discharging steady state analyses, the MISO thermal NUs and cost are listed in Table 4-2. No reactive power NUs are required.

Table 4-2: Summer Shoulder Thermal NUs

Constraint	Owner	Scenario	Mitigation	Cost (\$)
J1438 POI-J1467 POI 161 kV	ITCM	SH Charge	Sag limit between Windsor and J1467 POI, next limit 226 MVA.	\$0
Electric Jct-Nelson 345 kV	PJM	SH Charge	NU is not required unless it is identified as constraint in affected system study.	\$0
Montgomery-J1145 POI 345 kV	Ameren	SH Charge, Discharge	Rebuild 28 mi of 345 kV line for 3000A SE	\$28,000,000 ¹
Split Rock-White 345 kV	XEL WAPA	SH Charge	XEL: WAPA limiter. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$0
Franklin-Cedar Mountain 115 kV	XEL GRE	SH Charge	XEL: Fix line limiter, line is 795 ACSS and should have a rating of 397 MVA emergency. \$1,000,000 GRE: Owned by XEL. \$0	\$1,000,000
Lyon Co 115-69 kV xfmr	XEL	SH Charge	Replace transformer with 112 MVA transformer	\$1,500,000
Crystal Foods Tap-Arlington 69 kV	XEL	SH Charge, Discharge	Rebuild 6.9 miles to 477 ACSR	\$4,830,000
Riverton-Mud Lake 230 kV	MP GRE	SH Discharge	MP: MP limits are 414/414/478/484 SN/SE/WN/WE. \$0 GRE: Remediate with structures and get it up to 383 MVA.	\$600,000
Brainerd-Riverton 115 kV	MP	SH Discharge	Increase conductor clearance to 75C, new limits would be 90/99/114/125. Line is 13.12 miles long.	\$2,361,600
Windsor 161-69 kV xfmr	SMMPA ITCM	SH Charge, Discharge	New Transformer and associated equipment	\$3,500,000
Inman-Wing River 230 kV	GRE	SH Charge	Replace Line CT's	\$125,000
Wing River-Riverton 230 kV	GRE MP	SH Charge	GRE: Owned by MP. \$0 MP: MP's portion would require increase of conductor clearance to 65C, new limits would be 282/311/388/427 SN/SE/WN/WE. MP owns miles of conductor. \$3,237,300	\$3,237,300
Wing River-Verndale 115 kV	GRE MP	SH Charge	GRE: Owned by MP. \$0 MP: Replace (3) 115 kV switches, replace substation conductor at Verndale Substation, reconductor 1.07-mile line. New limits would be 126/126/126/126.	\$1,500,000

Constraint	Owner	Scenario	Mitigation	Cost (\$)
			\$1,500,000	
Thomastown Tap-Minnesota Pipeline Staples Tap 115 kV	GRE MP	SH Charge	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080,000	\$4,080,000
Thomastown Tap-Dog Lake 115 kV	GRE MP	SH Charge	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080,000	Incl.
Aldrich Tap-Minnesota Pipeline Staples Tap 115 kV	GRE MP	SH Charge	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080,000	Incl.
Aldrich-Verndale 115 kV	GRE MP	SH Charge	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080,000	Incl.
Greenfield Tap-Greenfield 69 kV	CIPCO	SH Charge	Rebuild 0.51 miles of 69 kV line	\$300,000
Greenfield Tap-Greenfield Municipal 69 kV	CIPCO	SH Charge	Rebuild 2 miles of 69 kV line	\$1,000,000
Greenfield-MENLOTP 69 kV	CIPCO	SH Charge	Rebuild 7.97 miles of 69 kV line	\$4,000,000
Anita-Casey REC 69 kV	CIPCO	SH Charge	Rebuild 13.5 miles of 69 kV line	\$6,750,000
Liberty 161-69 kV xfmr	CIPCO	SH Charge	NU for IR20 in CIPCO Queue. This will not be a constraint.	\$0
Luana-Monona 69 kV	ITCM	SH Charge, Discharge	Line rebuild 1.52 miles, relay upgrade	\$1,200,000
Luana-Postville 69 kV	ITCM DPC	SH Charge, Discharge	ITCM: Line rebuild 4.06 miles, \$3M DPC: Upgrade Luana-Postville Line. \$20,000	\$3,020,000
Albert Lea Westside-South Broadway 69 kV	ITCM	SH Charge, Discharge	MTEP13 project 4099	\$0
Orient M8-Orient REC 69 kV	CIPCO	SH Charge	Rebuild 3.35 miles of 69 kV line	\$1,700,000
Orient M8-North Summit Lake 69 kV	CIPCO	SH Charge	Rebuild 7.71 miles of 69 kV line	\$4,000,000

Constraint	Owner	Scenario	Mitigation	Cost (\$)
Riverside-Sharon Tap 69 kV	CIPCO ITCM	SH Charge	No mitigation required. Contingency would disconnect Sub 56 161 kV bus including J1477.	\$0
Orient REC-Greenfield Muni 69 kV	CIPCO	SH Charge	Rebuild 6 miles of 69 kV line	\$3,000,000
North Summit Lake-Creston J2 69 kV	ITCM CIPCO	SH Charge	ITCM: Relay thermal, new ITCM rating 79 MVA. \$100K CIPCO: ITCM constraint. \$0	\$100,000
South Summit Lake-Creston 69 kV	CIPCO WAPA	SH Charge	Reconductor 69 kV tie	\$250,000
Casey REC-MENLOTP 69 kV	CIPCO	SH Charge	Reconductor 3.96 miles of 69 kV line	\$2,000,000
Osceola-Osceola REC 69 kV	CIPCO ITCM	SH Charge	CIPCO: ITCM CONSTRAINT. \$0 ITCM: Rating 62/66 MVA. \$0	\$0
Sharon Tap-Hills 69 kV	MEC CIPCO	SH Charge	No mitigation required. Contingency would disconnect Sub 56 161 kV bus including J1477.	\$0
Hazleton-J1438 POI 161 kV	ITCM	SH Charge	Most limiting sag limit between Windsor and J1467 POI, next limit 226 MVA. \$0	\$0
Hazleton-Dundee 161 kV	ITCM CIPCO	SH Charge	CIPCO: Replace terminal equipment at Dundee. \$600,000 ITCM: reconductor 18 miles, line is DC with Hazleton - Hickory Creek 345. \$13.23M	\$13,830,000
Lore-Hickory Creek 161 kV	ITCM	SH Charge	Hickory Creek - Turkey river 13 miles 161 kV, DC with MVP 5	\$8,700,000
Lore 161 kV bus tie	ITCM	SH Charge	Hickory Creek - Turkey river 13 miles 161 kV, DC with MVP 5	Incl.
Traer-Dysart 161 kV	ITCM	SH Charge	MTEP19 project 15455	\$0
Liberty-Hickory Creek 161 kV	ITCM CIPCO	SH Charge	CIPCO: Replace terminal equipment at Liberty. \$250,000 ITCM: Reconductor 4 miles, DC with Hickory - Hazleton 345. \$2.94M	\$3,190,000
Arnold-J1284 POI 345 kV	ITCM	SH Charge	Sag mitigation, 2 spans brings rating to 1203 MVA	\$240,000
Morgan Valley-J1284 POI 345 kV	ITCM	SH Charge	Sag limits near Arnold end, rating 1321 MVA	\$0
Morgan Valley-Tiffin 345 kV	ITCM MEC	SH Charge	MEC: Structure Replacements. New rating predicted to be 1079 MVA. ITCM: ITCM terminal 1895 MVA. \$0	\$300,000
Ward-Bismark 230 kV	BEPC WAPA	SH Charge	NU is not required unless it is identified as constraint in affected system study.	\$0
Pleasant Lake-Leeds 115 kV	WAPA	SH Charge	NU is not required unless it is identified as constraint in affected system study.	\$0
Pleasant Lake-Rugby 115 kV	WAPA	SH Charge	NU is not required unless it is identified as constraint in affected system study.	\$0

Constraint	Owner	Scenario	Mitigation	Cost (\$)
Granite Falls-Marshall 115 kV	WAPA BEPC	SH Charge, Discharge	NU is not required unless it is identified as constraint in affected system study.	\$0
Wilton-Winger 230 kV	MPC	SH Charge	NU is not required unless it is identified as constraint in affected system study.	\$0
Mandan-Ward 230 kV	MDU BEPC	SH Charge	MDU: Limiter is on BEPC. \$0 BEPC: NU is not required unless it is identified as constraint in affected system study.	\$0

Note 1: Majority cost of the Network Upgrade will be shared by DPP 2019 Central projects J1191, J1376, J1488, J1490.

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Section

5

Local Planning Criteria Analysis

Local Planning Criteria (LPC) analyses were performed to identify additional constraints per Transmission Owning Companies' LPC.

5.1 GRE Local Planning Criteria Analysis

Great River Energy (GRE) determined with provided rationale that the GRE LPC should be applied to the projects listed in Table 5-1. The GRE LPC analysis consisted of steady-state contingency analysis for summer shoulder condition.

Table 5-1: DPP Projects with GRE LPC Applicable

MISO Project	Pmax (MW)	Fuel type	POI	Rationale
J1221/J1230	150/40	Solar/Battery	Riverton 230 kV	GRE is the Transmission Owner of the portion of the transmission line at the POI
J1426	100	Solar	Belton County 115 kV	GRE is the Transmission Owner of the substation

Based on geographic locations of the projects' Point of Interconnection (POI), the projects were separated into the following two groups for GRE LPC study:

- J1221/J1230 GRE LPC group: J1221, J1230
- J1426 GRE LPC group: J1426

Siemens PTI performed the GRE local planning criteria analysis based on GRE's LPC. The J1221/J1230 GRE local planning criteria analysis details can be found in Appendix F.1.1, and the J1426 GRE local planning criteria analysis details can be found in Appendix F.1.2.

5.1.1 Additional Network Upgrades Identified in J1221/J1230 GRE LPC Analysis

Additional Network Upgrades required in the J1221 & J1230 GRE LPC study are listed in Table 5-2.

Table 5-2: Additional Network Upgrades Required in the J1221 & J1230 GRE LPC Study

Constraint	Owner	Mitigation	Cost (\$)
Benton Co-Mud Lake 230 kV	GRE MP	Re-conductor to 795 DRAKE ACSS	\$21,100,000

5.1.2 Additional Network Upgrades Identified in J1426 GRE LPC Analysis

Additional Network Upgrades required in the J1426 GRE LPC study are listed in Table 5-3.

Table 5-3: Additional Network Upgrades Required in the J1426 GRE LPC Study

Constraint	Owner	Mitigation	Cost (\$)
Bunker Lake-Sherco 345 kV CKT #2	GRE XEL	GRE: XEL line	\$0
Bunker Lake-Coon Creek 345 kV CKT #2	GRE XEL	GRE: XEL line	\$0
Bunker Lake-Blaine 230 kV	GRE	Re-conductor to 795 DRAKE ACSS and replace switches and wave trap	\$6,962,690
Bunker Lake-Elk River 230 kV	GRE	Re-conductor to 795 DRAKE ACSS and replace switches	\$8,158,000
Dickinson-Dickinson SS 115 kV	GRE	Re-conductor to 795 DRAKE ACSS and replace wave trap	\$152,690
Elk River-Monticello 230 kV	GRE XEL	Re-conductor to 795 DRAKE ACSS	\$6,623,200
Sheridan Tap-Wabasso 69 kV	GRE ITCM	Re-conductor to 336 Linnet ACSR	\$1,566,000
Dickinson SS-Lake Pulaski 115 kV	GRE XEL	GRE: XEL line	\$0

5.2 OTP Local Planning Criteria Analysis

J1328, J1371, J1428, J1456, and J1485 are to be interconnected in Otter Tail Power (OTP) transmission system. In addition to MISO's standard DPP analysis, OTP determined that J1328, J1371, J1428, J1456, and J1485 are required for OTP LPC study. The OTP LPC analysis consisted of steady-state contingency analysis for summer shoulder, summer peak, and Light Load No Wind conditions.

Three additional scenarios were analyzed in the OTP LPC study:

1. OTP LPC Summer Peak (SPK)
2. OTP LPC Summer Shoulder (SH)
3. OTP LPC Light Load No Wind (LLNW)

Siemens PTI performed the local planning criteria analysis based on OTP's LPC. The OTP local planning criteria analysis details can be found in Appendix F.2.

5.2.1 Additional Network Upgrades Identified in OTP LPC Analysis

No thermal or voltage constraints were identified in the summer peak or summer shoulder scenario.

In the light load no wind scenario, high voltage violations were identified in the area of J1456. 1×15 Mvar switched reactor is required to be installed at the J1456 POI 230 kV substation. J1456 is responsible for for this Network Upgrade.

5.3 DPC Local Planning Criteria Analysis

The DPC LPC study is performed to adhere to DPC Local Planning Criteria. This DPC LPC analysis is required for J1314 project in addition to MISO's standard DPP analysis.

Two additional scenarios are analyzed in this DPC LPC study:

1. DPC LPC Summer Shoulder
2. DPC LPC Summer Peak

Siemens PTI performed the local planning criteria analysis based on DPC's LPC. The DPC local planning criteria analysis details can be found in Appendix F.3.

5.3.1 Additional Network Upgrades Identified in DPC LPC Analysis

No thermal or voltage constraints were identified in the DPC LPC summer shoulder or summer peak analysis.

No additional Network Upgrades are required in the DPC LPC study.

MWEX Voltage Stability Study

ATC performed steady state voltage stability analysis. Voltage stability analysis is required to determine if the initial conditions of the DPP system models under study are in a stable state as defined by Power-Voltage (PV) curves of the Minnesota Wisconsin Export Interface (MWEX) for the worst contingency.

As shown in Table 6-1, the Pre-DPP scenario in the 2024SH case is voltage stable and meets ATC planning criteria.

The Post-DPP scenario in the 2024SH case is voltage stable and meets ATC planning criteria.

Because the Post-DPP scenario meets the planning criteria, Network Upgrades related to voltage stability will not be assigned to the Interconnection Customers, based on the assumptions used in this analysis.

The MWEX voltage stability study details can be found in Appendix G.

Table 6-1: MWEX Margins to Collapse in the 2024SH Cases

	Real Power Flow (MW)							
	AHD-PAR ¹	MWEX			Post-Contingent Margin to Nose ²		MWEX Control Point	
Case	N-0 Initial Condition	N-0 I.C. ³	N-1 I.C. ³	N-1 Nose	(MW)	(%)	(MW)	Notes
Pre-DPP	442.7	1106.6	547.3	805	257.7	32.0	1436	Voltage Stable ^{4b} Sufficient Margin ^{4a}
Post-DPP	515.3	1298.2	663.5	794.7	131.2	16.5	1406	Voltage Stable ^{4b} Sufficient Margin ^{4a}

Notes:

- As described in the active MWEX Operating Guide, the AHD-SLK interface is a single element Power Transfer Distribution Factor (PTDF) interface measured at the Minnesota Power 230 kV side of the Arrowhead 230 kV phase shifter. Due to the addition of the Superior substation, the AHD-SLK interface has here been renamed to AHD-PAR to avoid confusion.
- Post-Contingent Margin to Nose is defined as:
 - "Post-Contingent Margin to Nose (MW)" = "MWEX N-1 Nose" – "N-1 Initial Condition"
 - "Post-Contingent Margin to Nose (%)" = "Post-Contingent Margin to Nose (MW)" / "MWEX N-1 Nose"

3. Initial Condition flows were measured in the base cases with an intact system and the worst contingency, plus operation of various control systems as needed with all transformer taps, switched shunts, and PARs locked.
4. ATC Planning Criteria requires:
 - a. The N-0 Initial Condition must be less than or equal to the Control Point (CP). The CP is 98% of the System Operating Limit (SOL). The SOL is the pre-contingent flow that results in a post-contingent flow that is 95% of the post-Contingency P-V curve nose.
 - b. $V_{nose} < V_{min}$.

Deliverability Study

7.1 Project Description

Interconnection requests requesting Network Resource Interconnection Services (NRIS) were considered for deliverability analysis.

7.2 Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS).

If the generator is determined as not fully deliverable, the customer can choose either to change his project to an Energy Resource (ER) project or proceed with the system upgrades that will make the generator fully deliverable.

Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind generators are tested at 100 % of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

7.3 Study Methodology

MISO Generation Deliverability Study method can be found in Appendix C of the MISO Generation Interconnection Business Practices Manual BPM-015-r22.

7.4 2024 Deliverability Study Result

7.4.1 J1190

J1190 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)
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7.4.2 J1212

J1212 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	60 MW (100%)
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7.4.3 J1218

J1218 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	44.84	0.0988	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$65,500	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	76.87	0.0988	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$1,242,773	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	200.00	0.0970	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$275,608	\$4,000,000

7.4.4 J1221

J1221 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Stone Lake 345/161 Transformer Upgrade	69.15	0.0518	J1221, J1230	\$3,157,895	\$4,000,000
Beaver Creek - Grimes 345 Line Rebuild	118.95	0.0800	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$714,377	\$26,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	124.88	0.0922	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$32,289	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	124.89	0.1205	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$21,535	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	132.75	0.0773	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$93,141	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	150.00	0.0860	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$9,707	\$800,000

7.4.5 J1229

J1229 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	22.42	0.0988	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$16,375	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	38.44	0.0988	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$310,693	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	50.00	0.0970	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$68,902	\$4,000,000

7.4.6 J1230

J1230 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Stone Lake 345/161 Transformer Upgrade	18.44	0.0518	J1221, J1230	\$842,105	\$4,000,000
Beaver Creek - Grimes 345 Line Rebuild	31.72	0.0800	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$190,500	\$26,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	33.30	0.0922	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$8,610	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	33.30	0.1205	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$5,743	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	35.40	0.0773	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$24,838	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	40.00	0.0860	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$2,588	\$800,000

7.4.7 J1270

J1270 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	92.2 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	145.74	0.1741	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$2,072,883	\$26,000,000
Remson-Raun Structure Upgrades	158.60	0.4245	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$352,210	\$1,100,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	166.50	0.0905	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$42,258	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	177.00	0.0755	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$17,991	\$800,000
Tiffin-Hills 345 kV Structure Replacments	200.00	0.0849	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$12,777	\$800,000

7.4.8 J1284

J1284 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	113.45 MW (45.38%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	181.05	0.0879	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$72,842	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	221.09	0.0879	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$1,382,082	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	224.13	0.0866	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$307,572	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	235.63	0.5073	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$296,100	\$2,700,000
Tiffin-Hills 345 kV Structure Replacments	250.00	0.4663	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$87,718	\$800,000

7.4.9 J1285

J1285 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	22.42	0.0879	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$14,568	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	38.44	0.0879	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$276,416	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	39.65	0.0866	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$61,514	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	44.25	0.5073	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$59,220	\$2,700,000
Tiffin-Hills 345 kV Structure Replacments	50.00	0.4663	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$17,544	\$800,000

7.4.10 J1298

J1298 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	8.71	0.0548	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$27,248	\$1,000,000
Murphy Creek Transformer Upgrade	67.26	0.1208	J1298, J1487	\$2,533,557	\$3,500,000
J1338 POI - J1363 POI 345 kV Rebuild	69.15	0.0548	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$516,984	\$19,000,000
Beaver Creek - Grimes 345 Line Rebuild	84.34	0.0859	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$767,062	\$26,000,000
Austin - Adams 161 kV Line Rebuild	109.30	0.3469	J1298, J1416, J1487	\$6,669,140	\$15,750,000
Remson-Raun Structure Upgrades	115.31	0.0532	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$33,105	\$1,100,000
Spencer-J1338 POI 345 kV Rebuild	118.95	0.0542	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$115,499	\$4,000,000

J1298 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	124.88	0.1481	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$51,866	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	132.75	0.2290	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$40,926	\$800,000
Tiffin-Hills 345 kV Structure Replacements	134.10	0.1377	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$15,542	\$800,000
Pleasant Valley-Byron 161 kV Rebuild	150.00	0.1173	J1298, J1341, J1413, J1416, J1487	\$4,096,187	\$24,900,000

7.4.11 J1300

J1300 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	55.32 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	76.74	0.1227	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$876,540	\$26,000,000
Brewster-Heron Lake 161 Structure Remediation	76.74	0.4259	J1300, J1325	\$373,653	\$960,000
Remson-Raun Structure Upgrades	95.16	0.1243	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$61,879	\$1,100,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	99.90	0.0898	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$25,159	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	106.20	0.0859	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$12,281	\$800,000
Tiffin-Hills 345 kV Structure Replacments	120.00	0.0842	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$7,603	\$800,000

7.4.12 J1312

J1312 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	23.05 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	39.65	0.0955	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$284,262	\$26,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	41.63	0.0755	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$8,814	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	44.25	0.0787	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$4,688	\$800,000
Tiffin-Hills 345 kV Structure Replacements	50.00	0.0708	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$2,664	\$800,000

7.4.13 J1313

J1313 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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7.4.14 J1314

J1314 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	41.89 MW (41.89%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	46.10	0.0503	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$16,673	\$1,000,000
Beaver Creek - Grimes 345 Line Rebuild	74.86	0.0676	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$402,432	\$26,000,000
J1338 POI - J1363 POI 345 kV Rebuild	79.30	0.0503	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$316,354	\$19,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	83.25	0.0896	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$20,919	\$2,700,000

J1314 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	41.89 MW (41.89%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1181 POI - Hazelton 345 Structure Replacements	88.50	0.1176	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$14,011	\$800,000
Tiffin-Hills 345 kV Structure Replacments	100.00	0.0834	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$6,276	\$800,000

7.4.15 J1315

J1315 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	276.6 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	437.22	0.0837	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$2,989,667	\$26,000,000
Remson-Raun Structure Upgrades	475.80	0.0581	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$144,618	\$1,100,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	499.50	0.0954	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$133,639	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	499.56	0.1194	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$85,355	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	531.00	0.0532	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$256,409	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	600.00	0.0890	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$40,181	\$800,000

7.4.16 J1321

J1321 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	29.96 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	47.36	0.1872	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$724,378	\$26,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	54.11	0.0925	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$14,037	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	57.53	0.0754	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$5,839	\$800,000
Tiffin-Hills 345 kV Structure Replacments	65.00	0.0868	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$4,245	\$800,000

7.4.17 J1325

J1325 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Elk-J1300 POI 161 kV Rebuild	80.92	0.9021	J1325	\$12,700,000	\$12,700,000
Wolf Tap-Rushmore 69 kV Rebuild	92.20	0.0978	J1325	\$217,500	\$217,500
Beaver Creek - Grimes 345 Line Rebuild	127.90	0.1133	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$1,348,982	\$26,000,000
Brewster-Heron Lake 161 Structure Remediation	145.74	0.4010	J1300, J1325	\$586,347	\$960,000
Remson-Raun Structure Upgrades	156.20	0.0783	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$64,966	\$1,100,000
Magnolia Transformer Terminal upgrade with ERIS Upgrade	158.60	0.0978	J1325	\$517,000	\$517,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	166.50	0.0844	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$39,410	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	177.00	0.0831	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$19,802	\$800,000

J1325 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Tiffin-Hills 345 kV Structure Replacements	200.00	0.0791	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$11,904	\$800,000

7.4.18 J1328

J1328 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	46.1 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	52.49	0.0902	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$536,973	\$26,000,000
Granite Falls-Minn Valley 230 terminal upgrades	79.30	0.0594	J1328, J1371, J1485	\$205,536	\$1,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	83.25	0.0875	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$20,429	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	83.26	0.1053	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$12,546	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	88.50	0.0688	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$55,266	\$2,000,000
Tiffin-Hills 345 kV Structure Replacements	100.00	0.0817	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$6,148	\$800,000

7.4.19 J1337

J1337 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	138.3 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	218.61	0.0819	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$1,462,687	\$26,000,000
Remson-Raun Structure Upgrades	237.90	0.0535	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$66,584	\$1,100,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	249.75	0.0985	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$68,991	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	249.78	0.1326	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$47,395	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	265.50	0.2679	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$645,602	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	300.00	0.0918	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$20,723	\$800,000

7.4.20 J1341

J1341 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	178.48	0.0719	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$94,857	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	305.94	0.0719	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$1,799,769	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	315.62	0.0704	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$398,057	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	352.24	0.2928	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$272,075	\$2,700,000
Tiffin-Hills 345 kV Structure Replacments	355.82	0.2706	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$81,039	\$800,000
Pleasant Valley-Byron 161 kV Rebuild	398.00	0.0585	J1298, J1341, J1413, J1416, J1487	\$5,420,377	\$24,900,000

7.4.21 J1349

J1349 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	154.43 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	244.12	0.1187	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$1,943,254	\$26,000,000
Remson-Raun Structure Upgrades	265.66	0.1690	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$192,803	\$1,100,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	278.89	0.0952	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$61,123	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	280.85	0.0961	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$31,487	\$800,000
ITCM Lakefield-XEL Lakefield 345 Structure Replacements	296.48	0.9975	J1349	\$2,000,000	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	335.00	0.0890	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$18,416	\$800,000

7.4.22 J1359

J1359 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	42.60	0.0863	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$27,176	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	73.02	0.0863	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$515,631	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	75.34	0.0849	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$114,583	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	84.08	0.2716	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$60,240	\$2,700,000
Tiffin-Hills 345 kV Structure Replacments	95.00	0.2508	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$17,928	\$800,000

7.4.23 J1365

J1365 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	75.96	0.0695	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$39,026	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	124.66	0.0695	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$740,463	\$19,000,000
Laurel - Jasper 161 kV Rebuild	130.22	0.3418	J1365	\$10,560,000	\$10,560,000
Spencer-J1338 POI 345 kV Rebuild	134.34	0.0692	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$166,536	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	149.92	0.1759	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$69,569	\$2,700,000
Tiffin-Hills 345 kV Structure Replacments	169.40	0.1686	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$21,491	\$800,000

7.4.24 J1371

J1371 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	69.15 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	78.73	0.0918	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$819,747	\$26,000,000
Granite Falls-Minn Valley 230 terminal upgrades	118.95	0.0594	J1328, J1371, J1485	\$308,304	\$1,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	124.88	0.0871	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$30,503	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	124.89	0.1038	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$18,551	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	132.75	0.0631	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$76,031	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	150.00	0.0814	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$9,188	\$800,000

7.4.25 J1395

J1395 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	92.2 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	145.74	0.0931	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$1,108,475	\$26,000,000
Remson-Raun Structure Upgrades	158.60	0.0520	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$43,145	\$1,100,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	166.50	0.0919	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$42,912	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	166.52	0.1125	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$26,807	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	177.00	0.0555	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$89,165	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	200.00	0.0858	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$12,912	\$800,000

7.4.26 J1413

J1413 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)		0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	134.52	0.0606	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$60,263	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	138.30	0.0606	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$1,143,402	\$19,000,000
Beaver Creek - Grimes 345 Line Rebuild	230.62	0.0597	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$1,066,207	\$26,000,000
Spencer-J1338 POI 345 kV Rebuild	237.90	0.0595	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$253,587	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	249.76	0.1922	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$134,620	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	265.50	0.5111	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$182,683	\$800,000

J1413 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distributio n Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Tiffin-Hills 345 kV Structure Replacments	268.20	0.1781	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$40,204	\$800,000
Pleasant Valley-Byron 161 kV Rebuild	300.00	0.1639	J1298, J1341, J1413, J1416, J1487	\$11,446,975	\$24,900,000

7.4.27 J1416

J1416 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	66.82	0.0549	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$27,115	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	68.69	0.0549	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$514,474	\$19,000,000
Beaver Creek - Grimes 345 Line Rebuild	83.78	0.0597	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$529,550	\$26,000,000
Austin - Adams 161 kV Line Rebuild	109.26	0.3469	J1298, J1416, J1487	\$6,624,679	\$15,750,000
Glenworth - Hayward 161 kV terminal upgrades	114.54	0.3790	J1416, J1417	\$118,453	\$130,000
Spencer-J1338 POI 345 kV Rebuild	118.16	0.0545	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$115,365	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	124.05	0.1520	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$52,877	\$2,700,000

J1416 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1181 POI - Hazelton 345 Structure Replacements	131.87	0.1390	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$24,676	\$800,000
Tiffin-Hills 345 kV Structure Replacements	133.21	0.1416	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$15,876	\$800,000
Pleasant Valley-Byron 161 kV Rebuild	149.00	0.0621	J1298, J1341, J1413, J1416, J1487	\$2,154,113	\$24,900,000

7.4.28 J1417

J1417 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	33.63	0.0511	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$12,704	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	34.57	0.0511	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$241,039	\$19,000,000
Beaver Creek - Grimes 345 Line Rebuild	54.56	0.1321	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$589,807	\$26,000,000
Remson-Raun Structure Upgrades	55.00	0.0583	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$18,139	\$1,100,000
Glenworth - Hayward 161 kV terminal upgrades	57.65	0.0734	J1416, J1417	\$11,547	\$130,000
Spencer-J1338 POI 345 kV Rebuild	59.48	0.0511	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$54,447	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	62.44	0.1358	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474,	\$23,779	\$2,700,000

J1417 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
			J1485, J1487		
J1181 POI - Hazelton 345 Structure Replacements	66.38	0.0905	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$8,087	\$800,000
Tiffin-Hills 345 kV Structure Replacments	75.00	0.1266	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$7,145	\$800,000

7.4.29 J1426

J1426 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	46.1 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	72.87	0.0820	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$488,157	\$26,000,000
Remson-Raun Structure Upgrades	79.30	0.0519	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$21,531	\$1,100,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	83.25	0.0971	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$22,670	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	83.26	0.1296	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$15,441	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	88.50	0.2159	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$173,430	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	100.00	0.0905	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$6,810	\$800,000

7.4.30 J1427

J1427 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	179.36	0.0847	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$112,305	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	307.48	0.0847	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$2,130,827	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	317.20	0.0833	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$473,363	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	354.00	0.4497	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$419,968	\$2,700,000
Tiffin-Hills 345 kV Structure Replacments	400.00	0.4138	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$124,547	\$800,000

7.4.31 J1428

J1428 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	92.2 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	98.75	0.0813	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$967,981	\$26,000,000
Buffalo-Castleton 115 kV Reconductor	156.17	0.3081	J1428, J1456	\$3,343,355	\$4,250,000
Mapleton-Castleton 115 kV Reconductor	158.60	0.3081	J1428, J1456	\$3,343,355	\$4,250,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	166.50	0.0873	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$40,764	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	166.52	0.1094	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$26,069	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	177.00	0.0737	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$118,405	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	200.00	0.0815	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$12,265	\$800,000

7.4.32 J1432

J1432 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	112.11	0.0792	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$65,633	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	192.18	0.0792	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$1,245,289	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	250.00	0.0760	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$269,925	\$4,000,000

7.4.33 J1435

J1435 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	44.84	0.0743	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$24,629	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	76.87	0.0743	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$467,298	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	97.55	0.0717	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$101,861	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	100.00	0.0935	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$21,830	\$2,700,000

7.4.34 J1438

J1438 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	40.31	0.0701	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$23,237	\$1,000,000
J1438 POI - Hazelton 161 kV Rebuild	44.84	1.0000	J1438, J1467	\$4,800,000	\$12,000,000
J1338 POI - J1363 POI 345 kV Rebuild	76.87	0.0743	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$467,298	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	79.30	0.0686	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$97,457	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	97.55	0.2403	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$56,103	\$2,700,000
Tiffin-Hills 345 kV Structure Replacments	100.00	0.2188	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$16,464	\$800,000

7.4.35 J1444

J1444 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	89.68	0.0847	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$56,152	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	153.74	0.0847	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$1,065,414	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	158.60	0.0833	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$236,682	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	177.00	0.4497	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$209,984	\$2,700,000
Tiffin-Hills 345 kV Structure Replacments	200.00	0.4138	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$62,273	\$800,000

7.4.36 J1445

J1445 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	46.1 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	72.87	0.0823	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$489,943	\$26,000,000
Remson-Raun Structure Upgrades	79.30	0.0518	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$21,489	\$1,100,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	83.26	0.0970	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$22,647	\$2,700,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	83.26	0.0737	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$59,202	\$2,000,000
J1181 POI - Hazelton 345 Structure Replacements	88.50	0.1292	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$15,393	\$800,000
Tiffin-Hills 345 kV Structure Replacments	100.00	0.0904	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$6,802	\$800,000

7.4.37 J1446

J1446 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	69.15 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	109.30	0.0831	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$742,059	\$26,000,000
Remson-Raun Structure Upgrades	118.95	0.0552	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$34,350	\$1,100,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	124.88	0.0996	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$34,881	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	124.89	0.1341	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$23,966	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	132.75	0.2115	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$254,843	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	150.00	0.0928	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$10,474	\$800,000

7.4.38 J1456

J1456 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	138.3 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	148.14	0.0779	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$1,391,249	\$26,000,000
Buffalo-Castleton 115 kV Reconductor	234.24	0.0557	J1428, J1456	\$906,645	\$4,250,000
Mapleton-Castleton 115 kV Reconductor	237.90	0.0557	J1428, J1456	\$906,645	\$4,250,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	249.75	0.0819	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$57,364	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	265.50	0.1000	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$35,743	\$800,000
Tiffin-Hills 345 kV Structure Replacments	300.00	0.0766	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$17,291	\$800,000

7.4.39 J1461

J1461 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	23.05 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	36.43	0.0862	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$256,580	\$26,000,000
Remson-Raun Structure Upgrades	39.65	0.0575	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$11,927	\$1,100,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	41.63	0.1005	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$11,732	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	41.63	0.1344	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$8,006	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	44.25	0.0671	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$26,950	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	50.00	0.0937	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$3,525	\$800,000

7.4.40 J1467

J1467 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	60.46	0.0698	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$34,706	\$1,000,000
J1438-Hazelton 345 kV Rebuild	67.26	1.0000	J1438, J1467	\$7,200,000	\$12,000,000
J1338 POI - J1363 POI 345 kV Rebuild	105.18	0.0698	J1438, J1467	\$658,494	\$19,000,000
J1467 POI-J1438 POI 161 kV Rebuild	115.31	0.8885	J1467	\$2,181,000	\$2,181,000
Spencer-J1338 POI 345 kV Rebuild	118.95	0.0684	J1218,J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$145,760	\$4,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	132.75	0.2362	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$82,719	\$2,700,000
Tiffin-Hills 345 kV Structure Replacments	150.00	0.2188	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$24,696	\$800,000

7.4.41 J1468

J1468 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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7.4.42 J1471

J1471 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	89.68	0.1299	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$86,118	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	153.51	0.1299	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$1,633,970	\$19,000,000
Burlington-Denmark 161 kV Rebuild	153.74	0.4446	J1471, J1478, J1479	\$8,203,444	\$13,850,000
Spencer-J1338 POI 345 kV Rebuild	200.00	0.1289	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$366,246	\$4,000,000

7.4.43 J1473

J1473 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	46.1 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	79.30	0.0862	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$513,161	\$26,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	83.25	0.0939	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$21,923	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	88.50	0.1256	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$14,964	\$800,000
Tiffin-Hills 345 kV Structure Replacments	100.00	0.0875	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$6,584	\$800,000

7.4.44 J1474

J1474 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	83.77 MW (41.9%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	137.35	0.0513	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$34,010	\$1,000,000
Beaver Creek - Grimes 345 Line Rebuild	149.73	0.0612	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$728,664	\$26,000,000
J1338 POI - J1363 POI 345 kV Rebuild	158.60	0.0513	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$645,286	\$19,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	166.50	0.0827	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$38,616	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	177.00	0.1056	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$25,163	\$800,000

J1474 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	83.77 MW (41.9%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Tiffin-Hills 345 kV Structure Replacements	177.50	0.0769	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$11,573	\$800,000
Holcombe Transformer and Terminal Equipment Upgrade	200.00	0.0513	J1474	\$1,700,000	\$1,700,000

7.4.45 J1476

J1476 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	25 MW (100%)
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7.4.46 J1477

J1477 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	22.42	0.0898	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$14,883	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	38.44	0.0898	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$282,391	\$19,000,000
Spencer-J1338 POI 345 kV Rebuild	50.00	0.0870	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$61,799	\$4,000,000

7.4.47 J1478

J1478 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	11.21	0.1260	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$10,442	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	19.19	0.1260	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$198,114	\$19,000,000
Burlington-Denmark 161 kV Rebuild	19.22	0.3498	J1471, J1478, J1479	\$806,783	\$13,850,000
Spencer-J1338 POI 345 kV Rebuild	25.00	0.1251	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$44,431	\$4,000,000

7.4.48 J1479

J1479 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	44.84	0.1288	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$42,694	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	76.76	0.1288	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$810,067	\$19,000,000
Burlington-Denmark 161 kV Rebuild	76.87	0.5246	J1471, J1478, J1479	\$4,839,774	\$13,850,000
Spencer-J1338 POI 345 kV Rebuild	100.00	0.1281	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$181,986	\$4,000,000

7.4.49 J1485

J1485 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	115.24 MW (46.1%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Beaver Creek - Grimes 345 Line Rebuild	131.22	0.0862	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$1,282,902	\$26,000,000
Granite Falls-Minn Valley 230 terminal upgrades	198.26	0.0562	J1328, J1371, J1485	\$486,159	\$1,000,000
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	208.14	0.0871	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$50,838	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	208.16	0.1038	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371,J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$30,918	\$800,000
Coon Creek-Terminal 345 Terminal Equipment Upgrade	221.26	0.0631	J1221, J1230, J1315, J1328, J1337, J1371, J1395, J1426, J1428, J1445, J1446, J1461, J1485	\$126,719	\$2,000,000
Tiffin-Hills 345 kV Structure Replacments	250.00	0.0814	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$15,313	\$800,000

7.4.50 J1487

J1487 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	6.96	0.0524	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$20,843	\$1,000,000
Murphy Creek Transformer Upgrade	53.81	0.0576	J1298, J1487	\$966,443	\$3,500,000
J1338 POI - J1363 POI 345 kV Rebuild	55.32	0.0524	J1218, J1229, J1284, J1285, J1298, J1314, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1474, J1477, J1478, J1479, J1487	\$395,474	\$19,000,000
Beaver Creek - Grimes 345 Line Rebuild	67.48	0.0996	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$711,519	\$26,000,000
Austin - Adams 161 kV Line Rebuild	87.44	0.1597	J1298, J1416, J1487	\$2,456,181	\$15,750,000
Remson-Raun Structure Upgrades	92.24	0.0668	J1270, J1298, J1300, J1315, J1325, J1337, J1349, J1395, J1417, J1426, J1445, J1446, J1461, J1487	\$33,255	\$1,100,000
Spencer-J1338 POI 345 kV Rebuild	95.16	0.0521	J1218, J1229, J1284, J1285, J1298, J1341, J1359, J1365, J1413, J1416, J1417, J1427, J1432, J1435, J1438, J1444, J1467, J1471, J1477, J1478, J1479, J1487	\$88,820	\$4,000,000

J1487 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Morgan Valley-Tiffin 345 Structure Replacements and Terminal Equipment	99.90	0.1375	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1435, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$38,523	\$2,700,000
J1181 POI - Hazelton 345 Structure Replacements	106.20	0.1674	J1221, J1230, J1270, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1428, J1445, J1446, J1456, J1461, J1473, J1474, J1485, J1487	\$23,934	\$800,000
Tiffin-Hills 345 kV Structure Replacements	107.28	0.1281	J1221, J1230, J1270, J1284, J1285, J1298, J1300, J1312, J1314, J1315, J1321, J1325, J1328, J1337, J1341, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1427, J1428, J1438, J1444, J1445, J1446, J1456, J1461, J1467, J1473, J1474, J1485, J1487	\$11,567	\$800,000
Pleasant Valley-Byron 161 kV Rebuild	120.00	0.0638	J1298, J1341, J1413, J1416, J1487	\$1,782,348	\$24,900,000

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Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) test for Network Upgrades driven by higher queued interconnection project was performed for this System Impact Study.

No SNUs were identified in this study.

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Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

9.1 Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

9.2 ERIS Network Upgrades Proposed for DPP West Area Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the MISO ERIS analysis, LPC analyses, and the affected system analysis. The ERIS network upgrades include voltage network upgrades and thermal network upgrades identified in the MISO steady-state analysis, network upgrades identified in the Local Planning Criteria analysis and affected system analysis, voltage network upgrades identified in the MWEX voltage stability analysis, stability network upgrades identified in the MISO transient stability analysis, and short circuit network upgrades identified in the MISO short circuit analysis. The total costs of ERIS network upgrades for the 2024 scenario are summarized in Table 9-1.

Table 9-1: Summary of ERIS Network Upgrades

Category of Network Upgrades	Cost (\$)
Network Upgrades Identified in MWEX Voltage Stability analysis	\$0
Thermal Network Upgrades Identified in MISO Steady-State Analysis	\$219,938,628
Reactive Power Network Upgrades for Voltage Constraints	\$0
Network Upgrades Identified in GRE LPC Analysis	\$44,562,580
Network Upgrades Identified in OTP LPC Analysis	\$2,500,000
Network Upgrades Identified in DPC LPC Analysis	\$0
Shared Network Upgrades	\$0
Total	\$267,001,208

ERIS network upgrades are listed below.

Table 9-2: Network Upgrades Required for MWEX Voltage Stability

NUs	Miles	Cost (\$)
No MWEX NUs		\$0

Table 9-3: Thermal Network Upgrades in MISO Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
J1438 POI-J1467 POI 161 kV	ITCM	Sag limit between Windsor and J1467 POI, next limit 226 MVA.	\$0
Montgomery-J1145 POI 345 kV	Ameren	Rebuild 28 mi of 345 kV line for 3000A SE	\$28,000,000 ¹
Niota 161-69 kV xfmr	Ameren	Replace with new 112 MVA 161/69 kV Transformer.	\$4,000,000
Split Rock-White 345 kV	XEL WAPA	XEL: WAPA limiter. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$0
Split Rock-Rock Co. 161 kV	XEL	Rebuild 13.7 miles of 477 ACSR to 795 ACSR	\$20,550,000
Rock Co.-Magnolia 161 kV	XEL ITCM	XEL: Rebuild XEL portion 5.27 miles of 477 ACSR to 795 ACSR. \$6M ITCM: line rebuild ITCM portion 11.4 miles. \$12M	\$18,000,000
Split Rock 161-115-36.2 kV xfmr #6	XEL	Replace existing with 336 MVA transformer	\$3,000,000
Franklin-Cedar Mountain 115 kV	XEL GRE	XEL: Fix line limiter, line is 795 ACSS and should have a rating of 397 MVA emergency. \$1,000,000 GRE: Owned by XEL. \$0	\$1,000,000
Lyon Co 115-69 kV xfmr	XEL	Repalce transformer with 112 MVA transformer	\$1,500,000
Crystal Foods Tap-Arlington 69 kV	XEL	Rebuild 6.9 miles to 477 ACSR	\$4,830,000
Riverton-Mud Lake 230 kV	MP GRE	MP: MP limits are 414/414/478/484 SN/SE/WN/WE. \$0 GRE: Remediate with structures and get it up to 383 MVA.	\$600,000
Brainerd-Riverton 115 kV	MP	Increase conductor clearance to 75C, new limits would be 90/99/114/125. Line is 13.12 miles long.	\$2,361,600
Windsor 161-69 kV xfmr	SMPA ITCM	New Transformer and associated equipment	\$3,500,000
Inman-Wing River 230 kV	GRE	Replace Line CT's	\$125,000
Wing River-Riverton 230 kV	GRE MP	GRE: Owned by MP. \$0 MP: MP's portion would require increase of conductor clearance to 65C, new limits would be 282/311/388/427 SN/SE/WN/WE. MP owns miles of conductor. \$3,237,300	\$3,237,300

Constraint	Owner	Mitigation	Cost (\$)
Wing River-Verndale 115 kV	GRE MP	GRE: Owned by MP. \$0 MP: Replace (3) 115 kV switches, replace substation conductor at Verndale Substation, reconductor 1.07 mile line. New limits would be 126/126/126/126. \$1,500,000	\$1,500,000
Brewster-J1300 POI 161 kV	GRE ITCM	GRE: Owned by ITCM. \$0 ITCM: Lakefield Corner station and J1300 - Lakfield Corner new line. \$62.6M	\$62,600,000
Brewster-Heron Lake 161 kV	GRE ITCM	GRE: Owned by ITCM. \$0 ITCM: Lakefield Corner station and J1300 - Lakfield Corner new line. \$62.6M	Incl.
Thomastown Tap-Minnesota Pipeline Staples Tap 115 kV	GRE MP	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080.000	\$4,080,000
Thomastown Tap-Dog Lake 115 kV	GRE MP	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080.000	Incl.
Aldrich Tap-Minnesota Pipeline Staples Tap 115 kV	GRE MP	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080.000	Incl.
Aldrich-Verndale 115 kV	GRE MP	GRE: Owned by MP. \$0 MP: Increase conductor clearance on Dog Lake - Thomastown Tap - Minnesota Pipeline Staples Tap - Aldrich Tap - Verndale 115 kV line to 80C, new limits would be 103/113/129/142 SN/SE/WN/WE. Line is 17 miles long. \$4,080.000	Incl.
Buffalo 345-115-41.6 kV xfmr	OTP	Replace the Buffalo #2 transformer with a larger transformer.	\$3,000,000
Maple River 345-230-13.8 kV xfmr #1	OTP MPC	Replace the Maple River #1 transformer with a larger transformer.	\$4,000,000
Maple River 345-230-13.8 kV xfmr #2	OTP MPC	Replace the Maple River #2 transformer with a larger transformer.	\$4,000,000
Greenfield Tap-Greenfield 69 kV	CIPCO	Rebuild 0.51 miles of 69 kV line	\$300,000
Greenfield Tap-Greenfield Municipal 69 kV	CIPCO	Rebuild 2 miles of 69 kV line	\$1,000,000
Greenfield-MENLOTP 69 kV	CIPCO	Rebuild 7.97 miles of 69 kV line	\$4,000,000
Anita-Casey REC 69 kV	CIPCO	Rebuild 13.5 miles of 69 kV line	\$6,750,000

Constraint	Owner	Mitigation	Cost (\$)
Liberty 161-69 kV xfmr	CIPCO	NU for IR20 in CIPCO Queue. This will not be a constraint.	\$0
Luana-Monona 69 kV	ITCM	Line rebuild 1.52 miles, relay upgrade	\$1,200,000
Luana-Postville 69 kV	ITCM DPC	ITCM: Line rebuild 4.06 miles, \$3M DPC: Upgrade Luana-Postville Line. \$20,000	\$3,020,000
Heron Lake 161-69 kV xfmr	ITCM	Lakefield Corner station and J1300 - Lakefield Corner new line	Incl.
Albert Lea Westside-South Broadway 69 kV	ITCM	MTEP13 project 4099	\$0
Orient M8-Orient REC 69 kV	CIPCO	Rebuild 3.35 miles of 69 kV line	\$1,700,000
Orient M8-North Summit Lake 69 kV	CIPCO	Rebuild 7.71 miles of 69 kV line	\$4,000,000
Orient REC-Greenfield Muni 69 kV	CIPCO	Rebuild 6 miles of 69 kV line	\$3,000,000
North Summit Lake-Creston J2 69 kV	ITCM CIPCO	ITCM: Relay thermal, new ITCM rating 79 MVA. \$100K CIPCO: ITCM constraint. \$0	\$100,000
South Summit Lake-Creston 69 kV	CIPCO WAPA	Reconductor 69 kV tie	\$250,000
Casey REC-MENLOTP 69 kV	CIPCO	Reconductor 3.96 miles of 69 kV line	\$2,000,000
East Creston Tap-Creston J1 69 kV	ITCM CIPCO	CIPCO: Not a constraint ITCM: Rebuild 1 mile, new rating 72 MVA. \$730K	\$730,000
PCI J2-Oakridge 69 kV	ITCM CIPCO	CIPCO: Not a constraint. \$0 ITCM: Relay thermal, replace sub taps (.08 miles), new rating 143 MVA. \$450K	\$450,000
Magnolia-Elk 161 kV	ITCM	Rebuild 18.56 miles, new rating 366 MVA	\$19,600,000
Heron Lake-Lakefield 161 kV	ITCM	MTEP15 project 8106, limit 233 MVA	\$0
Hazleton-J1438 POI 161 kV	ITCM	Most limiting sag limit between Windsor and J1467 POI, next limit 226 MVA. \$0	\$0
Hazleton-Dundee 161 kV	ITCM CIPCO	CIPCO: Replace terminal equipment at Dundee. \$600,000 ITCM: reconductor 18 miles, line is DC with Hazleton - Hickory Creek 345. \$13.23M	\$13,830,000
Lore-Hickory Creek 161 kV	ITCM	Hickory Creek - Turkey river 13 miles 161 kV, DC with MVP 5	\$8,700,000
Lore 161 kV bus tie	ITCM	Hickory Creek - Turkey river 13 miles 161 kV, DC with MVP 5	Incl.
Traer-Dysart 161 kV	ITCM	MTEP19 project 15455	\$0
Liberty-Hickory Creek 161 kV	ITCM CIPCO	CIPCO: Replace terminal equipment at Liberty. \$250,000 ITCM: Reconductor 4 miles, DC with Hickory - Hazleton 345. \$2.94M	\$3,190,000

Constraint	Owner	Mitigation	Cost (\$)
Denmark-Burlington 161 kV #1	ITCM	Sag mitigation, 4 spans brings rating to 263 MVA	\$480,000
Arnold-J1284 POI 345 kV	ITCM	Sag mitigation, 2 spans brings rating to 1203 MVA	\$240,000
Morgan Valley-J1284 POI 345 kV	ITCM	Sag limits near Arnold end, rating 1321 MVA	\$0
Morgan Valley-Tiffin 345 kV	ITCM MEC	MEC: Structure Replacements. New rating predicted to be 1079 MVA. ITCM: ITCM terminal 1895 MVA. \$0	\$300,000
Wever-Lee 69 kV	ITCM AECI	ITCM: AECI constraint AECI: NU is not required unless it is identified as constraint in affected system study.	\$0
Mandan-Ward 230 kV	MDU BEPC	MDU: Limiter is on BEPC. \$0 BEPC: NU is not required unless it is identified as constraint in affected system study.	\$0
Apple River 161-69 kV xfmr #2	DPC	Upgrade Apple River Transformer #2	\$2,636,000

Note 1: Majority cost of the Network Upgrade will be shared by DPP 2019 Central projects J1191, J1376, J1488, J1490.

Table 9-4: Reactive Power NUs Required for Voltage Constraints

Network Upgrades	Owner	Cost (\$)
No reactive power NUs		\$0

Table 9-5: J1221/J1230 GRE Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Benton Co-Mud Lake 230 kV	GRE MP	Re-conductor to 795 DRAKE ACSS	\$21,100,000

Table 9-6: J1426 GRE Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Bunker Lake-Blaine 230 kV	GRE	Re-conductor to 795 DRAKE ACSS and replace switches and wave trap	\$6,962,690
Bunker Lake-Elk River 230 kV	GRE	Re-conductor to 795 DRAKE ACSS and replace switches	\$8,158,000
Dickinson-Dickinson SS 115 kV	GRE	Re-conductor to 795 DRAKE ACSS and replace wave trap	\$152,690
Elk River-Monticello 230 kV	GRE XEL	Re-conductor to 795 DRAKE ACSS	\$6,623,200
Sheridan Tap-Wabasso 69 kV	GRE ITCM	Re-conductor to 336 Linnet ACSR	\$1,566,000

Table 9-7: OTP Local Planning Criteria Network Upgrades

Network Upgrades	Owner	Cost (\$)
1×15 Mvar switched reactor at J1456 POI 230 kV		\$2,500,000

Table 9-8: DPC Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
No DPC LPC NUs			\$0

Table 9-9: Shared Network Upgrades

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total Network Upgrade Cost (\$)	Cost Responsibility
No SNUs					\$0

9.3 Cost Allocation Methodology

9.3.1 Thermal Network Upgrade Cost Allocation

The costs of thermal Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case. For thermal constraints identified in the shoulder peak scenario, the MW impact is calculated using the shoulder peak post-DPP case. The MW impact on constraints identified in the summer peak scenario is calculated using the summer peak post-DPP case. With all Group Study generation projects

dispatched in the Study Case, all thermal constraints will be identified and a distribution factor from each project on each constraint will be obtained.

Thermal NU cost will be allocated based on the pro rata share of the MW impact on all constraints from each project, where MW impact = DF * Gen Output of the project in the model where the constraint occurs. If the Network Upgrade alleviates multiple constrained facilities the cost is allocated based on the sum of the highest MW contribution on all of the constrained elements for the DPP project under contingency. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max(Proj.A MW contribution on constraint)}}{\sum_i \text{Max(Proj.i MW contribution on constraint)}} \right)$$

9.3.2 Voltage Network Upgrade Cost Allocation

Voltage NU cost allocation will be determined by the pro rata share of the voltage impact each project has on the most constrained bus under the most constraining contingency. The voltage impact of each project will be calculated by locking all voltage regulating equipment in the model and then backing out each project one at a time to identify each project's impact to the constraint. In severe instances of voltage collapse where projects cannot be backed out one at a time, they will be added one at a time to determine their impact to the constraint.

9.3.3 Transient Stability Network Upgrade Cost Allocation

Transient stability driven Network Upgrades will be cost allocated based on the pro rata share of the total MW request of all the projects causing instability. The project(s) causing instability will be determined by backing out each project one at a time to identify each project's impact to the constraint.

9.4 Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

For each thermal constraint, the maximum MW contribution (increasing flow) from each DPP project is calculated. MW contribution from one DPP project is set as zero if the project doesn't violate DPP reliability criteria for a constrained element.

For voltage network upgrades, generators which degrade voltage at the most constrained bus under the most constraining contingency will be responsible for mitigating these constraints.

Transient stability Network Upgrades are allocated based on projects causing instability. If multiple projects are causing instability, cost allocation will be based on pro rata share of total MW of all projects causing instability.

Additional NRIS Network Upgrades are allocated to the impacting NRIS projects. ERIS Network Upgrades will be allocated to the impacting projects only based on the ERIS results.

The calculated DF results, voltage impact, and MW contribution on each constraint are in Appendix H.1 for the 2024 scenario.

Finally, the cost allocation for each NU is calculated based on the contribution of each generating facility, as detailed in Appendix H.2 for the 2024 scenario.

Assuming all generating facilities in the DPP 2019 West Area group advance, a summary of the costs for total NUs (NUs for ERIS, NRIS, and Interconnection Facilities) allocated to each generating facility is listed in Table 9-10.

Table 9-10: Summary of Total NU Costs Allocated to Each Generation Project

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J1190	200	\$1,500,000	\$7,500	0.22%
J1212	60	\$7,500,000	\$125,000	1.08%
J1218	200	\$12,783,881	\$63,919	1.84%
J1221	150	\$27,686,838	\$184,579	3.99%
J1229	50	\$3,195,970	\$63,919	0.46%
J1230	40	\$17,420,390	\$435,510	2.51%
J1270	200	\$18,413,636	\$92,068	2.65%
J1284	250	\$10,476,315	\$41,905	1.51%
J1285	50	\$2,212,418	\$44,248	0.32%
J1298	150	\$14,867,116	\$99,114	2.14%
J1300	120	\$47,258,319	\$393,819	6.80%
J1312	50	\$5,300,428	\$106,009	0.76%
J1313	100	\$2,000,000	\$20,000	0.29%
J1314	100	\$4,162,665	\$41,627	0.60%
J1315	600	\$25,345,869	\$42,243	3.65%
J1321	65	\$902,211	\$13,880	0.13%
J1325	200	\$105,854,706	\$529,274	15.24%
J1328	100	\$2,086,898	\$20,869	0.30%
J1337	300	\$34,055,871	\$113,520	4.90%
J1341	398	\$25,265,727	\$63,482	3.64%
J1349	335	\$14,450,022	\$43,134	2.08%
J1359	95	\$3,685,559	\$38,795	0.53%
J1365	169.4	\$16,597,084	\$97,976	2.39%
J1371	150	\$3,262,324	\$21,749	0.47%
J1395	200	\$14,123,416	\$70,617	2.03%
J1413	300	\$24,327,941	\$81,093	3.50%
J1416	149	\$15,177,176	\$101,860	2.19%
J1417	75	\$11,021,702	\$146,956	1.59%
J1426	100	\$4,728,039	\$47,280	0.68%
J1427	400	\$6,873,490	\$17,184	0.99%
J1428	200	\$20,102,193	\$100,511	2.89%
J1432	250	\$11,580,847	\$46,323	1.67%

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J1435	100	\$14,315,617	\$143,156	2.06%
J1438	100	\$10,460,559	\$104,606	1.51%
J1444	200	\$3,630,505	\$18,153	0.52%
J1445	100	\$5,615,477	\$56,155	0.81%
J1446	150	\$6,253,262	\$41,688	0.90%
J1456	300	\$10,814,937	\$36,050	1.56%
J1461	50	\$10,318,721	\$206,374	1.49%
J1467	150	\$27,782,186	\$185,215	4.00%
J1468	100	\$5,000,000	\$50,000	0.72%
J1471	200	\$16,078,046	\$80,390	2.31%
J1473	100	\$8,056,632	\$80,566	1.16%
J1474	200	\$10,683,312	\$53,417	1.54%
J1476	25	\$23,860,149	\$954,406	3.44%
J1477	50	\$459,073	\$9,181	0.07%
J1478	25	\$1,079,722	\$43,189	0.16%
J1479	100	\$13,046,300	\$130,463	1.88%
J1485	250	\$4,742,848	\$18,971	0.68%
J1487	120	\$8,100,309	\$67,503	1.17%
Total/Average	8126.4	\$694,516,708	\$111,909	100.00%

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A.1 DPP 2019 Generation Projects

Table A-1: DPP 2019 West Area Projects

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1190	IA	Clinton	ITCM	Calamus East 161 kV	200	200	Solar	NRIS
J1212	MN	Murray	XCEL	Chanarambie-Ellsborough 115 kV	60	60	Solar	NRIS
J1218	IA	Muscatine	MEC	Sub 92 345 kV	200	200	Solar	NRIS
J1221	MN	Wadena	MP	Riverton 230 kV	150	150	Solar	NRIS
J1229	IA	Muscatine	MEC	Sub 92 345 kV	50	50	Battery	NRIS
J1230	MN	Wadena	MP	Riverton 230 kV	40	40	Battery	NRIS
J1270	IA	Palo Alto	MEC	Kossuth - O'Brien 345 kV	200	200	Wind	NRIS
J1284	IA	Linn	ITCM	Duane Arnold - Tiffin 345 kV	250	250	Solar	NRIS
J1285	IA	Linn	ITCM	Duane Arnold - Tiffin 345 kV	50	50	Battery	NRIS
J1298	MN	Freeborn	SMMPA	Hayward - Murphy Creek 161 kV	150	150	Solar	NRIS
J1300	MN	Nobles	ITCM	Brewster - Elk 161 kV	120	120	Solar	NRIS
J1312	SD	McCook	XCEL	Grant 115 kV	50	50	Solar	NRIS
J1313	IA	Greene	ITCM	Karma 161 kV	100	100	Solar	NRIS
J1314	WI	Polk	DPC	Apple River 161 kV	100	100	Solar	NRIS
J1315	MN	Cottonwood, Murray, Redwood	GRE	Lyon County - Cedar Mountain 345 kV	600	600	Wind	NRIS
J1321	IA	Palo Alto	MEC	O'Brien - Kossuth 345 kV	65	65	Battery	NRIS
J1325	MN	Rock	ITCM	Magnolia 161 kV	200	200	Solar	NRIS
J1328	SD	Grant	OTP	Big Stone South 345 kV	100	100	Solar	NRIS
J1337	MN	Sherburne	XCEL	Sherburne County 345 kV	300	300	Solar	NRIS
J1341	IA	Buchanan	ITCM	Hazleton 345 kV	398	398	Wind	NRIS
J1349	MN	Cottonwood	XCEL	Crandall 345 kV	335 (60 PV, 275 Wind)	335	Hybrid	NRIS
J1359	IA	Linn	ITCM	Oakridge 69 kV	95	95	Gas	NRIS

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1365	IA	Tama	ITCM	Traer 161 kV	169.4	169.4	Wind	NRIS
J1371	MN	Yellow Medicine	OTP	Astoria 345 kV	150	150	Wind	NRIS
J1395	MN	Lyon	GRE	Nest Lake - Lyon County 345 kV	200	200	Solar	NRIS
J1413	IA	Howard, Mitchell	ITCM	Hazleton - Mitchell County 345 kV (5.4 mi from Mitchell Co, 65 mi from Hazleton)	300	300	Solar	NRIS
J1416	IA	Worth	ITCM	Worth County - Glenworth 161 kV	149	149	Solar	NRIS
J1417	IA	Worth	ITCM	Ledyard - Colby 345 kV	75	75	Battery	NRIS
J1426	MN	Benton	GRE	Benton County 115 kV	100	100	Solar	NRIS
J1427	IA	Benton	ITCM	Duane Arnold 345 kV	400	400	Wind	NRIS
J1428	ND	Cass	OTP	Buffalo 345 kV	200	200	Solar	NRIS
J1432	IA	Clinton	ITCM	Salem - Rock Creek 345 kV	250	250	Wind	NRIS
J1435	IA	Dubuque	ITCM	Center Grove 161 kV	100	100	Battery	NRIS
J1438	IA	Fayette	ITCM	Windsor - Hazelton 161 kV	100	100	Solar	NRIS
J1444	IA	Linn	ITCM	Duane Arnold 345 kV	200	200	Solar	NRIS
J1445	MN	Benton	XCEL	Mayhew Lake 115 kV	100	100	Solar	NRIS
J1446	MN	Wright	XCEL	Lake Pulaski 115 kV	150	150	Solar	NRIS
J1456	ND	Sheridan	OTP	Harvey - Underwood 230 kV	300	300	Wind	NRIS
J1461	MN	Carver	XCEL	McLeod - Blue Lake 230 kV	50	50	Solar	NRIS
J1467	IA	Fayette	ITCM	Windsor - Hazleton 161 kV	150	150	Wind	NRIS
J1468	MN	Dakota	XCEL	West Hastings 115 kV	100	100	Battery	NRIS
J1471	IA	Lee	ITCM	Wever - Burlington Generator 161 kV	200	200	Solar	NRIS
J1473	MN	Chisago	XCEL	Lawrence Creek - Shafer 115 kV	100	100	Solar	NRIS
J1474	WI	Chippewa	XCEL	Pine Lake - Eagle Point 115 kV	200	200	Solar	NRIS
J1476	IA	Union	ITCM	Creston East 69 kV (J1132)	25	25	Battery	NRIS
J1477	IA	Scott	MEC	Sub 56 161 kV (J1131)	50	50	Battery	NRIS
J1478	IA	Des Moines	ITCM	Agency 69 kV (J1135)	25	25	Battery	NRIS
J1479	IA	Des Moines	ITCM	Burlington 161 kV	100	100	Battery	NRIS
J1485	SD	Deuel	OTP	Astoria 345 kV (J493/J510)	250	250	Wind	NRIS
J1487	MN	Freeborn	ITCM	Freeborn 161 kV	120	120	Wind	NRIS

Table A-2: Dynamic Modeling, Collector System and Shunt Compensation Modeling for DPP West Area Projects

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1190	80 SMA 2.5 MW Inverter MVA = 2.75 MVA each	One 200 MW unit	PF: 0.909 Qmin = - 91.7 MVAR Qmax = +91.7 MVAR 220 MVA	R = 0.0022, X = 0.0024, B = NA	0.55 - 34.5 kV YG - D MVA = 2.75 MVA Z% = 5.75, X/R = 8 #Units = 80 DETC 2.5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 132 MVA Z% = 8.5, X/R = 32 DETC 2.5% w 5 taps	R = 0.0001, X = 0.0004, B = NA
J1212	19 HEM FS3350M Inverter MVA = 3.465 MVA each	One 60 MW unit	PF: 0.935 Qmin = - 23.349 MVAR Qmax = +23.349 MVAR 65.83 MVA	R = 0.0011, X = 0.0004, B = 0.0009 1×12 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 3.465 MVA Z% = 5.75, X/R = 10.5 #Units = 19 DETC 2.5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 42 MVA Z% = 9.5, X/R = 30 DETC 2.5% w 5 taps	R = 0.00002, X = 0.00006, B = 0.00036
J1218	54 TMEIC 3.78 MW Inverter MVA = 4.2 MVA each	Two 100 MW units	2× PF: 0.90 Qmin = - 49.42 MVAR Qmax = +49.42 MVAR 113.4 MVA	Collector 1: R = 0.00049, X = 0.00057, B = 0.00281 1×3 MVAR cap bank Collector 2: R = 0.00049, X = 0.00057, B = 0.00281 1×3 MVAR cap bank	2× 0.63 - 34.5 kV YG - D MVA = 113.4 MVA Z% = 5.75, X/R = 10.0 #Units = 54 DETC 2.5% w 5 taps	2× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	NA
J1221	40TMEIC 3.78 MW Inverter MVA = 4.2 MVA each	One 150 MW unit	PF: 0.90 Qmin = - 73.22 MVAR Qmax = +73.22 MVAR 168.0 MVA	R = 0.00065, X = 0.00080, B = 0.00416 1×4 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.20 MVA Z% = 5.75, X/R = 10.0 #Units = 40	34.5 - 230 - 13.8 kV YG - YG - D MVA = 102 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	R = 0.000112, X = 0.000468, B = 0.000409

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MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
					DETC 2.5% w 5 taps		
J1229	14 TMEIC 3.78 MW Inverter MVA = 4.2 MVA each	One 50 MW unit	PF: 0.90 Qmin = - 25.63 MVAR Qmax = +25.63 MVAR 58.8 MVA	R = 0.00186, X = 0.00218, B = 0.00150 1×2 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.20 MVA Z% = 5.75, X/R = 10.0 #Units = 14 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 36 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	NA
J1230	12 TMEIC 3.78 MW Inverter MVA = 4.2 MVA each	One 40 MW unit	PF: 0.90 Qmin = - 21.97 MVAR Qmax = +21.97 MVAR 50.4 MVA	R = 0.00228, X = 0.00242, B = 0.00125 1×2 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.20 MVA Z% = 5.75, X/R = 10.0 #Units = 12 DETC 2.5% w 5 taps	34.5 - 230 - 13.8 kV YG - YG - D MVA = 27 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	NA
J1270	49 Vestas V136 MVA = 5.1 MVA each	One 200 MW unit	PF: 0.95 Qmin = - 65.73 MVAR Qmax = +65.73 MVAR 211 MVA	R = 0.00239, X = 0.00382, B = 0.04383 2×13 MVAR cap bank	0.72 - 34.5 kV YG - D MVA = 5.10 MVA Z% = 9.90, X/R = 12.375 #Units = 49 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 144 MVA Z% = 9.5, X/R = 52.80 DETC 10% w 33 taps	R = 0.00020, X = 0.001208, B = 0.010995
J1284	68 TMEIC 3.78 MW Inverter MVA = 4.2 MVA each	Two 125 MW units	2× PF: 0.90 Qmin = - 62.245 MVAR Qmax = +62.245 MVAR 142.8 MVA	Collector 1: R = 0.00038, X = 0.00046, B = 0.00357 1×4 MVAR cap bank Collector 2: R = 0.00038, X = 0.00046, B = 0.00357 1×4 MVAR cap bank	2× 0.63 - 34.5 kV YG - D MVA = 142.8 MVA Z% = 5.75, X/R = 10 #Units = 68 DETC 2.5% w 5 taps	2× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 84 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	NA

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1285	14 TMEIC 3.78 MW Inverter MVA = 4.2 MVA each	One 50 MW unit	PF: 0.90 Qmin = - 25.63 MVAR Qmax = +25.63 MVAR 58.8 MVA	R = 0.00186, X = 0.00218, B = 0.00150 1×2 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.20 MVA Z% = 5.75, X/R = 10.0 #Units = 14 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 36 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	NA
J1298	51 PE HEM 3 MW Inverter MVA = 3.3 MVA each	One 150 MW unit	PF: 0.909 Qmin = - 70.15 MVAR Qmax = +70.15 MVAR 168.3 MVA	R = 0.003403, X = 0.003126, B = 0.005843	0.60 - 34.5 kV YG - D MVA = 3.3 MVA Z% = 8, X/R = 10.5 #Units = 51 DETC 2.5% w 5 taps	34.5 - 161 YG - YG - D MVA = 100 MVA Z% = 9.0, X/R = 50 DETC 2.5% w 5 taps	R = 0.00010, X = 0.00020, B = 0.00010
J1300	41 PE HEM 3 MW Inverter MVA = 3.3 MVA each	One 120 MW unit	PF: 0.909 Qmin = - 56.398 MVAR Qmax = +56.398 MVAR 135.3 MVA	R = 0.004114, X = 0.003779, B = 0.004585	0.60 - 34.5 kV YG - D MVA = 3.3 MVA Z% = 8, X/R = 10.5 #Units = 41 DETC 2.5% w 5 taps	34.5 - 161 YG - YG MVA = 80 MVA Z% = 9.0, X/R = 50 DETC 2.5% w 5 taps	R = 0.00010, X = 0.00020, B = 0.00010
J1312	16 PE HEMK 3.432 MW Inverter MVA = 3.432 MVA each	One 50 MW unit	PF: 0.95 Qmin = - 18.048 MVAR Qmax = +18.048 MVAR 54.91 MVA	R = 0.006143, X = 0.008598, B = 0.009158 2×7 MVAR cap bank	0.66 - 34.5 kV YG - D MVA = 4.0 MVA Z% = 9, X/R = 25 #Units = 16 DETC 2.5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 47 MVA Z% = 7.5, X/R = 25 DETC 10% w 33 taps	R = 0.000210, X = 0.000603, B = 0.00029
J1313	30 PE HEMK 3.59 MW Inverter MVA = 3.59 MVA each	One 100 MW unit	PF: 0.95 Qmin = - 33.629 MVAR Qmax = +33.629 MVAR 107.7 MVA	R = 0.003435, X = 0.004876, B = 0.018044 2×14 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 4.0 MVA Z% = 5.75, X/R = 8 #Units = 30 DETC 2.5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 100 MVA Z% = 7.0, X/R = 25 DETC 10% w 33 taps	R = 0.0001070, X = 0.0003080, B = 0.0000700

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MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1314	30 PE HEMK 3.59 MW Inverter MVA = 3.59 MVA each	One 100 MW unit	PF: 0.95 Qmin = - 33.629 MVAR Qmax = +33.629 MVAR 107.7 MVA	R = 0.003435, X = 0.004876, B = 0.018044 1×28 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 4.0 MVA Z% = 9, X/R = 25 #Units = 30 DETC 2.5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 100 MVA Z% = 7.0, X/R = 25 DETC 10% w 33 taps	R = 0.0001070, X = 0.0003080, B = 0.0000700
J1315	174 Vesta V126 3.45 MW MVA = 3.45 MVA each	Six 100 MW units	6× PF: +0.90/ -0.92 Qmin = - 42.599 MVAR Qmax = +48.432 MVAR 111.1 MVA	6× R = 0.0000613, X = 0.000883, B = 0.11121 1×25 MVAR cap bank	6× 0.69 - 34.5 kV YG - D MVA = 116 MVA Z% = 5.7, X/R = 8 #Units = 174 DETC 2.5% w 5 taps	6× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 75 MVA Z% = 9.0, X/R = 60 DETC 10% w 33 taps	R = 0.001461, X = 0.00419, B = 0.163137
J1321	18 TMEIC 3.78 MW Inverter MVA = 4.2 MVA each	One 65 MW unit	PF: 0.90 Qmin = - 32.95 MVAR Qmax = +32.95 MVAR 75.6 MVA	R = 0.00152, X = 0.00170, B = 0.00193 1×2 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.2 MVA Z% = 5.75, X/R = 10 #Units = 18 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 45 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	NA
J1325	60 PE HEMK 5.39 MW Inverter MVA = 3.59 MVA each	Two 100 MW units	2× PF: 0.95 Qmin = - 33.63 MVAR Qmax = +33.63 MVAR 107.7 MVA	2× R = 0.003435, X = 0.004876, B = 0.022654 1×26 MVAR cap bank	2× 0.69 - 34.5 kV YG - D MVA = 120 MVA Z% = 5.75, X/R = 8.0 #Units = 60 DETC 2.5% w 5 taps	2× 34.5 - 161 - 13.8 kV YG - YG - D MVA = 100 MVA Z% = 9.0, X/R = 25 DETC 10% w 33 taps	R = 0.000107, X = 0.000308, B = 0.00012

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1328	40 TMEIC 2.5 MW Inverter MVA = 2.7 MVA each	One 100 MW unit	PF: 0.925 Qmin = - 41.078 MVAR Qmax = +41.078 MVAR 108.1 MVA	R = 0.0041, X = 0.0038, B = 0.0142	0.60 - 34.5 kV YG - D MVA = 2.70 MVA Z% = 5.75, X/R = 8.62 #Units =40 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 100 MVA Z% = 9.34, X/R = 13 DETC 2.5% w 5 taps	R = 0.00296, X = 0.000852, B = 0.03311
J1337	90 HEMK PE 3.59 MW Inverter MVA = 3.59 MVA each	Three 100 MW units	3× PF: 0.95 Qmin = - 33.63 MVAR Qmax = +33.63 MVAR 107.7 MVA	3× R = 0.00247, X = 0.00355, B = 0.026794 1×26 MVAR cap bank	3× 0.69 - 34.5 kV YG - D MVA = 120 MVA Z% = 5.75, X/R = 8 #Units = 90 DETC 2.5% w 5 taps	3× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 100 MVA Z% = 9.0, X/R = 60 DETC 10% w 33 taps	R = 0.000175, X = 0.000503, B = 0.019576
J1341	160 2.52 MW GE WTG MVA = 2.8 MVA each	Two 199 MW units	2× PF: 0.90 Qmin = - 97.639 MVAR Qmax = +97.639 MVAR 224 MVA	Collector 1: R = 0.002338, X = 0.002597, B = 0.048 Collector 2: R = 0.002315, X = 0.002571, B = 0.048	2× 0.69 - 34.5 kV YG - D MVA = 224 MVA Z% = 6, X/R = 8.8 #Units = 160 DETC 2.5% w 5 taps	2× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 132 MVA Z% = 8.0, X/R = 44.3 LTC 10% w 33 taps	R = 0.00100, X = 0.0097, B = 0.0018

Model Development

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1349	24 2.5 MW TMEIC Inverters MVA = 2.7 MVA 98 2.82 MW GE WTG MVA = 3.13 MVA each	One 60 MW PV unit One 135 MW wind unit One 140 MW wind unit	TMEIC: 1× PF: 0.925 Qmin = - 24.647 MVAR Qmax = +24.647 MVAR 64.86 MVA GE WTG #1 PF: 0.90 Qmin = - 65.56 MVAR Qmax = +65.56 MVAR 150.4 MVA GE WTG #2 PF: 0.90 Qmin = - 68.291 MVAR Qmax = +68.291 MVAR 156.66 MVA	Solar: R = 0.00395, X = 0.00464, B = 0.03815 1x10 MVAR Cap Wind: Col1: R = 0.0120, X = 0.0105, B = 0.0409 1x15 MVAR Cap Col2: R = 0.0141, X = 0.0109, B = 0.0443 1x15 MVAR Cap	TMEIC: 1× 0.60 - 34.5 YG - D MVA = 2.7 MVA Z% = 5.75, X/R = 8.62 #Units = 24 DETC 2.5% w 5 taps Wind #1 0.69 - 34.5 YG - D MVA = 156 MVA Z% = 6, X/R = 7.50 Wind #2 0.69 - 34.5 YG - D MVA = 162.5 MVA Z% = 6, X/R = 7.50 #Units = 98 DETC 2.5% w 5 taps	TMEIC: 1× 34.5 - 345-13.8 kV YG - YG-D MVA = 51 MVA Z% = 7.50, X/R = 13 DETC 2.5% w 5 taps Wind: 2× 34.5 - 345-13.8 kV YG - YG-D MVA = 100 MVA Z% = 8.85, X/R = 39.3 DETC 2.5% w 5 taps	R = 0.001053, X = 0.009609, B = 0.162
J1359	Siemens ABB 2 60 MW Gas Units MVA = 70.59 MVA each	Two 60 MW unit	2× PF: +0.85/ -0.90 Qmin = - 29.06 MVAR Qmax = +37.18 MVAR 70.59 MVA	NA	NA	2× 12.5 - 69 kV YG - D MVA = 50 MVA Z% = 12.0, X/R = 23.70 DETC 2.5% w 5 taps	NA

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1365	77 Vestas 2.2 MW V120 WTG MVA = 2.326 MVA each	One 169.4 MW unit	PF: 0.95 Qmin = - 55.679 MVAR Qmax = +55.679 MVAR 179.1 MVA	R=0.00595, X=0.01015, B=0.03735	0.69 - 34.5 kV YG - D MVA = 2.326 MVA Z% = 9.62, X/R = 10.9 #Units = 77 DETC 2.5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 107.50 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	R=0.000033, X=0.000138, B=0.00005
J1371	37 Vestas V150 4.2 MW MVA = 5.1 MVA each	One 150 MW unit	Qmin = - 61.4 MVAR Qmax = +77.9 MVAR 188.7 MVA	R=0.00355, X=0.00501, B=0.01390	0.72 - 34.5 kV YG - D MVA = 4.70 MVA Z% = 9.0, X/R = 11.25 #Units = 37 DETC 2.5% w 5 taps	34.5 - 345 - 13.2 kV YG - YG - D MVA = 101 MVA Z% = 10.0, X/R = 43 LTC 10% w 33 taps	R=0.00088, X=0.00694, B=0.0685
J1395	70 HEMK PE 3.3 MW Inverter MVA = 3.3 MVA each	Two 100 MW unit	2× PF: 0.90 Qmin = - 50.345 MVAR Qmax = +50.345 MVAR 115.5 MVA	Collector 1: R=0.00303, X=0.00239, B=0.01192 Collector 2: R=0.00463, X=0.00366, B=0.01496	2× 0.66 - 34.5 kV YG - D MVA = 126 MVA Z% = 5.75, X/R = 9.70 #Units = 70 DETC 2.5% w 5 taps	2× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 75 MVA Z% = 12.0, X/R = 45 DETC 10% w 33 taps	R=0.00001, X=0.00006, B=0.00067
J1413	80 TMEIC Ninja 3.78 MW Inverter MVA = 4.2 MVA each	Two 150 MW unit	2× PF: 0.90 Qmin = - 73.23 MVAR Qmax = +73.23 MVAR 168 MVA	2× R=0.00509, X=0.00607, B=0.15801 1×4 MVAR cap bank	2× 0.63 - 34.5 kV YG - D MVA = 4.20 MVA Z% = 5.75, X/R = 10 #Units = 80 DETC 2.5% w 5 taps	2× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 105 MVA Z% = 9, X/R = 45 DETC 2.5% w 5 taps	NA

Model Development

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1416	40 TMEIC Ninja 3.78 MW Inverter MVA = 4.2 MVA each	One 149 MW unit	PF: 0.90 Qmin = - 73.23 MVAR Qmax = +73.23 MVAR 168 MVA	R=0.00701, X=0.00810, B=0.08127 1×4 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.20 MVA Z% = 5.75, X/R = 10 #Units = 40 DETC 2.5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 105 MVA Z% = 9, X/R = 45 DETC 2.5% w 5 taps	NA
J1417	20 TMEIC Ninja 3.78 MW Inverter MVA = 4.2 MVA each	One 75 MW unit	PF: 0.90 Qmin = - 36.61 MVAR Qmax = +36.61 MVAR 84 MVA	R=0.00454, X=0.00548, B=0.20033	0.63 - 34.5 kV YG - D MVA = 4.20 MVA Z% = 5.75, X/R = 10 #Units = 20 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 54 MVA Z% = 9, X/R = 35 DETC 2.5% w 5 taps	R=0.00005, X=0.0007, B=0.00624
J1426	27 TMEIC 3.71 MW Inverter MVA = 4.05 MVA each	One 100 MW unit	PF: 0.916 Qmin = - 43.796 MVAR Qmax = +43.796 MVAR 109.35 MVA	R=0.00074, X=0.00136, B=0.003941 2×7 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.05 MVA Z% = 5.75, X/R = 7 #Units = 27 DETC 2.5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 8, X/R = 35 DETC 2.5% w 5 taps	NA
J1427	134 GE 3.03 MW WTG MVA = 3.36 MVA each	Two 200 MW units	2× PF: 0.90 Qmin = - 98.322 MVAR Qmax = +98.322 MVAR 225.12 MVA	Collector 1: R=0.005, X=0.020, B=0.085 2×12 MVAR cap bank Collector 2: R=0.005, X=0.020, B=0.085	2× 0.69 - 34.5 kV YG - D MVA = 288.1 MVA Z% = 5.75, X/R = 7.5 #Units = 134 DETC 2.5% w 5 taps	2× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 135 MVA Z% = 8.5, X/R = 45 DETC 2.5% w 5 taps	R=0.00121, X=0.0102, B=0.041129

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1428	54 TMEIC 3.71 MW Inverter MVA = 4.05 MVA each	One 200 MW unit	PF: 0.916 Qmin = - 87.741 MVAR Qmax = +87.741 MVAR 218.7 MVA	R=0.00037, X=0.00068, B=0.00394 2×14 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.05 MVA Z% = 5.75, X/R = 7 #Units = 54 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 135 MVA Z% = 8.5, X/R = 45 DETC 2.5% w 5 taps	NA
J1432	83 GE 3.03 MW WTG MVA = 3.36 MVA each	One 250 MW unit	PF: 0.90 Qmin = - 121.8 MVAR Qmax = +121.8 MVAR 278.88 MVA	R=0.00268, X=0.00395, B=0.0466 2×15 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 3.30 MVA Z% = 5.75, X/R = 7.5 #Units = 83 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 165 MVA Z% = 8.5, X/R = 45 DETC 2.5% w 5 taps	R=0.000002, X=0.00002, B=0.00017
J1435	84 PE 1.29 MVA inverters Total MVA = 108.4 MVA	One 100 MW unit	PF: 0.923 Qmin = - 41.84 MVAR Qmax = +41.84 MVAR 108.4 MVA	R = 0.00138, X = 0.00253, B = 0.001 1×14 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 105 (2.5*42) MVA Z% = 6.84, X/R = 7.5 #Units =84 DETC 5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 8, X/R = 40 DETC 5% w 5 taps	NA
J1438	27 TMEIC 4.05 MVA inverters Total MVA = 108 MVA	One 100 MW unit	PF: 0.926 Qmin = - 40.792 MVAR Qmax = +40.792 MVAR 108 MVA	R = 0.0013, X = 0.0012, B = 0.0025 2×7 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 109.35 (4.05*27) Z% = 5.75, X/R = 7 #Units =27 DETC 5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 8, X/R = 40 DETC 5% w 5 taps	R = 0.000025, X = 0.000146, B = 0.000036

Model Development

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1444	54 TMEIC 4.05 MVA inverters Total MVA = 216 MVA	One 200 MW unit	PF: 0.926 Qmin = - 81.584 MVAR Qmax = +81.584 MVAR 216 MVA	R = 0.00037, X = 0.00068, B = 0.00394 2×14 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 218.7 (4.05*54) Z% = 5.75, X/R = 7 #Units =54 DETC 5% w 5 taps	34.5 -345 - 13.8 kV YG - YG - D MVA = 135 MVA Z% = 8.5, X/R = 45 DETC 5% w 5 taps	R = 0.00003, X = 0.00023, B = 0.00232
J1445	27 TMEIC 4.05 MVA inverters Total MVA = 109 MVA	One 100 MW unit	PF: 0.917 Qmin = - 43.37 MVAR Qmax = +43.37 MVAR 109 MVA	R = 0.00074, X = 0.00136, B = 0.00197 2×7 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 109.35 (4.05*27) Z% = 5.75, X/R = 7 #Units =27 DETC 5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 8, X/R = 35 DETC 5% w 5 taps	R = 0.000048, X = 0.000286, B = 0.000019
J1446	126 PE 1.29 MVA inverters Total MVA = 162 MVA	One 150 MW unit	PF: 0.926 Qmin = - 61.188 MVAR Qmax = +61.188 MVAR 162 MVA	R = 0.00138, X = 0.00253, B = 0.001 2×11 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 157.5 (2.5*63) Z% = 6.84, X/R = 7.5 #Units =126 DETC 5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 105 MVA Z% = 8, X/R = 35 DETC 5% w 5 taps	NA

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1456	150 Vestas 2.0 MVA WTG Total MVA = 105.26*3 MVA	Three 100 MW units	3× PF: 0.95 Qmin = - 32.8684 MVAR Qmax = +32.8684 MVAR 105.26 MVA	Collector 1: R = 0.0052, X = 0.00601, B = 0.16775 1×3 MVAR cap bank Collector 2: R = 0.00395, X = 0.00454, B = 0.13017 1×3 MVAR cap bank Collector 3: R = 0.0023, X = 0.00261, B = 0.07866 1×3 MVAR cap bank	3× 0.69 - 34.5 kV YG - D MVA = 103 (2.06*50) Z% = 9.8, X/R = 10.5 #Units =150 DETC 5% w 5 taps	3× 34.5 - 230 - 13.2 kV YG - YG - D MVA = 75 MVA Z% = 9.0, X/R = 35 DETC 10% w 33 taps	R = 0.00001, X = 0.00008, B = 0.00018
J1461	14 TMEIC 4.05 MVA inverters Total MVA = 56 MVA	One 50 MW unit	PF: 0.893 Qmin = - 25.219 MVAR Qmax = +25.219 MVAR 56 MVA	R = 0.00138, X = 0.00253, B = 0.001 1×7 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 56.7 (4.05*14) Z% = 5.75, X/R = 7 #Units =14 DETC 5% w 5 taps	34.5 - 230 - 13.8 kV YG - YG - D MVA = 36 MVA Z% = 8, X/R = 35 DETC 5% w 5 taps	R = 0.000121, X = 0.000698, B = 0.000760
J1467	50*GE DFIG 3.36 MVA WTG Total MVA = 168 MVA	One 150 MW unit	PF: 0.9 Qmin = - 72.648 MVAR Qmax = +72.648 MVAR 168 MVA	R = 0.00138, X = 0.00253, B = 0.00125 2×9 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 165 (3.3*50) Z% = 5.75, X/R = 7 #Units =50 DETC 5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 105 MVA Z% = 8.5, X/R = 40 DETC 5% w 5 taps	R = 0.000049, X = 0.000307, B = 0.000069

Model Development

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1468	84 PE 1.29 MVA inverters Total MVA = 108.4 MVA	One 100 MW unit	PF: 0.9 Qmin = - 48.432 MVAR Qmax = +48.432 MVAR 108.4 MVA	R = 0.00138, X = 0.00253, B = 0.001 1×14 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 105 (2.5*42) Z% = 6.84, X/R = 7.5 #Units =84 DETC 5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 8, X/R = 35 DETC 5% w 5 taps	NA
J1471	54 TMEIC 4.05 MVA inverters Total MVA = 216 MVA	One 200 MW unit	PF: 0.926 Qmin = - 81.584 MVAR Qmax = +81.584 MVAR 216 MVA	R = 0.00037, X = 0.00068, B = 0.00394 2×14 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 218.7 (4.05*54) Z% = 5.75, X/R = 7 #Units =54 DETC 5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 138 MVA Z% = 8.5, X/R = 40 DETC 5% w 5 taps	R = 0.00004, X = 0.00021, B = 0.0001
J1473	27 TMEIC 4.05 MVA inverters Total MVA = 109 MVA	One 100 MW unit	PF: 0.917 Qmin = - 43.37 MVAR Qmax = +43.37 MVAR 109 MVA	R = 0.00074, X = 0.00136, B = 0.003941 2×7 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 109.35 (4.05*27) Z% = 5.75, X/R = 7 #Units =54 DETC 5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 8, X/R = 35 DETC 5% w 5 taps	R = 0.000048, X = 0.000286, B = 0.00002
J1474	54 TMEIC 4.05 MVA inverters Total MVA = 216 MVA	One 200 MW unit	PF: 0.926 Qmin = - 81.584 MVAR Qmax = +81.584 MVAR 216 MVA	R = 0.00138, X = 0.00253, B = 0.001 2×14 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 218.7 (4.05*54) Z% = 5.75, X/R = 7 #Units =54 DETC 5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 138 MVA Z% = 8.5, X/R = 40 DETC 5% w 5 taps	NA
J1476	9 PE FS3000CU15 Inverter Total MVA = 29.17 MVA	One 25 MW unit	PF: 0.86 Qmin = - 15.03 MVAR Qmax = +15.03 MVAR 29.17 MVA	R = 0.00776, X = 0.00951, B = 0.02263	0.69 - 34.5 kV YG - D MVA = 31.5 (3.5*9) Z% = 5.75, X/R = 11 #Units = 9	34.5 - 69 kV YG - D MVA = 51 MVA Z% = 8.5, X/R = 31 DETC 5% w 5 taps	NA

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
					DETC 5% w 5 taps		
J1477	17 PE FS3000CU15 Inverter Total MVA = 58.33 MVA	One 50 MW unit	PF: 0.86 Qmin = - 30.06 MVAR Qmax = +30.06 MVAR 58.33 MVA	R = 0.00776, X = 0.00951, B = 0.02263	0.69 - 34.5 kV YG - D MVA = 59.5 (3.5*17) Z% = 5.75, X/R = 11 #Units = 17 DETC 5% w 5 taps	34.5 - 161 kV YG - D MVA = 99 MVA Z% = 9.5, X/R = 32.5 DETC 5% w 5 taps	NA
J1478	9 PE FS3000CU15 Inverter Total MVA = 29.17 MVA	One 25 MW unit	PF: 0.86 Qmin = - 15.03 MVAR Qmax = +15.03 MVAR 29.17 MVA	R = 0.00776, X = 0.00951, B = 0.02263	0.69 - 34.5 kV YG - D MVA = 31.5 (3.5*9) Z% = 5.75, X/R = 11 #Units = 9 DETC 5% w 5 taps	34.5 - 69 kV YG - D MVA = 51 MVA Z% = 8.5, X/R = 31 DETC 5% w 5 taps	NA
J1479	34 PE FS3000CU15 Inverter Total MVA = 116.67 MVA	One 100 MW unit	PF: 0.86 Qmin = - 60.099 MVAR Qmax = +60.099 MVAR 116.67 MVA	R = 0.00776, X = 0.00951, B = 0.02263	0.69 - 34.5 kV YG - D MVA = 119 (3.5*34) Z% = 5.75, X/R = 11 #Units = 34 DETC 5% w 5 taps	34.5 - 161 kV YG - D MVA = 66 MVA Z% = 9.5, X/R = 32.5 DETC 5% w 5 taps	R = 0.00217, X = 0.0091, B = 0.011315
J1485	100 2.52 MW GE turbines Total MVA = 138.89*2 MVA (2.78 MVA each)	Two 125 MW unit	2× PF: 0.90 Qmin = - 60.5403 MVAR Qmax = +60.5403 MVAR 138.89 MVA	2× R = 0.000008, X = 0.000094, B = 0.000907 1×3 MVAR cap bank	2× 0.69 - 34.5 kV YG - D MVA = 140 (2.8*50) Z% = 5.75, X/R = 10 #Units = 50*2 DETC 5% w 5 taps	2× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 84 MVA Z% = 9.0, X/R = 45 DETC 5% w 5 taps	R = 0.00508, X = 0.00607, B = 0.000907

Model Development

MISO Project #	Turbine / Inverter	Generator Model	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1487	49 2.52 MW GE turbines Total MVA = 137.2 MVA (2.8 MVA each)	One 120 MW unit	PF: 0.90 Qmin = - 59.804 MVAR Qmax = +59.804 MVAR 137.2 MVA	R = 0.006707, X = 0.006266, B = 0.061328 1×11 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 147 (3*49) Z% = 5.75, X/R = 7 #Units = 49 DETC 5% w 5 taps	34.5 - 161 kV YG - YG MVA = 80 MVA Z% = 10.0, X/R = 36 DETC 5% w 5 taps	R = 0.002238, X = 0.019604, B = 0.020399

Table A-3: DPP 2019 Central Area Projects

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1191	MO	Boone	City of Columbia, MO	Bolstad 69 kV	64	64	Solar	NRIS
J1197	IL	LaSalle	AMIL	Oglesby 138 kV	75	75	Battery Storage	NRIS
J1198	IL	Jasper	AMIL	Newton 345 kV	51.8	51.8	Hybrid	NRIS
J1199	IL	Fulton	AMIL	Duck Creek 345 kV	20	20	Hybrid	NRIS
J1200	IL	Putnam	AMIL	Hennepin 138 kV	24.2	24.2	Hybrid	NRIS
J1201	IL	Montgomery	AMIL	Coffeen 345 kV	44.2	44.2	Hybrid	NRIS
J1202	IL	Randolph	AMIL	Baldwin Plant 345 kV Switchyard - Baldwin Unit 2	68.4	68.4	Hybrid	NRIS
J1204	IL	Cumberland	AMIL	Casey West 345 kV	199.3	199.3	Solar	NRIS
J1208	IL	Saline	AMIL	Muddy - Crab Orchard 138 kV	80	80	Solar	NRIS
J1209	IL	Saline	AMIL	Muddy - Crab Orchard 138 kV	80	80	Solar	NRIS
J1213	MO	Iron	AMMO	Taum Sauk 138kV Switchyard, Bus 1 & 2	60	60	Hydro	NRIS
J1216	IL	Washington	AMIL	Ashley 138 kV	250	250	Solar	NRIS
J1222	MO	Ralls	AMMO	Spencer Creek 345 kV	200	200	Solar	NRIS
J1225	IL	Washington	AMIL	Ashley-Turkey Hill Line 1482, between the Okawville and North Nashville substations	160	160	Solar	NRIS
J1227	IN	Spencer	SIGE	Dubois - FB Culley 138 kV	125	125	Solar	NRIS
J1231	KY	Lyon	BREC	Barkley - Caldwell 161 kV	125	125	Solar	NRIS
J1232	IL	Coles	AMIL	Chesterville - Mattoon West 138 kV	99	99	Solar	NRIS
J1233	IN	Pike	SIGE	Cato 138 kV	32.5	32.5	Battery Storage	NRIS
J1234	IN	Knox	DEI	Edwardsport - Washington Mun, Linton 138 kV	100	100	Solar	NRIS
J1235	IN	Knox	DEI	Edwardsport - Washington Mun, Linton 138 kV	50	50	Solar	NRIS
J1241	IL	Jefferson	AMIL	West Mt Vernon - Xenia 345 kV	200	200	Solar	NRIS
J1242	IL	Hancock	AMIL	Hamilton Line 796 Tap to Audrian 69 kV Line	25	25	Solar	External NRIS

Model Development

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1243	IL	McDonough	AMIL	Macomb West #2 - Blandinsville Tap 69 kV	25	25	Solar	External NRIS
J1254	IN	Knox	DEI	Merom - Gibson - Wheatland 345 kV	150	150	Solar	NRIS
J1263	IL	Clark	AMIL	Casey - Kansas West 345 kV	408.6	408.6	Hybrid	NRIS
J1265	IN	LaGrange	NIPSCO	Wolcottville - South Milford 69 kV	43	43	Solar	NRIS
J1266	IL	Marion	AMIL	Kinmundy - Salem 138 kV	99	99	Solar	NRIS
J1268	MO	Lincoln	AMMO	Pike - Troy 161 kV	150	150	Solar	NRIS
J1269	IN	Warrick	SIGE	Elliott 69kV	100	100	Hybrid	NRIS
J1271	IN	Vanderburgh	SIGE	Oak Grove 138 kV	100	100	Battery Storage	NRIS
J1272	IN	Vanderburgh	SIGE	Angel Mound 138 kV	50	50	Battery Storage	NRIS
J1274	IN	Posey	SIGE	Point 69 kV	100	100	Hybrid	NRIS
J1275	KY	Henderson	BREC	Henderson 161 kV	100	100	Solar	NRIS
J1276	IN	Posey	SIGE	Point 69 kV	100	100	Hybrid	NRIS
J1289	IL	Morgan	AMIL	Turner 345 kV	200	200	Wind	NRIS
J1295	IN	Gibson	DEI	Gibson - Francisco 345 kV	280	280	Solar	NRIS
J1296	IN	Gibson	SIGE	Francisco 138 kV	200	200	Solar	NRIS
J1299	MO	Scott	AMMO	Miner - Kelso 161 kV	200	200	Solar	NRIS
J1301	IL	Washington	AMIL	Prest 138 kV	150	150	Solar	NRIS
J1302	IL	Saline	AMIL	Muddy - Norris City 138 kV	150	150	Solar	NRIS
J1303	IL	Jackson	SIPC	Campbell Hill - Jackson 161 kV	150	150	Solar	NRIS
J1306	IL	Perry	AMIL	Pickneyville - St John 230 kV	200	200	Solar	NRIS
J1308	IN	Posey	SIGE	AB Brown - Gibson 345 kV	300	300	Solar	NRIS
J1311	IL	St. Clair	AMIL	Fayetteville Bee Hollow Road 138 kV	150	150	Solar	NRIS
J1318	IN	Rush	DEI	Rushville 69 kV	25	25	Solar	NRIS
J1323	IN	LaGrange	NIPSCO	Lagrange 138 kV	25	25	Battery Storage	NRIS
J1330	IL	Randolph	AMIL	Baldwin 345 kV	200	200	Solar	NRIS
J1332	IN	Jasper	NIPSCO	Schahfer 345 kV	200	200	Solar	NRIS
J1333	IN	Jasper, Starke	NIPSCO	Schahfer - Burr Oak 345 kV	199.5	199.5	Hybrid	NRIS
J1334	IN	Jasper,	NIPSCO	Schahfer - Burr Oak 345 kV	199.5	199.5	Hybrid	NRIS

Model Development

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
		Starke						
J1335	IN	Jasper, Starke	NIPSCO	Schahfer - Burr Oak 345 kV	199.5	199.5	Hybrid	NRIS
J1336	IN	Jasper, Starke	NIPSCO	Schahfer 345 kV	199.5	199.5	Hybrid	NRIS
J1338	MO	Audrain	AMMO	Spencer Creek - Montgomery 345 kV	150	150	Wind	NRIS
J1339	IN	Jasper, Starke	NIPSCO	Schahfer 345 kV	199.5	199.5	Hybrid	NRIS
J1340	IN	Jasper, Starke	NIPSCO	Schahfer 345 kV	199.5	199.5	Hybrid	NRIS
J1342	IN	Starke County	NIPSCO	Starke 138 kV	99	99	Solar	NRIS
J1348	IN	Vermillion	DEI	Cayuga - Sugar Creek 345 kV	200	200	Solar	NRIS
J1352	MO	Montgomery	AMMO	Montgomery 345 kV	100	100	Solar	NRIS
J1353	IN	Lake	NIPSCO	Green Acres 138 kV	150	150	Battery Storage	NRIS
J1354	IN	Decatur	HE	Decatur County 138 kV	146	146	Battery Storage	NRIS
J1355	IN	LaPorte	NIPSCO	Stillwell 138 kV	160	160	Battery Storage	NRIS
J1356	IN	Martin	DEI	Whitfield 138 kV	160	160	Solar	NRIS
J1357	IN	Madison	DEI	Madison 138 kV	150	150	Battery Storage	NRIS
J1358	IN	LaPorte	NIPSCO	Luchtman 138 kV	125	125	Battery Storage	NRIS
J1360	IL	Piatt	AMIL	Goose Creek 345 kV	300	300	Wind	NRIS
J1363	MO	Audrain	AMMO	Spencer Creek - Montgomery 345 kV	150	150	Solar	NRIS
J1369	IN	Dubois	SIGE	Dubois 138 kV	100	100	Solar	NRIS
J1376	MO	Cooper	AMMO	Barnett - Georgetown Overton 161 kV	100	100	Wind	NRIS
J1378	IN	Fountain	DEI	Veedersburg West 230 kV	200	200	Solar	NRIS
J1381	IN	Howard	DEI	Greentown 230 kV	200	200	Solar	NRIS
J1382	IL	Clark	AMIL	Hutsonville - Neoga 138 kV	150	150	Solar	NRIS
J1383	IL	Fulton	AMIL	Ipava 138 kV	150	150	Solar	NRIS
J1385	IN	Hendricks	DEI	Stilesville 138 kV	100	100	Solar	NRIS

Model Development

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1386	IN	Lake	NIPSCO	Green Acres 138 kV	100	100	Solar	NRIS
J1387	IN	Jasper	NIPSCO	Circuit 6958 69 kV Line	60	60	Solar	ERIS
J1388	IN	Gibson	DEI	Oakland City - Old Ben Coal 138 kV	150	150	Solar	NRIS
J1390	IN	Bartholomew	DEI	Gallagher Station - Columbus 230 kV	150	150	Solar	NRIS
J1391	IN	Dubois	SIGE	Duff 138 kV	100	100	Solar	NRIS
J1392	IN	LaGrange	NIPSCO	Lagrange 138 kV	100	100	Solar	NRIS
J1393	IN	Pike	SIGE	Cato 138 kV	130	130	Solar	NRIS
J1396	IN	Pike	HE	Ratts-Victory 161 kV	150	150	Solar	NRIS
J1398	IN	Pike	HE	Ratts 161 kV	150	150	Solar	NRIS
J1405	IN	Greene	DEI	Sanborn - Linton 138 kV	140	140	Solar	NRIS
J1407	IN	Porter	NIPSCO	South Valparaiso 138 kV	100	100	Solar	NRIS
J1408	IL	Macoupin	AMIL	Virden South 138 kV	100	100	Hybrid	NRIS
J1409	KY	Henderson	BREC	Henderson 161 kV	25	25	Battery Storage	NRIS
J1419	IN	Greene	DEI	Edwardsport - Amo 345 kV	250	250	Solar	NRIS
J1422	IL	Edwards	AMIL	Albion - Olney 138 kV	150	150	Hybrid	NRIS
J1447	IN	Porter	NIPSCO	Schahfer - Green Acres 345 kV	250	250	Solar	NRIS
J1448	KY	Caldwell	BREC	Hopkins - Caldwell County - Barkely 161 kV	150	150	Solar	NRIS
J1450	KY	Webster	BREC	Hopkins - Reid 161 kV	150	150	Solar	NRIS
J1453	IL	Mason	AMIL	Havana - Shockey 138 kV	188.3	188.3	Wind	NRIS
J1454	IL	Mason	AMIL	Mason 138 kV	58.5	58.5	Wind	NRIS
J1457	IN	Vermillion	DEI	Sugar Creek - Cayuga 345 kV	200	200	Solar	NRIS
J1464	IL	Macoupin, Morgan, Sangamon	AMIL	Meredosia - Austin 345 kV	592.8	592.8	Solar	NRIS
J1466	KY	Henderson	BREC	Reid 161 kV	250	250	Solar	NRIS
J1470	IN	Benton, Warren	DEI	Cayuga CT 345 kV	30	30	Wind	NRIS Only
J1475	IL	Clark	AMIL	Casey West - Sullivan 345 kV	75	75	Solar	NRIS
J1481	IN	Howard	DEI	Greentown 230 kV	200	200	Solar	NRIS

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MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1482	IN	White	NIPSCO	Monticello - Springboro 138 kV	150	150	Solar	NRIS
J1488	MO	Randolph	AMMO	McCredie - Montgomery 345 kV	500	500	High Voltage DC	NRIS
J1490	MO	Randolph	AMMO	McCredie - Montgomery 345 kV	1000	1000	High Voltage DC	NRIS

Table A-4: DPP 2019 Michigan Area Projects

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1192	MI	Eaton	METC	Oneida 138 kV	35	35	Solar	NRIS
J1194	MI	Midland, Saginaw	METC	Murphy 138 kV	225	225	Wind	NRIS
J1196	MI	Sanilac	ITCT	Greenwood - Rapson (Banner) 345 kV	200	200	Solar	NRIS
J1203	MI	Genesee	METC	Thetford 345 kV	125	125	Solar	NRIS
J1206	MI	Branch	METC	Newton 138 kV	50	50	Gas	NRIS
J1207	MI	Branch	METC	Newton 138 kV	50	50	Gas	NRIS
J1210	MI	Iosco	METC	Whittemore - Twining 138 kV	50	50	Solar	NRIS
J1211	MI	Isabella	CE	Vernon - METC Tap 138 kV	50	50	Solar	NRIS
J1224	MI	Washtenaw	ITCT	Majestic - Milan 345 kV	170	170	Solar	NRIS
J1226	MI	Lenawee	METC	Morocco 345 kV	200	200	Solar	NRIS
J1248	MI	Calhoun	METC	Marshall - Black Stone 138 kV	100	100	Solar	NRIS
J1255	MI	Ionia	METC	Marquette - Vergennes 138 kV	200	200	Solar	NRIS
J1262	MI	Wayne	ITCT	Waterman 120 kV	25	25	Gas	ERIS
J1292	MI	Ionia	METC	Thompson Road 138 kV	198	198	Wind	NRIS
J1297	MI	Shiawassee	METC	Cornell - Bell Road 138 kV	170	170	Solar	NRIS
J1307	MI	Eaton	METC	Oneida - Battle Creek 345 kV	150	150	Solar	NRIS
J1310	MI	Jackson	METC	Blackstone - Marshall 138 kV	125	125	Solar	NRIS
J1319	MI	Huron	ITCT	Rapson 120 kV	117.5	117.5	Wind	NRIS
J1320	MI	Branch	METC	Coldwater - Batavia 138 kV	150	150	Solar	NRIS
J1322	MI	Eaton	METC	Battle Creek - Oneida 345 kV	200	200	Solar	NRIS
J1327	MI	Ionia	METC	Marquette - Vergennes 138 kV	20	20	Battery Storage	NRIS
J1329	MI	Washtenaw	ITCT	Majestic - Milan 345 kV	20	20	Battery Storage	NRIS
J1331	MI	St. Clair	ITCT	Greenwood 120 kV	100	100	Solar	NRIS
J1350	MI	Monroe	METC	Milan - Monroe - Morocco 345 kV Point	150	150	Solar	NRIS
J1375	MI	Muskegon	METC	Du Pont - Cobb -White Lake 138 kV	100	100	Solar	NRIS

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MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1379	MI	Montcalm	METC	Eureka - Vestaburg 138 kV	200	200	Solar	NRIS
J1389	MI	Saginaw	METC	Murphy Station - Nelson Road 345 kV	120	120	Solar	NRIS
J1394	MI	Shiawassee	METC	Cornell - Layton 138 kV	150	150	Solar	NRIS
J1399	MI	Ingham	METC	Delhi - Churchill - Tompkins 138 kV	90	90	Solar	NRIS
J1401	MI	Branch	METC	Batavia - Wagner 138 kV	150	150	Solar	NRIS
J1403	MI	Ionia	METC	Marquette 138 kV	200	200	Solar	NRIS
J1430	MI	Calhoun	METC	Verona - Foundary 138 kV	100	100	Solar	NRIS
J1449	MI	Monroe	METC	Morocco 138 kV	100	100	Solar	NRIS
J1472	MI	Jackson	METC	Leoni - Parr Raod 138 kV	100	100	Solar	NRIS

Table A-5: DPP 2019 ATC Area Projects

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1214	WI	Dane	ATC	Rockdale 345 kV	300	300	Solar	NRIS
J1244	MI	Houghton	ATC	Winona - Toivola - Portage Tap - Atlantic Mine 69 kV	38.4	38.4	Wind	NRIS
J1251	MI	Marquette	ATC	Arnold - Perkins 138 kV	100	100	Solar	NRIS
J1252	MI	Marquette	ATC	Arnold - Perkins 138 kV	20	20	Battery Storage	NRIS
J1253	WI	Fond du Lac	ATC	South Fond Du Lac to Fitzgerald 345 kV	100	100	Solar	NRIS
J1304	WI	Rock	ATC	Paddock - Townline Road 138 kV	65	65	Solar	NRIS
J1305	WI	Green	ATC	Albany - Bass Creek 138 kV	49.9	49.9	Solar	NRIS
J1316	WI	Kenosha	ATC	Paris 138 kV	50	50	Battery Storage	NRIS
J1326	WI	Dane	ATC	Rockdale 345 kV	75	75	Battery Storage	NRIS
J1345	WI	Dodge	ATC	North Beaver Dam 69 kV	25	25	Battery Storage	NRIS
J1370	MI	Chippewa	ATC	9 Mile SW STA - Dafter 69 kV	50	50	Gas	NRIS
J1374	WI	Grant	ATC	ATC Ebenezer Sub on Lancaster-Eden 138 kV	95	95	Wind	NRIS
J1377	WI	Rock	ATC	Wempletown - Rockdale 345 kV	98.56	98.56	Solar	NRIS
J1410	WI	Dane	ATC	N. Madison 345 kV	300	300	Solar	NRIS
J1411	WI	Dane	ATC	N. Madison 345 kV	75	75	Battery Storage	NRIS
J1460	WI	Rock	ATC	Sunrise 138 kV	200	200	Solar	NRIS
J1483	WI	Iowa	ATC	Hill Valley 345 kV	99	99	Wind	NRIS

A.2 DPP 2018 April West Area Phase 2 Network Upgrades

Table A-6: DPP 2018 April West Phase 2 NUs

Network Upgrades	Owner	Mitigation
Hazel Creek-Scott County 345 kV	XEL	Base Case NU
Big Stone South-Twin Brooks 345 kV 2nd Circuit	OTP	DPP 2017 Aug. Ph2 NU
New J628 POI– Prairie 230 kV 2nd Circuit	GRE	DPP 2017 Aug. Ph2 NU
J1024 POI-Clarinda 161 kV	MEC	MEC: substation terminal equipment upgrades. New rating predicted to be 410 MVA.
J1181 POI-Hazleton 345 kV	MEC ITCM	MEC: MEC owns portion of line conductor. Structure replacements. New MEC Only rating expected to be 1094/1094 MVA. \$600,000 ITCM: ITCM records show a rating of 1006 MVA summer. \$0
J611 POI-Maryville 161 kV	MEC GMO	MEC: Existing MEC only rating expected to be 410 MVA after DPP 2016 AUG West line reconductor network upgrade. GMO: NU is not required unless it is identified as constraint in affected system study.
J720 POI-Lakefield 345 kV	ITCM	ITCM records show a rating of 932 MVA summer limit due to MEC facilities.
G16-017 Tap-Ft. Thompson 345 kV	WAPA	NU is not required unless it is identified as constraint in affected system study.
Wilmarth-Field North 345 kV	XEL	Fieldon-Wilmarth 345 rebuild
Wilmarth-Sheas Lake 345 kV	XEL	Wilmarth-Sheas Lake 345 kV Rebuild. NU in DPP 2017 Aug West Ph2
Split Rock-White 345 kV	XEL WAPA	XEL: Limiter is on WAPA facility. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.
Blue Lake-Scott Co 345 kV	XEL	Blue Lake-Scott County 345 kV Rebuild. NU in DPP 2017 Aug West Ph2
Field South-Field North 345 kV	XEL	bypassing the Fieldon series cap
Field South-Crandal 345 kV	XEL	Crandal-Fieldon 345 kV Rebuild. NU in DPP 2017 Aug West Ph2
Lyon Co-Hazel Creek 345 kV	XEL	Upgrade some sub equipment at Hazel that would put the rating to 1790 MVA normal and emergency
Helena-Sheas Lk 345 kV	XEL	Helena-Sheas Lake 345 kV Rebuild. NU in DPP 2017 Aug West Ph2
Sheyenne-Lake Park 230 kV	XEL MPC OTP	Sheyenne-Lake Park 230 kV Upgrade. NU in DPP 2017 Aug West Ph2

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Network Upgrades	Owner	Mitigation
Sheyenne-Mapleton 115 kV	XEL OTP	Sheyenne-Mapleton 115 kV Rebuild. NU in DPP 2017 Aug West Ph2
M.E. International-Westwood 115 kV	XEL GRE	XEL: Rebuild ME International to Westwood tap (2.1 miles) with 795 ACSS conductor and replace line switches. \$5,000,000 GRE: XEL facility
Austin-Murphy 161 kV	SMMPA	Austin-Murphy Creek 161 kV Rebuild. NU in DPP 2017 Aug West Ph2
Murphy-J1128 POI 161 kV	SMMPA	Murphy Creek – Hayward 161 kV Rebuild. NU in DPP 2017 Aug West Ph2
Hubbard-Badoura 230 kV	MP	Increase conductor clearance for 55C operation (15 miles)
West St. Cloud-Westwood 115 kV	GRE XEL	Rebuild 0.6 mi to 2x795 ACSS
Chub Lake 345-115-34.5 kV xfmr	GRE	Add second 345/115 kV transformer at Chub Lake
STSTPHNT-Fishill 115 kV	GRE	GRE: XEL owns equipment and line. \$0 XEL: Uprate line to 795 ACSS. \$5.1M
CSLTNET-Mapleton 115 kV	OTP	CSLTNET-Mapleton 115 kV Uprate. LPC NU in DPP 2017 Aug West Ph2
Buffalo 345-230-41.6 kV xfmr #2	OTP	Replace Buffalo transformer #2 with larger unit.
Hankinson-Forman 230 kV	OTP	Hankinson-Forman 230 kV Uprate. LPC NU in DPP 2017 Aug West Ph2
Wahpeton-Fergus Falls 230 kV	OTP MRES	Wahpeton-Fergus Falls 230 kV Uprate. NU in DPP 2017 Aug West Ph2
Audubon-Lake Park 230 kV	OTP	Audubon-Lake Park 230 kV Uprate. NU in DPP 2017 Aug West Ph2
Big Stone South 345-230-34.5 kV #1	OTP	Big Stone South Transformer #1 Upgrade. LPC NU in DPP 2017 Aug West Ph2
Big Stone South 345-230-34.5 kV #2	OTP	Big Stone South Transformer #2 Upgrade. LPC NU in DPP 2017 Aug West Ph2
Ellendale-County Line 69 kV	ITCM	Rebuild 5.79 miles
Hayward-County Line 69 kV	ITCM	Rebuild 13.32 miles
Osceola-Osceola REC 69 kV	ITCM CIPCO	ITCM: ITCM rating 42/44 MVA SN/SE CIPCO: NU is not required unless identified in affected system study
Adams-Hayward 161 kV	ITCM	Adams-Hayward 161 kV Uprate. NU in DPP 2017 Aug West Ph2
Hazleton-Arnold 345 kV	ITCM MEC	MEC: MEC owns a portion of line conductor. Structure replacements. New MEC Only rating expected to be

Network Upgrades	Owner	Mitigation
		1139/1139 MVA. \$800K ITCM: Structure replacements. New ITCM rating 1285 MVA/SN/SE. \$480K
Hazleton-Hickory Crk 345 kV	ITCM	ITCM rating 1569 MVA SN/SE
Adams-Creston 161 kV	MEC WAPA	Structure replacements. New rating expected to be 182/182 MVA
Webster-Wright 161 kV	MEC	Reconductor line and substation terminal equipment upgrades. New rating predicted to be 315/335 MVA.
Franklin-Wall Lake 161 kV	MEC	Reconductor line. New rating predicted to be 335/335 MVA.
Red Willow-Mingo 345 kV	NPPD SUNC	NU is not required unless it is identified as constraint in affected system study.
Ward-Bismark 230 kV	BEPC WAPA	NU is not required unless it is identified as constraint in affected system study.
Ft. Thompson 345 kV bus 1-3 TIE	WAPA	NU is not required unless it is identified as constraint in affected system study.
Wilton-Winger 230 kV	MPC OTP	OTP: Sufficient for the flows of 395.2 MVA. \$0
Ellendale-Aberdeen 115 kV	MDU NWE	Ellendale-Aberdeen Jct 115 kV Upgrade. LPC NU in DPP 2017 Aug West Ph2
Heskett-J302 POI 230 kV	MDU	Add a breaker at Merricourt and build 2nd Mandan-Napoleon SW 230 kV line w/ 1272 ACSS (includes river crossing).
Heskett-Mandan 230 kV	MDU	The Heskett 230 kV sub has an estimated retirement date of 7/23/2021. When the Heskett 230 kV sub is retired, this constraint will no longer exist.
Heskett 230-115-13.8 kV xfmr	MDU	The Heskett 230/115 kV transformer is planned to be moved to Mandan to function in parallel to the existing Mandan 230/115 kV transformer. Once the Heskett 230/115 kV transformer is moved, this constraint will no longer exist.
Mandan-Ward 230 kV	MDU BEPC	MDU's rating at Mandan is 956 MVA. MPC owns equipment at Mandan. WAPA owns the line. BEPC owns Ward.
Fox Tail-Tatanka North 230 kV	MDU	Major substation upgrades at Tatanka North 230 (new rating: 610/610 MVA [N/E]).
Fox Tail-Ellendale 230 kV	MDU	Foxtail-Ellendale 230 kV Rebuild. LPC NU in DPP 2017 Aug West Ph2
Merricourt-Wishek 230 kV	MDU	Rebuild/reconductor line with 1272 ACSS (new rating: 797/797 MVA [N/E]).

Model Development

Network Upgrades	Owner	Mitigation
Merricourt-Tatanka North 230 kV	MDU	Rebuild line to ACSS. New Rating: 478/478 MVA.
Merricourt-Ellendale 230 kV	MDU	Rebuild/reconductor Merricourt-Ellendale 230 kV line w/ 1272 ACSS (new rating: 776/776 MVA [N/E]).
Wishek-J302 POI 230 kV	MDU	J302 POI-Wishek 230 kV Rebuild. LPC NU in DPP 2017 Aug West Ph2
Wabaco-Rochester 161 kV	DPC	Rebuild line with 795 ACSS conductor. \$11M. MTEP Appendix A project
Wabaco-Alma 161 kV	DPC	Rebuild line with 795 ACSS conductor
Voltage constraints	OTP	Add 1×40 Mvar switched capacitor at Oakes 230 kV (620362)
150 MVAR capacitor at Bison 345 kV	XEL	150 MVAR capacitor at Bison 345 kV
Three 50 MVAR capacitor at Maple River 230 kV	MPC	Three 50 MVAR capacitor at Maple River 230 kV
Two 75 MVAR SVCs at J873 POI 345 kV	MEC	Two 75 MVAR SVCs at J873 POI 345 kV
Two 50 MVAR capacitors at Wahpeton 230 kV	OTP	Two 50 MVAR capacitors at Wahpeton 230 kV
Build Alexandria - Twin Brooks 345 kV line	OTP MRES	Build Alexandria - Twin Brooks 345 kV line
Four 50 MVAR capacitors at Alexandria 345 kV	MRES	Four 50 MVAR capacitors at Alexandria 345 kV
200 MVAR SVC at Wahpeton 230 kV	OTP	200 MVAR SVC at Wahpeton 230 kV
Murray-I35 Tap 69 kV	CIPCO	Rebuild 5.68 miles with T2-4/0 ACSR at \$350k per mile.

A.3 DPP 2017 August West Area Phase 2 Network Upgrades

Table A-7: DPP 2017 August West Phase 2 NUs

Network Upgrades	Owner
Murphy Creek – Hayward 161 kV Rebuild	SMMPA
Austin-Murphy Creek 161 kV Rebuild	SMMPA
Merricourt-Wishek 230 kV Rebuild	MDU
J302 POI-Heskett 230 kV Rebuild	MDU
J302 POI-Wishek 230 kV Rebuild	MDU
Heskett-Mandan 230 kV Uprate	MDU
Sheyenne-Lake Park 230 kV Uprate	XEL, OTP, MPC
Sheyenne-Mapleton 115 kV Rebuild	XEL, OTP
Wilton-Bemidji 115 kV Uprate	OTP
Crookston-Fertile 115 kV Uprate	OTP
Crookston-Falconer 115 kV Uprate	OTP
Blackhawk 345/161 kV Transformer Uprate	MEC
Hubbard-Audubon 230 kV Uprate	MP, GRE, XEL, OTP
Walters-Alden 69 kV Rebuild	ITCM
Alden-Aleawst 69 kV Rebuild	ITCM
Rice-Cresco 69 kV Rebuild	ITCM
Rice-Saratoga 69 kV Uprate	ITCM
Decorah-Madison 69 kV Uprate	ITCM
Glenworth-Hayward 161 kV Uprate	ITCM
Adams-Hayward 161 kV Uprate	ITCM
New J628 POI– Prairie 230 kV 2nd Circuit	GRE
J628 POI-Prairie 230 kV Rebuild	GRE
Helena-Chub Lake 2nd Circuit	GRE/CAPX
Canby-Granite Falls 115 kV Rebuild	OTP
Wahpeton-Fergus Falls 230 kV Uprate	OTP
Audubon-Lake Park 230 kV Uprate	OTP
Big Stone-Big Stone South 230 kV Circuit 1 Rebuild	OTP
Big Stone-Big Stone South 230 kV Circuit 2 Rebuild	OTP
Big Stone-Twin Brooks 2nd Circuit	OTP
Big Stone-Blair 230 kV Rebuild	OTP/NWE

Model Development

Network Upgrades	Owner
Adams 345/161 kV Transformer Upgrade	XEL
Wilmarth-Sheas Lake 345 kV Rebuild	XEL
Blue Lake-Scott County 345 kV Rebuild	XEL
Crandal-Fieldon 345 kV Rebuild	XEL
Helena-Sheas Lake 345 kV Rebuild	XEL
Jerico-Howard Tap 69 kV Uprate	DPC
Two J873 POI 75 MVAR SVC's	MEC
Coyote Transformer Upgrade	MDU
Coyote-Beulah 115 kV Uprate	MDU
Six 50 Mvar Capacitors at Wahpeton 230 kV	OTP
One 150 Mvar Capacitor at Bison 345 kV	XEL
Three 50 Mvar Capacitors at Maple River 230 kV	MPC

A.4 Model Review Comments

Table A-8: Model Review Comments

Company	Python/ Idev File Name	2024 SH Benchmark	2024 SH Study Discharging	2024 SH Study Charging	2024 SPK Benchmark	2024 SPK Study Discharging
PJM	Master_Remove PJM withdrawn prjs.py	X	X	X	X	X
SPP	Master_Remove SPP withdrawn prjs.py	X	X	X	X	X
SPP	DISIS-2017-001_West_Master.py	X	X	X	X	X
SPP	SH-Dispatch DISIS-2017-001.py	X	X	X		
SPP	PK-Dispatch DISIS-2017-001.py				X	X
ICs	J1190_upd.py	X	X	X	X	X
ICs	J1221_POI_WinRiver_to_Riverton.py	X	X	X	X	X
ICs	J1298_POI_change.py	X	X	X	X	X
ICs	J1312_upd.py	X	X	X	X	X
ICs	J1366_Out_of_Service.py	X	X	X	X	X
ICs	J1343_Out_of_Service.py	X	X	X	X	X
ICs	J1344_Out_of_Service.py	X	X	X	X	X
ICs	Bus_Code.py	X	X	X	X	X
ICs	J1313_upd.py	X	X	X	X	X
ICs	J1314_upd.py	X	X	X	X	X
ICs	J1315_upd.py	X	X	X	X	X
ICs	J1325_upd.py	X	X	X	X	X
ICs	J1337_upd.py	X	X	X	X	X
ICs	J1193_Out_of_Service.py	X	X	X	X	X
ICs	J1480_Out_of_Service.py	X	X	X	X	X
ICs	Update Vref_347832.py	X	X	X	X	X
ICs	correct J1262.py	X	X	X	X	X
ICs	J1371 correction.py	X	X	X	X	X
ICs	J1371 Dispatch_SH.py	X	X	X		
ICs	DPP-2019-C1_West_J1395_2020-10-22.idv	X	X	X	X	X
ICs	PK-J1395 Dispatch.py				X	X
ICs	DPP-2019-C1_West_J1456_2020-10-22.idv	X	X	X	X	X
ICs	J1467_PF_upd.py	X	X	X	X	X
ICs	J1468_PF_upd.py	X	X	X	X	X
ICs	SUM_PGEN_PMIN_Fix.py	X	X	X	X	X

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Company	Python/ Idev File Name	2024 SH Benchmark	2024 SH Study Discharging	2024 SH Study Charging	2024 SPK Benchmark	2024 SPK Study Discharging
ICs	MBASE_fix.py	X	X	X	X	X
ICs	Rating_Fix.py	X	X	X	X	X
ICs	LowVoltage_fix.py	X	X	X	X	X
MEC	MEC_DPP_2019_West_Ph1_SH_Updates.py	X	X	X		
MEC	MEC_DPP_2019_West_Ph1_SUM_Updates.py				X	X
MEC	Add J449 Offline.idv	X	X	X	X	X
MEC	RMV Duplicated J614.py	X	X	X	X	X
MEC	ADD J530 Offline.idv	X	X	X	X	X
MEC	Correct J534 area.py	X	X	X	X	X
MEC	Correct J1341 gentie.py	X	X	X	X	X
MEC	Correct 631144-41814-44137-631139.py	X	X	X	X	X
MEC	RMV prior DPP NUs.py	X	X	X	X	X
MEC	dispatch Gen-SH Dis Study.py		X			
MEC	dispatch Gen-SH Ch Study.py			X		
MEC	dispatch Gen-SH Bench.py	X				
MEC	dispatch Gen-PK Study.py					X
MEC	dispatch Gen-PK Bench.py				X	
MDU	DPP19-West-Ph1-2024SH_MDU-Updates.idv	X	X	X		
MDU	DPP19-West-Ph1-2024SUM_MDU-Updates.idv				X	X
MDU	DPP-2019-West-Ph1_MDU-Updates.con	X	X	X	X	X
ATC	Remove_SFL_FIZ.idv	X	X	X	X	X
GRE	Correct J897 POI.py	X	X	X	X	X
SPTI	J1359_Control_Mode_Change.py	X	X	X	X	X
SPTI	Correct J1122 branches.py	X	X	X	X	X
SPTI	Correct J1122 modeling.py	X	X	X	X	X
SPTI	Correct J1468 Modeling.py	X	X	X	X	X
SPTI	Dispatch J1468_SH-Discharge.py		X			
SPTI	Dispatch J1468_SH-Charge.py			X		
MPC	MPC-DPP-2019-Cycle-Phase 1 Updates.idv	X	X	X	X	X
MRES	2019 West Hutch Voltages.idv	X	X	X	X	X
MRES	MRES OJ Line Rating Change.idv	X	X	X	X	X
OTP	OTP_2019_West_Ph1_corrections.idv	X	X	X	X	X

A.5 MISO North as the Study Sink

Table A-9: MISO North as the Study Sink

Area #	Area Name	Area #	Area Name
207	HE	608	MP
208	DEI	613	SMMPA
210	SIGE	615	GRE
216	IPL	620	OTP
217	NIPS	627	ALTW
218	METC	633	MPW
219	ITC	635	MEC
295	WEC	661	MDU
296	MIUP	663	BEPC-MISO
314	BREC	680	DPC
333	CWLD	694	ALTE
356	AMMO	696	WPS
357	AMIL	697	MGE
360	CWLP	698	UPPC
361	SIPC	701	North Prior
600	Xcel		

A.6 PJM Market as PJM Projects Sink

Table A-10: PJM Market as PJM Projects Sink

Area #	Area Name	Area #	Area Name
201	AP	230	PECO
202	ATSI	231	PSE&G
205	AEP	232	BGE
209	DAY	233	PEPCO
212	DEO&K	234	AE
215	DLCO	235	DP&L
222	CE	236	UGI
225	PJM	237	RECO
226	PENELEC	320	EKPC
227	METED	345	DVP
228	JCP&L	363	LGEE
229	PPL		

A.7 SPP Market as SPP Projects Sink

Table A-11: SPP Market as SPP Projects Sink

Area #	Area Name	Area #	Area Name
515	SWPA	541	KCPL
520	AEPW	542	KACY
523	GRDA	544	EMDE
524	OKGE	545	INDN
525	WFEC	546	SPRM
526	SPS	640	NPPD
527	OMPA	645	OPPD
531	MIDW	650	LES
534	SUNC	652	WAPA
536	WERE	659	BEPC-SPP

A.8 Contingency Files used in Steady-State Analysis

Table A-12: List of Contingencies used in Steady-State Analysis

Contingency File Name	Description	Shoulder	Peak
Automatic single element contingencies	Single element outages at buses 60 kV and above in the study region	x	x
CC Bipole Events.con	Specified category P1, P7 contingencies in GRE Coal Creek	x	x
HVDC_Red_2024SH.con	Contingencies with HVDC reduction	x	
HVDC_Red_2024PK.con	Contingencies with HVDC reduction		x
MEC-DPP2019WESTPH1-Cat P1-2020.11.06.con	Specified category P1 contingencies in MEC	x	x
MEC-DPP2019WESTPH1-Cat P2-2020.11.06.con	Specified category P2 contingencies in MEC	x	x
MEC-DPP2019WESTPH1-Cat P5-2020.11.06.con	Specified category P5 contingencies in MEC	x	x
MEC-DPP2019WESTPH1-Cat P7-2020.11.06.con	Specified category P7 contingencies in MEC	x	x
MISO19_2024_SUM_TA_P1_ATC.con	Specified category P1 contingencies in ATC	x	x
MISO19_2024_SUM_TA_P1_IOWA.con	Specified category P1 contingencies in Iowa	x	x
MISO19_2024_SUM_TA_P1_MINN-DAKS.con	Specified category P1 contingencies in Minnesota, Dakotas	x	x
MISO19_2024_SUM_TA_P1_P2_P4_P5_NoLoadLoss.con	Specified category P1, P2, P4, P5 contingencies in MISO cLASSIC	x	x
MISO19_2024_SUM_TA_P2_P4_P5_P7_LoadLoss.con	Specified category P2, P4, P5, P7 contingencies in MISO Classic	x	x
MISO19_2024_SUM_TA_P1_AMEREN.con	Specified category P1 contingencies in Ameren	x	x
2020_RTEP_Single_2017Aug-Updated-3-4-19.con	Specified single contingencies in PJM	x	x
2020_RTEP_Bus_AC2-updated-3-4-19.con	Specified bus contingencies in PJM	x	x
2020_RTEP_Line_FB_2017Aug-updated-3-4-19.con	Specified breaker failure contingencies in PJM	x	x
2020_RTEP_Tower_AC2-updated-3-4-19.con	Specified common structure contingencies in PJM	x	x
160303-KACY_P1.con	Specified category P1 contingencies in KACY	x	x
160303-KACY_P2.con	Specified category P2 contingencies in KACY	x	x
AECI-AMMO.CON	Specified contingencies between AECI and AMMO	x	x
KCPL_P1.con	Specified category P1 contingencies in KCPL	x	x
KCPL_P2.con	Specified category P2 contingencies in KCPL	x	x
KCPL_P4.con	Specified category P4 contingencies in KCPL	x	x

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Contingency File Name	Description	Shoulder	Peak
KCPL_P5.con	Specified category P5 contingencies in KCPL	x	x
KCPL_P7.con	Specified category P7 contingencies in KCPL	x	x

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Model Data

B.1 Power Flow Model Data

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B.2 2024 Slider Diagrams

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Appendix
C

Reactive Power Requirement Analysis Results (FERC Order 827)

Table C-1: Reactive Power Requirement Analysis Results

			Lagging Power Factor Results				Leading Power Factor Results					
Project #	TYPE	HV Side Bus #	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?	Inverter Inherent Power Factor	Shunt Compensation
J1190	Solar	41903	196.70	50.92	0.9681	No	196.28	-138.72	-0.8166	Yes	SMA 2.5 MW Inverter PF: +/- 0.909	None
J1212	Solar	42123	59.32	23.51	0.9297	Yes	59.29	-36.63	-0.8507	Yes	HEM 3.465 MVA Inverer PF: +/- 0.935	1×12 MVAR
J1218	Solar	42299	198.06	64.25	0.9512	No	197.82	-146.60	-0.8034	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	2×3 MVAR
J1221	Solar	42306	148.45	46.40	0.9545	No	148.26	-109.57	-0.8042	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	1×4 MVAR
J1229	Battery	42299	49.49	17.33	0.9438	Yes	49.40	-37.66	-0.7953	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	1×2 MVAR
J1230	Battery	42306	39.59	15.35	0.9324	Yes	39.53	-32.11	-0.7762	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	1×2 MVAR
J1270	Wind	42703	197.11	48.06	0.9715	No	196.81	-112.35	-0.8685	Yes	Vestas 5.1 MVA WTG PF: +/- 0.95	2×13 MVAR
J1284	Solar	42859	247.56	80.86	0.9506	No	247.24	185.92	-0.7992	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	2×4 MVAR
J1285	Battery	42859	49.49	17.33	0.9438	Yes	49.40	-37.67	-0.7952	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	1×2 MVAR
J1298	Solar	41283	147.59	36.15	0.9713	No	147.14	-111.39	-0.7973	Yes	PE HEM 3.0 MW Inverter PF: +/- 0.909	None
J1300	Solar	43003	118.10	29.21	0.9707	No	117.74	-89.41	-0.7964	Yes	PE HEM 3.0 MW Inverter PF: +/- 0.909	None
J1312	Solar	43123	49.49	24.53	0.8960	Yes	49.50	-25.60	-0.8882	Yes	PE HEMK 3.432 MW Inverter PF: +/- 0.95	2×7 MVAR
J1313	Solar	43133	98.70	51.89	0.8851	Yes	98.54	-46.52	-0.9043	Yes	PE HEMK 3.59 MW Inverter PF: +/- 0.95	2×14 MVAR
J1314	Solar	43143	99.02	49.15	0.8957	Yes	99.00	-47.76	-0.9007	Yes	PE HEMK 3.59 MW Inverter PF: +/- 0.95	1×28 MVAR
J1315	Wind	43213	593.70	355.38	0.8580	Yes	594.12	-305.46	-0.8893	Yes	Vestas 3.45 MW WTG PF: + 0.90/ - 0.92	6×25 MVAR
J1321	Battery	43213	64.33	21.25	0.9495	Yes	64.25	-48.59	-0.7976	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	1×2 MVAR
J1325	Solar	43256	197.14	93.20	0.9041	Yes	196.82	-97.70	-0.8957	Yes	PE HEMK 5.39 MW Inverter PF: +/- 0.95	2×26 MVAR
J1328	Solar	43283	98.17	26.87	0.9645	No	98.01	-56.72	-0.8655	Yes	TMEIC 2.50 MW Inverter PF: +/- 0.925	None
J1337	Solar	43378	296.73	135.66	0.9095	Yes	296.85	-137.49	-0.9074	Yes	PE HEMK 3.59 MW Inverter PF: +/- 0.95	3×26 MVAR
J1341	Wind	43416	391.11	125.26	0.9524	No	390.18	-276.38	-0.8160	Yes	GE 2.52 MW WTG PF: +/- 0.90	None
J1349	Hybrid	43499	325.28	141.13	0.9174	Yes	324.34	-219.86	-0.8277	Yes	TMEIC 2.5 MW Inverter PF: +/- 0.925 GE 2.82 MW WTG PF: +/- 0.90	1×10 MVAR 2×15 MVAR
J1365	Wind	43653	165.72	16.72	0.9949	No	165.24	-100.24	-0.8550	Yes	Vestas 2.2 MW WTG PF: +/- 0.95	None
J1371	Wind	43713	152.52	36.59	0.9724	No	152.27	-107.24	-0.8176	Yes	Vestas 4.2 MW WTG PF: +0.893 /-0.93	None

Table C-1: Reactive Power Requirement Analysis Results

			Lagging Power Factor Results				Leading Power Factor Results					
Project #	TYPE	HV Side Bus #	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?	Inverter Inherent Power Factor	Shunt Compensation
J1395	Solar	43956	199.52	56.17	0.9626	No	199.28	-149.97	-0.7990	Yes	PE HEMK 3.3 MW Inverter PF: +/- 0.90	None
J1413	Solar	44136	294.52	119.18	0.9270	Yes	294.28	-183.62	-0.8484	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	2×4 MVAR
J1416	Solar	44163	145.93	53.68	0.9385	Yes	145.52	-101.13	-0.8212	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	1×4 MVAR
J1417	Battery	44173	73.94	40.16	0.8787	Yes	74.07	-30.25	-0.9258	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	None
J1426	Solar	44263	98.70	-39.46	0.9285	Yes	98.66	-63.90	-0.8393	Yes	TMEIC 3.71 MW Inverter PF: +/- 0.916	2×7 MVAR
J1427	Wind	44275	391.58	134.37	0.9459	Yes	389.77	-307.34	-0.7852	Yes	GE 3.03 MW WTG PF: +/- 0.90	2×12 MVAR
J1428	Solar	44283	197.52	74.25	0.9360	Yes	197.39	-131.59	-0.8321	Yes	TMEIC 3.71 MW Inverter PF: +/- 0.916	2×14 MVAR
J1432	Wind	44323	245.16	99.49	0.9266	Yes	244.70	-179.27	-0.8067	Yes	GE 3.03 MW PF: +/- 0.90	2×15 MVAR
J1435	Battery	44353	98.56	34.82	0.9429	Yes	98.50	-63.69	-0.8397	Yes	PE 1.29 MVA Inverter PF: +/- 0.923	1×14 MVAR
J1438	Solar	44383	98.71	36.55	0.9378	Yes	98.68	-60.37	-0.8530	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.926	2×7 MVAR
J1444	Solar	44443	197.55	68.55	0.9447	Yes	197.48	-123.96	-0.8470	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.926	2×14 MVAR
J1445	Solar	44453	98.71	38.87	0.9305	Yes	98.66	-63.61	-0.8405	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.917	2×7 MVAR
J1446	Solar	44463	147.71	53.65	0.9399	Yes	147.59	-93.47	-0.8448	Yes	PE 1.29 MVA Inverter PF: +/- 0.926	2×11 MVAR
J1456	Wind	44673	294.91	77.35	0.9673	No	294.54	-132.65	-0.9118	Yes	Vestas 2.0 MW WTG PF: +/- 0.95	3×3 MVAR
J1461	Solar	44613	49.34	22.38	0.9107	Yes	49.31	-35.66	-0.8103	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.893	1×7 MVAR
J1467	Wind	44673	147.72	58.49	0.9298	Yes	147.64	-106.90	-0.8100	Yes	GE 3.36 MVA WTG PF: +/- 0.9	2×9 MVAR
J1468	Battery	44683	98.42	40.55	0.9246	Yes	98.33	-72.04	-0.8067	Yes	PE 1.29 MVA Inverter PF: +/- 0.90	1×14 MVAR
J1471	Solar	44713	197.62	71.59	0.9402	Yes	197.43	-123.10	-0.8486	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.926	2×14 MVAR
J1473	Solar	44733	98.71	39.07	0.9298	Yes	98.66	-63.38	-0.8414	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.917	2×7 MVAR
J1474	Solar	44743	197.11	70.72	0.9413	Yes	196.93	-124.38	-0.8455	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.926	2×14 MVAR
J1476	Battery	41323	24.76	14.46	0.8635	Yes	24.73	-16.13	-0.8376	Yes	PE 3.24 MVA Inverter PF: +/- 0.86	None
J1477	Battery	41313	49.35	25.52	0.8883	Yes	49.35	-34.42	-0.8202	Yes	PE 3.43 MVA Inverter PF: +/- 0.86	None
J1478	Battery	41353	24.73	13.99	0.8704	Yes	24.73	16.13	-0.8376	Yes	PE 3.24 MVA Inverter PF: +/- 0.86	None
J1479	Battery	44793	97.96	37.59	0.9336	Yes	97.43	-89.23	-0.7375	Yes	PE 3.43 MVA Inverter PF: +/- 0.86	None
J1485	Wind	44856	247.68	75.08	0.9570	No	247.36	-182.87	-0.8041	Yes	GE 2.52 MW WTG PF: +/- 0.90	2×3 MVAR

Table C-1: Reactive Power Requirement Analysis Results

			Lagging Power Factor Results				Leading Power Factor Results					
Project #	TYPE	HV Side Bus #	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?	Inverter Inherent Power Factor	Shunt Compensation
J1487	Wind	44873	120.76	47.19	0.9314	Yes	120.38	-87.63	-0.8085	Yes	GE 2.52 MW WTG PF: +/- 0.90	1×11 MVAR

2024 Summer Peak Contingency Analysis Results

D.1 Constraints in 2024 Summer Peak (SPK) Discharging Condition

Table D-1: 2024 SPK Discharging System Intact Thermal Constraints

Table D-2: 2024 SPK Discharging System Intact Voltage Constraints

Table D-3: 2024 SPK Discharging Category P1 Thermal Constraints

Table D-4: 2024 SPK Discharging Category P1 Voltage Constraints

Table D-5: 2024 SPK Discharging Category P2-P7 Thermal Constraints

Table D-6: 2024 SPK Discharging Category P2-P7 Voltage Constraints

Table D-7: 2024 SPK Discharging Non-Converged Contingencies

Table D-8: 2024 SPK Discharging Non-Converged Contingencies DCCC Results

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2024 Summer Shoulder Contingency Analysis Results

E.1 Constraints in 2024 Summer Shoulder (SH) Discharging Condition

Table E-1: 2024 SH Discharging System Intact Thermal Constraints

Table E-2: 2024 SH Discharging System Intact Voltage Constraints

Table E-3: 2024 SH Discharging Category P1 Thermal Constraints

Table E-4: 2024 SH Discharging Category P1 Voltage Constraints

Table E-5: 2024 SH Discharging Category P2-P7 Thermal Constraints

Table E-6: 2024 SH Discharging Category P2-P7 Voltage Constraints

Table E-7: 2024 SH Discharging Non-Converged Contingencies

Table E-8: 2024 SH Discharging Non-Converged Contingencies DCCC Results

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E.2 Constraints in 2024 Summer Shoulder (SH) Charging Condition

Table E-9: 2024 SH Charging System Intact Thermal Constraints

Table E-10: 2024 SH Charging System Intact Voltage Constraints

Table E-11: 2024 SH Charging Category P1 Thermal Constraints

Table E-12: 2024 SH Charging Category P1 Voltage Constraints

Table E-13: 2024 SH Charging Category P2-P7 Thermal Constraints

Table E-14: 2024 SH Charging Category P2-P7 Voltage Constraints

Table E-15: 2024 SH Charging Non-Converged Contingencies

Table E-16: 2024 SH Charging Non-Converged Contingencies DCCC Results

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Local Planning Criteria Analysis Results

F.1 GRE Local Planning Criteria Analysis Results

F.1.1 J1221/J1230 GRE Local Planning Criteria Analysis

Below is the J1221/J1230 GRE local planning criteria analysis report.

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F.1.2 J1426 GRE Local Planning Criteria Analysis

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F.2 OTP LPC Analysis

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F.3 DPC LPC Analysis

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MWEX Voltage Study Details

Below is the MWEX voltage stability study report provided by ATC.

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2024 Cost Allocation Results

H.1 Distribution Factor (DF) and MW Contribution Results for Cost Allocation in 2024

Table H-1: Distribution Factor and MW Contribution on Constraints for Thermal NU Cost Allocation

Table H-2: Voltage Impact on MISO Voltage NUs Cost Allocation

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H.2 Cost Allocation Details

Table H-3: Network Upgrades Cost Allocation in 2024

Network Element		Regional Name		Owner	Unit	2190	2212	2218	2221	2227	2228	2229	2234	2235	2236	2237	2241	2242	2243	2244	2245	2246	2247	2248	2249	2250	2251	2252	2253	2254	2255	2256	2257	2258	2259	2260	2261	2262	2263	2264	2265	2266	2267	2268	2269	2270	2271	2272	2273	2274	2275	2276	2277	2278	2279	2280	2281	2282	2283	2284	2285	2286	2287	2288	2289	2290	2291	2292	2293	2294	2295	2296	2297	2298	2299	2300	2301	2302	2303	2304	2305	2306	2307	2308	2309	2310	2311	2312	2313	2314	2315	2316	2317	2318	2319	2320	2321	2322	2323	2324	2325	2326	2327	2328	2329	2330	2331	2332	2333	2334	2335	2336	2337	2338	2339	2340	2341	2342	2343	2344	2345	2346	2347	2348	2349	2350	2351	2352	2353	2354	2355	2356	2357	2358	2359	2360	2361	2362	2363	2364	2365	2366	2367	2368	2369	2370	2371	2372	2373	2374	2375	2376	2377	2378	2379	2380	2381	2382	2383	2384	2385	2386	2387	2388	2389	2390	2391	2392	2393	2394	2395	2396	2397	2398	2399	2400	2401	2402	2403	2404	2405	2406	2407	2408	2409	2410	2411	2412	2413	2414	2415	2416	2417	2418	2419	2420	2421	2422	2423	2424	2425	2426	2427	2428	2429	2430	2431	2432	2433	2434	2435	2436	2437	2438	2439	2440	2441	2442	2443	2444	2445	2446	2447	2448	2449	2450	2451	2452	2453	2454	2455	2456	2457	2458	2459	2460	2461	2462	2463	2464	2465	2466	2467	2468	2469	2470	2471	2472	2473	2474	2475	2476	2477	2478	2479	2480	2481	2482	2483	2484	2485	2486	2487	2488	2489	2490	2491	2492	2493	2494	2495	2496	2497	2498	2499	2500	2501	2502	2503	2504	2505	2506	2507	2508	2509	2510	2511	2512	2513	2514	2515	2516	2517	2518	2519	2520	2521	2522	2523	2524	2525	2526	2527	2528	2529	2530	2531	2532	2533	2534	2535	2536	2537	2538	2539	2540	2541	2542	2543	2544	2545	2546	2547	2548	2549	2550	2551	2552	2553	2554	2555	2556	2557	2558	2559	2560	2561	2562	2563	2564	2565	2566	2567	2568	2569	2570	2571	2572	2573	2574	2575	2576	2577	2578	2579	2580	2581	2582	2583	2584	2585	2586	2587	2588	2589	2590	2591	2592	2593	2594	2595	2596	2597	2598	2599	2600	2601	2602	2603	2604	2605	2606	2607	2608	2609	2610	2611	2612	2613	2614	2615	2616	2617	2618	2619	2620	2621	2622	2623	2624	2625	2626	2627	2628	2629	2630	2631	2632	2633	2634	2635	2636	2637	2638	2639	2640	2641	2642	2643	2644	2645	2646	2647	2648	2649	2650	2651	2652	2653	2654	2655	2656	2657	2658	2659	2660	2661	2662	2663	2664	2665	2666	2667	2668	2669	2670	2671	2672	2673	2674	2675	2676	2677	2678	2679	2680																												
Buckley Creek 150' x 150' (150' x 150')		150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	150'	1

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